

Professional
Road Race
Tactics
For
Amateur
Cycling Teams
By David Harrison

Professional Road Race Tactics For Amateur Cycling Teams

©2004 David Harrison
All Rights Reserved

Published by RBR Publishing Company, 3141 Forest Knolls Dr., Chapel Hill, NC 27516, USA. 2001 RBR Publishing Company. All rights reserved. No part of this publication may be reproduced or transmitted in any form or by any means, electronic, mechanical, magnetic, and photographic including photocopying, recording or by any information storage or retrieval system without the prior written permission of RBR Publishing Company or David Harrison. No patent liability is assumed with respect to the use of the information contained in this publication. Neither is any liability assumed for damages resulting from the use of the information contained in this publication.

<http://www.RoadBikeRider>.

Photo, page 43, courtesy of Fitchburg Longsjo Classic, @2002, used with permission.

New to Acrobat Reader or eBooks?

Now that your eBook is open, here are a few tips for a great reading experience.

First, use Acrobat Reader's features to create the screen view you like best.

1. Maximize the window size to fill your screen.
2. Look at the top row of toolbar icons. Near the right end, you'll see representations of three pages, each with a corner turned down. Click on them to find a view you like.
3. Look in the bar at the bottom of the Reader screen. You'll see tiny representations of a single page, a scrolling page, and side-by-side scrolling pages. Click on the first two to find the scrolling action you like.

TIP! For the most book-like reading experience on a large monitor, click the middle page icon on top and the side-by-side page icon at the bottom.

4. If necessary, use the toolbar zoom menu or enter your own percentage value in the box (then press enter) to help the eBook pages fit on your screen.
5. Use the pull-down View menu for another way to activate viewing options.

How to Navigate This eBook

- For quick navigation, all [blue](#) titles in the text are linked to their respective definitions in the glossary. (And [blue](#) URLs will open web pages if you're reading this eBook while online.)
- To move through the eBook in clean one-page increments, click the triangular symbols in the Reader top or bottom bars.
- Click on the Hand Tool (hand icon) to turn your mouse pointer into a page grabber. Hold down your left mouse button and move the mouse to scroll the page.
- Use the **Find** function (binocular icon) to instantly locate each use of a word, term or name in the eBook.

Experiment with Reader tools to get the look and reading experience you like. If you have difficulty, e-mail to CustomerService@roadbikerider.com and we will do our best to help you.

TIP! If you want to print the eBook, set your printer to draft mode for speed and economy. Then punch holes in the pages and store them in a loose-leaf binder.

Thank you to Raj, Shaun, Matt, and Nathan who
endured my tactical experiments and helped me find the strategies that work best.

Table of Contents

Chapter 1 Basic Strategy Introduced	7
Chapter 2 Building a Team	9
Joining a Team	9
Leading a Team	11
Yearly Schedule	12
Base Miles and Meeting #1	14
Team Meeting #2: Building Speed, and Learning Tactics	16
Early Season Races	22
Mid-Season Races	23
Transition Time	23
Fall and Cyclocross	24
Chapter 3 Building Team Skills	26
Pacelines	27
How to Paceline	28
Two Riders	28
Adding More Riders	29
Starting the Paceline	31
Maintaining the Paceline	32
Changing Terrain	33
Additional Paceline Techniques	35
The Gatekeeper	38
Paceline Tactics: Applying the Paceline to Race Situations	40
Pacelines Involving other Teams	44
Riding Within the Peloton and Protecting a Team Leader	49
Protecting your Territory	50
Reading the Peloton	51
Changing the Amoeba into the Train	54
The Chase	56
The Leadout	56
Other Duties of the Domestiques	62
Small Group Strategies	64
Teammates in the Small Group	68
Practicing	70
Track Practice	72
Training Races	75
Chapter 4 Communication	78
Chapter 5 Winning the race	83
Rider Responsibilities	83
Defensive Tactics	85
Positioning Tactics	85
Offensive Tactics	86
Blocking the Chase	91
Race Progression	92

Understanding and Beating Individual Racers	94
Specific Strategies for Typical Race Courses	95
The Flat Road Race	95
Rolling Hills	96
The Big Hill	97
Technical Criterium	99
Wide and Fast Criterium	99
Chapter 6 Rider Upgrades and Team Success	101
Handling Each Win	101
Choosing the Next Team Leader	102
Rider Upgrades	103
Success	104
Appendix 1 Team Questionnaire	105
Appendix 2 Winter Riding	108
Appendix 3 Glossary	110

Chapter 1 Basic Strategy Introduced

With a single twenty mile lap to go, my teammate Shaun pulls in front of me. Two hours into the race, we have forgotten the rain and the cold. Both of us glide along near the front of the peloton; I keep a running verbal commentary of my position while Shaun keeps his nose in the wind. The group surges forward as we ascend a short incline. I let Shaun know my position as he claws our way back to the front. Ten miles later, the rest of the team surrounds me, ready for the big pull. I have been resting for sixty miles preparing for this sprint. A corner marks the five kilometer point and I follow my three teammates to the front. Every thirty seconds, the lead rider pulls off, slips back, and I let him return to the paceline. Our team speed is high enough that no challengers appear while I prepare for my first efforts exposed to the wind.

A cycling team should be more than a group of riders wearing similar jerseys. Training together is fun, but using the skills of each rider on the team to increase the chances of winning is even better. The simple tactics outlined in this book will show you how to organize your group into a team and how to use team tactics to win bicycle races.

Team tactics save energy during a bicycle race by dividing up race responsibilities between team members. Unlike fitness gains or a new set of wheels, team tactics can be developed with only a small amount of practice and no expense. The advantage is significant enough that every professional team uses some form of teamwork to increase their chances of winning. Unfortunately, because amateur teams rarely have a coach, always have limited time commitments, and may consist of riders living miles apart, lower level teams rarely understand the advantages available to teams and thus never practice the skills necessary to gain these advantages over their rivals.

Tactics can be grouped into three basic categories: *defensive*, *positioning*, and *offensive*. A defensive tactic is any method that attempts to save the energy of a particular rider. For a two-person team, one rider would spend the entire race riding in front of the protected rider keeping the protected rider out of the wind, saving energy for an attack or the final sprint. At the end of a race, positioning is just as important as having a strong sprint. Teammates that work together to make sure their best sprinter is at the front of the *peloton* are using a positioning tactic known as a *leadout*. Offensive tactics force other teams or other riders to use energy. By attacking and accelerating away from the field the rest of the pack is forced to use energy in a *chase* effort.

The offensive, defensive, and positioning tactics that aid a two-man team work on an even grander scale when applied to larger teams. A team of ten riders for instance might choose two *team leaders* to save energy for the final sprint while four riders (known as *domestiques*) ride in front of the team leaders protecting them from the wind. During a technical corner, just before a large hill, or before the final sprint, the domestiques will pull the entire team to the front of the peloton and use a *paceline train* to hold everyone in position. The remaining four riders on the team attack the peloton throughout the race and try to create a *breakaway*. If the break is successful, the rest of

the peloton is forced to use energy to chase down the attack. All the while, the team leader is saving energy for a final sprint if the break is caught.

Professional teams usually have one team leader chosen for each race. The names Tyler Hamilton, Lance Armstrong, and Chris Horner come to mind. Yet, even in these high-profile teams, different riders are selected to lead the group during races that match their distinctive abilities and training goals. Lance Armstrong often protects George Hincapie in the early spring classics. Chris Horner has been known to give a world class leadout to Victor Rapsinski. In each of these cases, the training year and the different skills of each rider affect who leads the group and who is the domestique for each race. The same should be true for lower level teams. Working together doesn't mean throwing your season away to help one of your friends take the glory. A servant one week is a king the next.

Even beginner riders will benefit from practicing [small group](#) situations and working to help their teammates. A lot of [category](#) five races have smaller fields and, because of the extreme differences in fitness levels, small groups can form in any race. Practicing these situations will increase the chances of your team placing first. Practicing pacelines and leadouts develops safer riding skills and helps new riders achieve quicker upgrades. Finally, novice riders can work in a big race and help the team by setting the pace as long as he or she is able. The thrill of being involved in the race and the increased effort required creates a better workout while gaining tactical experience.

Amateur cycling teams also benefit tactically by looking beyond the immediate racing season and creating a year-round approach to racing and having fun. Cycling teams should be able to switch from winter base rides to spring speed building and adjust to summer races without losing members or sacrificing team interest. After the summer racing season ends, transitioning to cyclocross or additional team miles can keep everyone riding throughout the year and can be just as fun as the summer races. By providing an increased level of organization, a team can attract new members, achieve a new level of focus at practices, and still have a good time.

The skills necessary to protect your teammates and put them in winning position include pacelines, leadouts, attacking, and blocking. Understanding how and why the peloton changes, as well as knowing what tactics are available when riding in small groups can lead to more wins and better team cohesion. This book addresses each of these skills in their simplest form, outlines how to gain these skills on the limited time budget available to recreational racers, and includes specific strategies for different types of races.

Chapter 2 Building a Team

Almost every town in America has at least one group of cyclists that ride together on occasion; larger cities have many racing teams. Cycling groups make riding more enjoyable and reduce the effort of each mile. Racing teams organize training rides, put on races for other teams to attend, and generally provide an excellent resource of experience and racing knowledge. Any new rider will progress faster if he or she joins a team.

The first hurdle for an established team of riders hoping to develop a tactical advantage is to organize practices focused on building and utilizing team skills. Creating a weekly organized team ride with a training plan and specific skills practice requires excellent team leadership and a tradition of team effort. Given the extra energy required to train as a team it is no wonder that most groups consider team practice a simple Saturday morning group ride and team tactics a carpool to the race.

Teams that ignore the tactical advantages of working together in races merely lack the leadership and knowledge of an experienced or highly motivated member willing to organize a few practices or lead a few discussions about tactics. Cycling tactics are not complicated, but they are not obvious either. After studying the methods and reasoning of team tactics, even you could be voted the next team leader and lead your team to victory.

After you have helped your team create new skills and win a couple of races, even the most determined individualistic riders on your team will want to have the group working for them. No longer will your group be satisfied with logging a few miles on the weekend, but everyone will want to practice pacelines and work together on a group ride to win a sprint. For part of the year learning and perfecting the basic team skills of leadouts, protecting the team leader, and paceline trains adds a welcome change to the regular workout schedule.

Joining a Team

Even beginner riders should join a cycling team. Joining a cycling team is not necessary to enjoy cycling or even racing, but being a part of a team has many advantages. Even if the team does not use any of the tactics explained in this book, the training, social, and organizational advantages of a cycling squad are numerous. The advantage of having other people to ride with throughout the year is alone worth the effort.

In larger cities, there will be many teams from which to choose. Social teams ride together and occasionally race. Competitive teams have harder training sessions and race nearly every weekend. In addition to possessing different training levels and goals, teams

take on different characteristics according to the personalities of their members. Finding the right team is the rider's first step to gaining a distinct tactical advantage in [road races](#).

If your state has a cold winter, the best time to find a cycling team is in November, December or January. This is when teams are starting to prepare for a racing season which starts in March or April. Racers in warmer climates might try to find a team in August or September as the racing season in the south can start as early as December or January. Most teams will accept members year around, however, so don't hesitate to find a team as soon as you want to start training or racing.

To find your way into the underground network of cycling you must first know where to look. A first step would be to look on the internet using search terms such as, "[your city] bike racing," or, "[your city] bicycle shops." Larger bike shops and race teams have internet sites that allow the team to communicate and display their recent adventures. Team names are listed with race results so finding a local results list makes the search easier. A more personal method is to stop by a large cycling shop and ask about local cycling teams. More than likely the shop will have their own team or at least some knowledge about other groups in the area. The most comprehensive resource available is to contact the United States Cycling Federation (USCF, www.usacycling.org) which is the governing body for bicycle racing in the United States.

While you are searching for a team, be on the lookout for any organized rides advertised on websites or at local bike shops. Attending a Saturday morning group ride is the most personal way to meet the cyclists in your region. During the ride talk to other cyclists and ask them if they race; then ask them about their group. If you see a pack of riders similar to you in ability, interview them about their team.

In some areas of the country, there is a distinction between racing teams and racing clubs. Racing teams are invitation-only groups while racing clubs will allow anyone to join. The difference is often the level of commitment to the team. A racing club will allow you to come and go from group rides while a racing team might have mandatory practices or other requirements. It is sometimes possible for beginners with a high level of interest or commitment to ride on racing teams.

Which team is right for you depends on your experience and your commitment. If you are a beginner, look for a team with a lot of beginning riders or a development program designed to teach new riders basic racing skills. If you are more experienced and interested in team tactics, you will want to find a team that sticks together on group rides and regularly rides throughout the week. If you find a team that is organized, has a good social atmosphere, and rides together more than once a week, that team is probably ready to start working on team tactics.

After meeting a few of the riders in the area, try to gather some more details about each team. Some teams make money by organizing races and then use this money to reimburse their riders for race entry fees. A larger team with many different categories or levels of riders will allow you to stay involved as you progress through the different skill

levels in cycling. Any team with experienced riders will also be able to answer questions and act as mentors to new members. Finally, a team whose group rides start near your house can be the difference between staying involved with a team, and fading out after a few months of training.

Leading a Team

Effective team tactics require trust and practice to be effective. The proper execution of team tactics depends on high-speed coordination between team members without much communication. The only way to build these skills is to ride and practice as a team.

If you have ever watched a [team time trial](#) in the Tour de France you have seen the result of team practices. The good teams practice their team time trial as often as possible, while other teams fail to organize their riders and do not practice together. It is common for at least one favored team to lose the entire race on this stage because the group failed to coordinate their efforts effectively. Every local peloton has the same problem. Some teams work together, other teams do not. Creating a team that works together requires leadership. Having a strong understanding of basic team tactics places you in the position to motivate your team to work together.

Using the skills from this book it is possible to lead your team and gain useful advantages during each and every road race you enter. But the rewards of team tactics are not free. It will take organization, practice, and a willingness to stop in the middle of a ride and talk about teamwork. It will take discipline to practice the skills of pacelines, leadouts, attacking, blocking, and small group situations until everyone on the team understands the principles of road racing tactics. The entire group has to focus on team tactics during pre-season training to effectively execute tactics during the racing season. Giving up a few weekends of the typical group ride for structured training will bring success.

Joining a team is the first step. If that team does not seem interested in working together or developing the team's tactical skills, there is little you can do immediately. Unless the team is new, a beginning rider will have to build her influence before she can jump in and start suggesting training schedules and group practices. If you are an experienced rider, quietly begin to explain how team tactics could be useful in specific situations and how you could help develop those skills. If you want to influence the organization and leadership of the team, first focus on building your own cycling skills then begin to involve yourself in the team as much as possible. Balancing team traditions and new ideas is a delicate operation.

The fastest way to build a racing team is to ride together as much as possible. Organizing group rides is worth the effort because riding together quickly builds trust and creates a strong social atmosphere between the team members. The shared experience of a good three-hour ride is the foundation of team tactics. After building a certain level of

camaraderie between team members, a group of riders used to training by themselves and racing for their own good will begin to accept the idea of working together during a race.

A professional team has a coach. In most cases, what an amateur team needs is not a paid coach, but simply a leader who understands team tactics. The skills required to lead an amateur racing team are not hard to master. Studying the sport, riding as much as possible, and having a desire to help your friends learn is all it takes to lead a team.

Successful cycling involves a lot of training, equipment, and time. Most riders do not want to add to their commitment the time to organize meetings, lead discussions about tactics, organize carpools to races, and lead practice sessions. A great team has an elected [team captain](#) who is responsible for keeping everyone coordinated and organized. If their job is done well, everything but the racing is easier for the entire team.

The team captain might be the most experienced member of the team or simply the one with available time and energy. In either case, the typical responsibilities of the team captain should include:

- Calling team meetings.
- Managing the team email list.
- Scheduling practices.
- Planning workouts that suit the team.
- Scheduling team-organized training camps or early season team workouts.
- Introduction of new members.
- Organizing a team contact list.
- Organizing carpools to races.
- Knowing driving directions.
- Choosing a place to meet at the race.
- Choosing a time to meet to warm up for the race.
- Knowing good restaurants near each race.
- Motivating the team.
- Helping the team work through team conflicts.

If one or two people perform the duties of team captain, the rest of the team is free to focus on training. Although organization alone will not win a race, it can make the team experience much more valuable. Someone willing to do all of this work for the team also earns a lot of respect and can begin to shape the team into a group that uses tactics to their advantage.

Yearly Schedule

When you have convinced your teammates that working together is an important part of road racing and have been elected team captain, you must begin to organize the team. A strong yearly schedule will become a team tradition and will lead to increased

fitness. Helping to keep the team on schedule and organized will make it easier for your riders to attend races. Creating an email list and leading discussions will guarantee that your team gets stronger everyday.

The first task of a team captain is to set up a communication tool available to each member. This probably includes an email list and a telephone list. It is very easy to use Yahoo! Groups (www.yahogroups.com) or a personal email server to create a central email system that your teammates can use to contact each other. Yahoo! Groups also has a database feature that allows each rider to input their address and phone number. Once the email list is created, it is easy to call an organizational meeting or organize weekly group rides.

The team activity schedule must meet a lot of different needs. First and foremost, it must coincide with the basic training schedules of the riders. If your training schedule includes hard efforts when half of your riders are just trying to build distance, the plan will fail. Additionally, the plan must:

- Cope with temperature changes
- Adjust to daylight savings time
- Offer activities outside of road racing at times
- Be flexible enough to allow people to join in the middle of the season.

A schedule that meets all of these requirements can be repeated year after year with great success. In fact, after a year or two, the entire team will come to expect a certain progression throughout the season.

A typical schedule for a team could be as follows:

Pre-season:

- Regular weekend rides regardless of the weather
- Early season [time trial](#) or team time trial
- Yearly banquet featuring awards from the previous year
- Winter team meetings
- Weekly social meetings at a local restaurant or bar
- Spring training camp a few weeks before the first race of the season
- [Training Races](#)

Early Season:

- Races
- Regular weekend rides for people not able to race that weekend
- Weekday evening rides
- Occasional century or tour to create a non-racing break

Late Season

- More races, weekend rides, and evening rides
- Out-of-town trip to a stage race
- Non-racing event such as bike polo or a city tour

Off-season or Cyclocross

- Cyclocross training camp or clinic
- Cyclocross training rides
- Weekend training rides
- Cyclocross races combined with a cookout or other social event

Base Miles and Meeting #1

The first team meeting of the year should occur well before the first race of the year. The best time for the first meeting is usually near the beginning of your base training, usually two or three months prior to the first races. The team captain should organize the meeting using the communication system and lead the team's discussions.

The goal of this meeting is to kickoff the year (hopefully with some new members) and start the training cycle in preparation for the racing season. At this meeting the following details should be arranged:

- Introduction of all members and new members.
- Time to listen to team member's ideas about the season.
- Questionnaire about each rider.
- Organization of team ride locations and times.
- Description of paceline skills necessary for a strong team.
- Discussion of opportunities for a training camp weekend.
- Social event organization.

This first meeting should be a simple introduction to the team and only focus on training ride organization. New riders will be self conscious about any detailed technical discussions and will need to be encouraged to join the team workouts especially if they have not ridden in a group before or are physically out of shape. Don't discourage new members by describing, in great detail, how much pain and how hard your mid-season workouts will be. Similarly, don't confuse beginners with long technical detail about aero wheels or confusing team tactics. Keep this meeting simple and the team's size will grow more quickly.

At the start of the meeting, everyone present should be introduced. This simple way to learn everyone's name and get everyone acquainted should not be skipped. Also include some time to let everyone talk and put out their ideas about the season. If the team captain does all the talking the meeting will be very boring for the other participants.

Sometime during the meeting, the team captain should hand out the team questionnaire. This can happen as people are arriving at the meeting, or as people are leaving. The team captain could also facilitate a questionnaire online when people join the email list. The written questionnaire gives the team captain a written example of each rider's experience level, goals for the season, and interests. A sample questionnaire is included in Appendix 1.

A quick discussion should be enough to organize a good meeting place for team rides. A time should be chosen that allows the majority of riders to attend every week. If necessary, have two rides each week of the year so everyone can attend. Although email allows teams to organize rides at a moment's notice, at least one ride a week should consistently be on the calendar.

The main skill the team should practice first is simple paceline training. The portion of the early season team rides devoted to paceline training should be high enough to learn the skill, but low enough to allow the group to talk and get to know the new members. Somewhere between 25% and 50% is great. When the team is going slow and thinking about how long it will be before the first race, remember that the rest of the

peloton is inside watching television or racing each other through the countryside and ignoring their training schedule.

The group training aspect of your team will encourage people to ride even if it is extremely hot or cold outside. Having regular rides will allow new riders to build their fitness and they too will be encouraged to maintain a consistent training schedule. By keeping these initial rides low-key, the intimidation factor is low for new riders. Providing a skill such as pacelines for the team to develop raises the interest level of the outing. After each ride, advertise how much fun the day was even the temperature was below freezing or in the 100's. Most riders only need a little encouragement to get out of bed and brave the weather. All of these factors guarantee the team has a good base of experience before the racing begins.

A pre-season time trial or practice race will keep the strength and speed of the team high and will balance with all of the slow training miles. A time trial is the easiest kind of practice race to organize and can be a simple affair within your own team. Once everyone has been involved in a paceline practice session, holding a team time trial is a good idea to emphasize and promote the idea of working together while building trust and perfecting the paceline skills absolutely necessary to execute complicated leadouts and breakaways later in the year during racing season. The effort doesn't have to be long (5-10 miles) and can include an extended warm-up to reduce stress during the base phase.

Team Meeting #2: Building Speed, and Learning Tactics

After six to twelve weeks of team base training with a time trial or practice race mixed in to add intensity, the team is ready for a second meeting to organize faster training rides and introduce basic team tactics. A few weeks before racing starts, riders usually increase the intensity of their training, therefore the team rides should allow for harder efforts to meet the training needs of team members. Team skills are developed through organized group sessions that lead to a weekend training camp or similar event where the team focuses on tactics. In some areas of the country there will be practice races on the weekend allowing the team to practice specific tactical situations against real competition.

The second team meeting is held four to six weeks before the first race of the season. The agenda for this meeting should focus on building the tactical knowledge of the team and should include:

- Introduction of all members (again).
- Tactical basics (Chapter 3).
- Communication (Chapter 4).
- Race progression and specific strategies (Chapter 5).
- Race scheduling.
- Training camp organization.
- Practice schedule and training ride schedule.
- Race week organization.
- Purse strategy.

The second meeting is the most important meeting of the year because it is the only time when the entire team will discuss how to best work together. This is when the system is put into place. Time restraints prevent calling more meetings with amateur teams because, quite simply, cyclists would rather ride than sit down and discuss organization.

Team communication and tactics are covered in chapters 3, 4, and 5. Review these chapters with the team and add in any other useful tactics developed from your team experience. Make sure everyone understands his or her role in the team's success. Explain how a typical race progresses and detail how you think your group could increase their chances of success this year. The questionnaires filled out months earlier allow the team leader to tailor the group's training and commitment to everyone's skills and motivation.

Twelve racers at one race a weekend is better than six racers at two races. To make sure this happens, discuss the year's racing schedule at the meeting. Allow the group to talk about which races are most important to them and purposefully leave a few races off the "official" team calendar to maintain a balanced time commitment for the group. Explain to the team that anyone can attend non-official races but he or she should not expect an entire team to be present.

With the overall team schedule finalized, take time to have each member choose one or two races where she will be the protected team leader. The races chosen for each rider should coincide with that person's strengths or correspond with a personal reason for wanting to win that particular race. Personal reasons include past experience at the race, a race in their hometown, or a love for the particular race course. Focusing a teammate on one or two races a year motivates their training and allows him or her to perform at a higher level.

Race Motivation—The Maine State Criterium Championships

My team in Boston had a rider that grew up in Maine who was an excellent sprinter. That rider always wanted to win the Maine State Criterium Championships so the team always worked for him at that race. The rider's extreme focus on the one event paired with the team being excited to help him win always led to a strong finish.

The person chosen for each race will become the protected rider for that race and will have the best chance at winning. If you have a large team (12-15 riders at each race), it is possible to choose two or more protected riders for each event. If the team has a wide disparity in skill levels, some riders should be the team leader more often. New riders are excited that they can at least help the team during a race, and may be relieved that they do not have to be the protected rider, though this is not always the case. Managing the appointment of team leaders for each race is a team captain's hardest job, but is easier if the team has developed a social atmosphere and each rider is committed to the team.

A team-organized training camp is one of the best times to have the team ride together and work on the specific skills necessary to execute effective team tactics. A whole weekend of riding, rather than a couple of hours on a weeknight, convinces most people that stopping to evaluate the performance of the team is acceptable. Organized practice sessions during the camp can be arranged to isolate specific skills and quickly bring new riders into the system.

A training camp does not need to be fancy or take place out of town to be effective. For instance, simply having a team ride on both Saturday and Sunday could be successful. The main idea is to get the entire team together a couple of times within a short time period to focus on the skills necessary for effective tactics.

Training Camp--A Mini Stage Race

Boston Road Club training camps started Friday morning at 6:30 AM. The two-hour ride was mainly social except for the two hot spot sprints along the route. By following the same route as weekly fast training rides later in the season, new riders gained familiarity with the winding New England roads.

Saturday's practice occurred at the Well's Avenue training race site, again, a course used many times later in the season. The 0.8 mile circuit only had one real corner and no traffic on a Saturday morning. Here we ran a track practice focusing on protecting one rider amidst a small peloton and timing our leadouts so everyone understood the concept. Before the practice ended, we talked about the strategy for tomorrow's practice race so everyone knew what to expect.

Sunday's practice race at Gill, MA was the first race of the year for our team. In this race we decided to attempt to maintain a paceline train at the front of the peloton for the entire race. Focusing on only one goal made sure everyone knew what was happening. If at any time the train broke apart, the entire team should immediately attempt to gather together and regain the front. We started the race in the front row and pushed forward into a paceline. With the entire team working together we stayed in front until a wide section of road allowed aggressive attacks from all sides. Ten miles later the team regained the front and held it until the next wide section. Not many riders held on for the finish, but our team left knowing we could control any race in the regular season.

Monday was supposed to have another 6:30 AM ride. After three days of up-tempo training everyone was beat and no one showed. My legs felt like they had been through a stage race even though the first scheduled competition was a month away. Waking up each morning and riding my bike added a professional feel to the workouts and increases my confidence for races later in the year.

Weekend rides other than the training camp should begin to include more intensity to prepare the riders for upcoming races. If possible, the team should ride out to a good practice spot and work on the basic team skills of pacelines, gatekeeping, small group strategies, and protecting a rider in a peloton, then cruise home. An excellent practice spot has the following attributes:

- Little or no traffic
- Wide, smooth pavement.
- Lights, if possible.
- Fairly flat circuit.
- Central location to all team members.

The best locations are office parks with a central connecting road. These office parks often have no stop signs and limited traffic on the weekends or late evenings. Some even have enough street and building lights to allow late evening practices in the early spring.

Two or three sessions at the practice site should develop team trust and skills. At the same time, on-bike communication among the team members will increase. Once each member of the team understands how and when to work together, start a leadout, play [gatekeeper](#), and win small group situations, it is time for the team to enter a training race and test the group's skills.

A practice race should be utilized to build a team skill, and should not be a time where riders can go out and push their early season limits. At no other time in the season will there be a situation where the team may practice leading a peloton with a paceline train or focus solely on keeping one person out of the wind. It is also a chance to attempt a full speed leadout without fear of losing an important early season race. Pick a beginning rider to be the protected team leader and challenge the experienced riders to keep the new guy out of the wind. If the training race is a [Criterium](#), consider having the team sit out a lap in the middle of the race to evaluate the team's performance and to keep the race effort under control.

A key point to make during the second team meeting is that the next few weeks are the only time of the year where the team has organized skill-based practices. Workouts will continue all year, but a long group ride is more fun than laps in a business park. One or two months is not too much to ask and should provide plenty of time for everyone on the team to build the skills necessary to go into the first regular season race ready to work together. Putting in more time at this stage in the season to coordinate the team will be rewarded with good race results later in the year.

One of the easiest and most important organizational aspects of team cycling is to have a repeatable race week progression that prepares the team for each race. The second meeting should clarify the progression plan so everyone can utilize it when the season starts. The following information is absolutely required for a team to maintain organization and continue learning each race week.

Monday	Race leader and other riders send out race reports to all members detailing experiences from the previous weekend. This feedback increases the learning from each race and allows the team to develop throughout the season.
Tuesday	Riders planning on racing the following weekend email the group indicating their plans and if they can drive other teammates to the race.
Thursday	The team captain sends a race report email for the upcoming weekend. The email should include: team roster, protected riders, drivers, meeting time, course description, and basic tactical outline.
Friday	Car pool details are finalized.
Weekend	Race

A good team captain keeps the team organized and maintains a sense of momentum by maintaining a predictable schedule for each race. This allows each rider to relax about car pools, start times, race parking etc. and each rider can spend more energy training and thinking about winning the race. Team organization is an invisible base for team success.

Team Organization—One Step Further

After running the Boston team for two years I noticed that the Thursday emails concerning team rosters, protected riders, drivers, meeting times, course descriptions, and tactics were similar from year to year. At that point I created a form that allowed me to send out this data faster by copying the course descriptions, basic strategy, and driving directions from the year before while adding the names of teammates from the current year.

Eventually I created two more details about each race: bathroom stops on the drive to the race and local restaurant directions and reviews. In the attempt to stay hydrated before a race someone in the car always needed to stop before we reached the start. Sometimes we spent thirty minutes looking for a public restroom off the highway. The same amount of time was spent after a race looking for a restaurant in the area. The new information could save an hour of frustration from each weekend. Now the team often meets at the mentioned rest stop and some riders have been known to attend a race only to eat at their favorite restaurant.

The last discussion held during team meeting #2 should be about purse strategy. If a team wins a race it is likely that the team leader will win some prize money or merchandise. Having a team-developed method for distributing the winnings keeps the team working together. Organizing this part of the team early in the year prevents surprises later in the season.

The second team meeting of the year starts the busy season of preparing the team for the upcoming races. Team workouts, practice races, and the training camp weekend require a lot of planning by the team captain. If your team makes it through this part of the year with a disciplined attitude and a focus on basic riding skills, the race wins are just around the corner.

Early Season Races

Once your team has mastered pacelines, practiced leadouts, and has proven it can work together within a large peloton the group is ready for racing season. Unfortunately, when the racing season starts, there is little time to work on specific skills. If the team hasn't mastered a particular skill such as gatekeeping or leadouts, it will be hard to plan a practice session outside of the busy race calendar. The team should instead focus on execution and attempt to learn as much as possible through the Monday race reports.

Spring races are excellent opportunities to take advantage of the skill and fitness built through the disciplined preseason preparation. The first few weekends of racing are a lot of fun and should be used to allow the team to transition from organized practice to execution. Team discussions about the effectiveness of the team in each race are important, but stay away from early season blaming or other harsh reactions. Remember that you are a part of an amateur team and all members are volunteering their services.

Continue to use your race week progression to keep team logistics under control. A new training schedule can be created when daylight savings time starts and rides can be organized after work. A mid-week team ride or training race can help everyone build speed and acceleration abilities. By now the team is experienced enough to change the schedule and discuss tactics without an organized team meeting. Just use the email group to keep everyone up-to-date on any changes that are necessary.

If there is a weekend during the season without a scheduled race, the team might plan a long training ride or century. Driving a couple of hours to the mountains or at least to some nearby hills for a long group trek will build the team's spirit. A long, hard mountain ride will hone basic skills such as pacelines, and the extended distance will increase the confidence of each rider involved.

Training Ride—The Kancamagus

It is the tradition of the Boston Road Club to choose a weekend in May to ride a century in the White Mountains of New Hampshire. The ride includes five mountain passes (known in the Northeast as notches), with the first climb starting on the Kancamagus highway and the last brutal climb just ten miles before the finish. After the ride, the prime rib special at Truant's Restaurant in Lincoln, NH is payback for six hours in the saddle. The long day of pedaling requires pacelines to battle long inclines and tough headwinds. The steep headwalls split the team into small groups and the top of each climb is treated as a king-of-the-mountain sprint. The trip provides important training for the upcoming Fitchburg stage race and is a fun, non-race use of fitness built throughout the winter.

Mid-Season Races

As the racing season extends later into the year, most riders on the team will have had the experience of being the protected rider in a real race. At this point, everyone will have developed paceline skills, and the team will have improved its results. The communication between team members during the races will have increased and each rider's intuition of the correct tactic for each race situation will have become more ingrained. At this point the team should be on auto-pilot for the rest of the season. Use the Monday race report to keep the team learning and adjust the training schedule to keep riders from over-training.

Transition Time

Sometime during the end of the season the riders that have been training since the first team meeting will begin to lose interest in racing. Other riders on the team will have just joined and are ready for more races. With so many goals throughout the team, the team leader must balance the team's efforts to keep the group from disintegrating.

The racers that are tired of training should be allowed to take a few weeks off from the team. Don't worry when a training ride has a much smaller turnout. Relax, the season is over and enjoy the feeling of accomplishment that started months ago.

If enough new riders join the team, a second training camp held at this time of the year will develop the skills of the new teammates as soon as possible. Since they just started training and racing they will be enthusiastic about learning new skills and can apply their knowledge to the remaining races in the season. If new riders are

immediately organized and trained to understand the group's organizational system, the group will start next year ahead of schedule.

Fall and Cyclocross

Unless a portion of the team is truly focused on cyclocross success, the main goal of the team for the off-season will be to keep people riding. Activities at this point in the year need to be matched to the team's energy level. Put together a couple of events that are more social and less physical to encourage tired riders to get out on their bicycles. A good ride out to a coffee shop or restaurant is fun and low key. The main goal is to get people back out on the road before they lose interest in the sport. These rides should be short and very casual to break even the most stubborn off-season slump. Involving significant others and non-team-members increases the draw of such events.

Even rides of 5-15 miles can be effective at this time of year. A short ride can end the off-the-bike inertia that sets in after a few weeks on the couch. Once on the road, most teammates will remember again why they enjoy cycling and will start to build miles according to their training schedule.

If cyclocross races are available, it is most important to find a bike for each member of your team. Mountain bikes work great and encourage more people from your team onto the start line. While mid-week rides fade with the available light, promote early morning jaunts of an hour or two to again keep the team moving. Do not include big hills or long intervals at this time of the season, let weekend cross races be the venue for the week's hard efforts. A last ditch method to promote cyclocross is to provide food and drinks at the cross race for your team. Riding to the starting line knowing 45 minutes of pain is all that is between you and a grilled bratwurst is motivation few can ignore.

Some teams stop road racing in July and begin a second training season specifically for cyclocross races. The entire training season is repeated with cross skills replacing leadouts and team practices. After a lap or two, cyclocross is essentially a bunch of small groups negotiating the course. Small group tactics, such as [relay](#) attacking and group attacking, work just as well off the road as they do in the spring. Practicing dismounts and hurdles gives everyone on your team a big advantage every lap. Just like earlier in the year, having a team practice motivates everyone to attend and assures the team of strong fitness and skills.

Cyclocross Tactics

Cyclocross is a mix of time trial efforts combined with small group situations. Like time trials, the fastest riders monitor their effort to carry their momentum throughout the course. The fastest time trial riders are the ones with the most speed at the top of each hill. The same is true for cyclocross. Saving energy and putting effort into the top section of each climb and throughout any flat sections after climbs will gain seconds on every lap.

After a lap or two the pack is often split into multiple groups of two or more riders. The same options are available in cyclocross as in road racing: group attacking, relay attacking, or waiting for the sprint (see chapter 3). Having teammates in the group allows one person to continuously attack while the other rider rests and waits for the rest of the group to tire. As the group becomes smaller, solo attacks will create gaps and allow you to move up into the next group of racers.

Positioning for a section of the trail is often important as well. If the course has some technical corners work to be in the front of the group to avoid falling riders. Slow riders can break apart a group in a technical section by taking turns at a slower pace or sliding out and crashing, so being in front has an advantage.

Whew--the season is long and sometimes tedious. Implementing a yearly plan and creating team traditions are excellent steps to keeping a race team in top shape for winning races. Eventually, the social aspect of the team will keep people motivated every weekend, even throughout the freezing winter and the burning summer. Riding every week of the year is a big challenge, but a group can be great motivation. A good team captain can reduce the organizational burden for each rider and allows the team to make great strides in training, especially compared to riding alone.

Chapter 3 Building Team Skills

Cycling teams use two tactics that require team coordination. First, a paceline train allows a team to position a specific rider (the team leader) near the front of the peloton while forcing opposing riders to match the speed of your team. Second, when a specific position at the front of the peloton is not necessary, the team leader should be surrounded by teammates who work to help the team leader save energy by staying out of the wind. These two tactics should become the focus of most practices and are skills every rider on the team must learn.

Although there are only two main tactics requiring the efforts of an entire team, other advantages may be gained by individual efforts. Attacking and blocking require individual riders to work alone to help the team. During an attack, the peloton splits into small groups forcing an individual to race without the help of teammates. After the paceline train and protecting the leader are perfected, additional individual skills may be developed.

Finally, everyone on the team needs to completely understand and memorize the proper evolution of a bike race. The goal of team tactics is always to increase the chances of your team winning the race and it is essential to know which actions help this cause and which efforts do not. Educated riders know when to attack, when to rest, which attacks need to be [marked](#), and when to chase down opposing breakaways. Group practices that simulate specific racing situations build the tactical knowledge of the team.

In just a few practice sessions, a team can create the basic skills necessary to work together and will be ready to apply these skills to race situations. Setting aside one month of weekend rides a year is plenty, especially if one of the weekends is a team-organized training camp, or two-day practice session focused on learning about team tactics. Practices can be integrated with any training schedule for maximum team development.

The volunteer nature of bicycling teams is the first and greatest obstacle affecting team organization. Membership can change throughout the year as people lose interest in the sport or have other priorities arise. Spending a month trying to practice skills then having your team roster change completely can be discouraging. A second issue is that no one is required to attend practice. So the first session might draw 50% of the team members, while the next practice will involve an entirely different set of riders. The third reason amateur cyclists are against team practices is that they view the start-and-stop nature of a practice session as boring, especially when compared to the long countryside tours that the team has been organizing all winter. If one of these problems arises in your team, have patience and try to identify why the barrier formed, then design practice sessions that meet the needs of your riders.

If you are a team captain, point out that there is a specific progression to how the training calendar is organized and that the best time to practice team tactics is a few

weeks before the racing season begins. Let everyone know that these types of practices occur for only a few weeks out of the year, leaving plenty of time for different types of training rides later in the season. Keep the sessions flowing and fun by keeping people riding as much as possible and always explain why the specific exercise will help the team. Having a practice session the day before an early season training race is another particularly effective method of motivating the team to work together. Even if only a few riders show up, always hold a practice session. The riders that do attend will form a base for the team to build upon as the year progresses.

If you are a team member, help out your team captain by committing to attendance at these practice sessions and supporting the team's outline of skill clinics. Missing a race in the middle of the season means that you were not able to help the team that weekend, while skipping a practice session means that you will be less prepared to help the team the entire year. Do your best to study the background of how team tactics work and go into these sessions ready to practice the skills over and over.

If the team is committed to trying to learn the basic skills of pacelines, leadouts, attacking, blocking, chasing, and small group situations, then putting together a few weekends of practice sessions will not be difficult. Find a good practice spot without much traffic, set up a meeting time, outline the practice online with the email chat group, and get the team together.

Pacelines

The paceline is the basic skill used in the paceline train required to position the team leader for a final attack or sprint. In addition to the paceline train, a paceline is used in nearly every aspect of road racing: breaks will not form without good paceline skills, chases will not be effective without working together, and leadouts will be short if a paceline is not adopted. Pacelines are also useful on training rides; they allow the group to move at a faster pace, especially into the wind.

Unfortunately, many of the cyclists in any amateur peloton are not familiar with how to start a paceline, how to ride in a paceline, and when to ride in a paceline. Consequently, most lower-level races do not have breaks, chases, or leadouts, making the races less exciting. If your team practices pacelines throughout the off-season the team will gain a solid tactical advantage in every race.

Paceline training will aid both beginning and advanced riders. Beginners gain immediate pack riding skills by working with experienced riders in a controlled paceline. Riding in a paceline forces a beginning rider not only to control his or her own speed, but also gauge the speed of the nearby riders. Pacelines also reinforce the idea of riding a straight line, increasing a beginner's confidence in the peloton. Riders with lower fitness levels than the rest of the group can train with a faster group by working within a paceline as a gatekeeper.

Advanced riders will want to practice their paceline skills because no break is ever victorious without group organization. At the other end of the peloton, no chase effort after being **dropped** will be successful without immediate use of paceline skills. During the winter, paceline rides keep pack riding skills sharp making the first race of the year a little easier and a lot safer.

A team will benefit from paceline practice because it builds trust between new members and visible success is apparent from the first team ride. A team can learn paceline skills in about ten minutes of practice, and then perfect their skills on the rest of the ride. The synergy from a paceline is exciting at any speed. It is always fun to look down at the speedometer and see a speed five miles an hour greater than a similar solo effort would produce.

How to Paceline

The fundamental idea behind a paceline (and all other team tactics) rests on aerodynamics. With one person riding in front, the rest of the team is enjoying a 30% reduction in the energy necessary to keep pace. If the group shares the effort by switching the lead in an organized manner, the group's speed will be much greater than one person could generate alone. The method of changing the lead depends on the number of people in the paceline.

Two Riders

A group of two riders is the easiest paceline to describe, but it is sometimes the hardest to maintain. Two riders will not have the advantage of three or four people sharing the work, so the efforts are harder and longer. When two riders are switching leads, both riders are **in the wind**. Compared to pacelines with more riders, the lead rider in a two person paceline will stay in front for longer periods of time to reduce the loss of energy from lead changes. The most effective method for riding with two people also depends on the fitness level of each rider; the stronger rider staying in front for longer periods than the less fit rider.

Assuming that both riders are of equal strength, are working to try to maintain the highest speed (as in a team time trial), and the ground is flat, here is the general progression of events:

One rider is in front with the other rider either directly behind or slightly off to the downwind side due to a crosswind. The front rider stays there until she feels her speed dropping slightly (maybe 0.5 mph). The average "pull" in front lasts between 20-60 seconds in a two-person group. When the speed drops a little, the front rider moves sharply to the side into the wind and may stop pedaling for one or two strokes or twitch her elbow signaling a lead change. The rear rider sees the front rider pull over and then maintains her speed while accepting the larger effort of being in front.

There are some basic rules to remember:

- The only way to start a paceline is to move to the back of the group.
- The second rider never accelerates, but simply accepts the greater load as the front rider moves to the side.
- The front rider moves over first then slows slightly. A common mistake is slowing down before moving over, causing the whole group to slow.
- Group speed changes are accomplished very slowly and smoothly so that riders can stay as close as possible.
- The front rider bears full responsibility for avoiding road hazards and must make every effort to avoid objects without sudden swerving.
- The second rider should rest as much as possible in the slipstream of the lead rider.
- The lead rider should focus on maintaining the group speed and adjusting to hills and turns.
- Each rider needs to be aware of her own speed as well as the speed of each rider around her. Speeds need to be coordinated throughout corners and hills.

Adding More Riders

With three, four, or even five riders, the progression is much the same as for a two-person group. A single file line of riders stays together until the front rider moves over and goes to the back of the pack. The second rider does not accelerate, but simply maintains his speed as the former front runner drifts to the back. Everyone else is resting in the slipstream of the front rider preparing for their effort leading the group. Compared with a two-rider paceline, lead changes in larger groups may occur more often (every 5-20 seconds) because a lead change does not necessarily mean an increase in the group's effort. The length of each pull at the front of the group varies according to fitness, group dynamics, and terrain. Table 3.1 provides an estimate of group rotation speed for beginner riders. Advanced cyclists constantly study the other riders in the group to find a common speed for each situation.

Table 3.1 Paceline Pull Lengths for Different Sized Groups

GROUP SIZE	LENGTH OF PULL	LEAD CHANGE STRATEGY
2	20-60 seconds	If group speed drops by 0.5 mph, lead changes
3	10-30 seconds	If group speed drops by 0.5 mph, lead changes. Stronger riders take longer pulls.
4	5-20 seconds	Front rider pulls longer if he or she is stronger, otherwise equal pulls.
5	2-10 seconds	Rhythm of group adjusts to keep highest speed possible.
6+	1-2 seconds	Rotation begins and each rider pulls for a few seconds, then moves out of the way.

All of the skills described for two person pacelines also apply to larger groups. Each rider should maintain a steady speed so that the group can move along as smoothly as possible. Each rider should stay as close as possible to the next to improve efficiency. The front rider is always responsible for avoiding potholes while the rest of the group rests. Everyone should react to terrain changes such as hills and downhills by carrying their momentum and adjusting to the speed of the group.

With six or more people, the paceline becomes what is known as a rotation. The front rider no longer stays in front for a period of time, but immediately moves to the side once the former lead rider starts to drift to the back. Eventually, there are two lines of riders, one group moving at a slightly faster speed than the other group. When the last rider in the slower line is even with the last rider in the faster line, he moves over, accelerates slightly, and joins the faster. Figure 3.1 shows a typical rotation paceline at work.

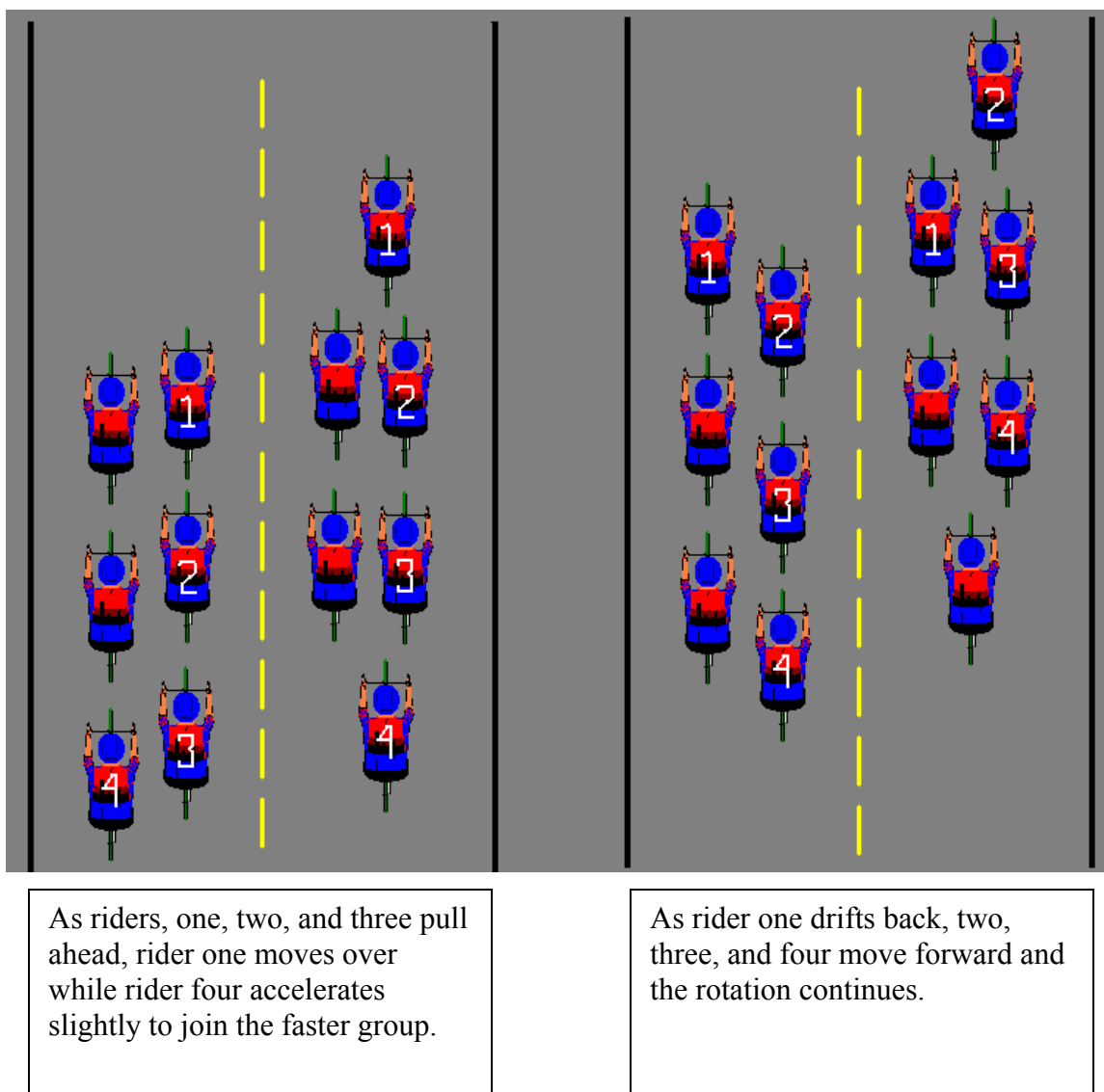


Figure 3.1 Typical Paceline Rotations

Starting the Paceline

A paceline requires teamwork among all riders involved. As mentioned before, most riders in the amateur peloton do not understand how to start a paceline. The most common mistake is for a person to go to the front of the group and stay there as if he were the only one interested in breaking away or chasing a break. This rider accelerates to their top speed and holds it as long as possible (usually no longer than 30 seconds) eventually allowing their speed to fade without moving over to let the paceline begin.

If a rider in front of you is slowing, but not moving out of the way to start a paceline the only effective way to keep the team effort is to slowly accelerate around the

front rider. There is no tactical advantage to jumping around the front rider unless you are trying to disrupt the paceline with a sudden acceleration. When working as a group all accelerations must be smooth. Smooth accelerations allow every rider in the group to react to changes and adjust her speed accordingly without opening gaps between riders.

The correct way to initiate a breakaway or start a chase is to accelerate to top speed then move over to start the paceline. Immediately starting the paceline guarantees that no one spends too much time in the front and the group's energy levels and speed remain high.

Maintaining the Paceline

Because the riders are trying to keep the highest possible speed and maintain the smallest distance between their wheels, any sudden acceleration or braking will disrupt the flow of two (or more) riders working together. It is common for the second rider to want to accelerate as soon as the front rider moves over. This acceleration is a reaction to the increase in effort required by the increased wind resistance and is reinforced by the notion that the second rider should "pull through" to the front. Pulling through is a misnomer. The front rider only maintains his speed as the front rider pulls over and slows slightly. If you are trying to paceline for the first time, reduce your tendency to accelerate by watching your speedometer and maintaining the same speed you had before taking the lead position.

Any gap that forms between two riders in a paceline should be closed as smoothly as possible. An inexperienced rider will jump (accelerate wildly) to close the gap starting a chain reaction down the entire paceline, Figure 3.2. The third rider will not have time to react to the sudden acceleration of the second rider creating another gap. One jump eventually affects the entire paceline as the gap moves down the group, decreasing everyone's energy when the riders should be resting. By contrast, each rider in a paceline is focused on the speed of the person in front of him, a gradual acceleration (as opposed to a jump) will be matched by the rest of the group and will allow the initial gap to close without opening other holes.

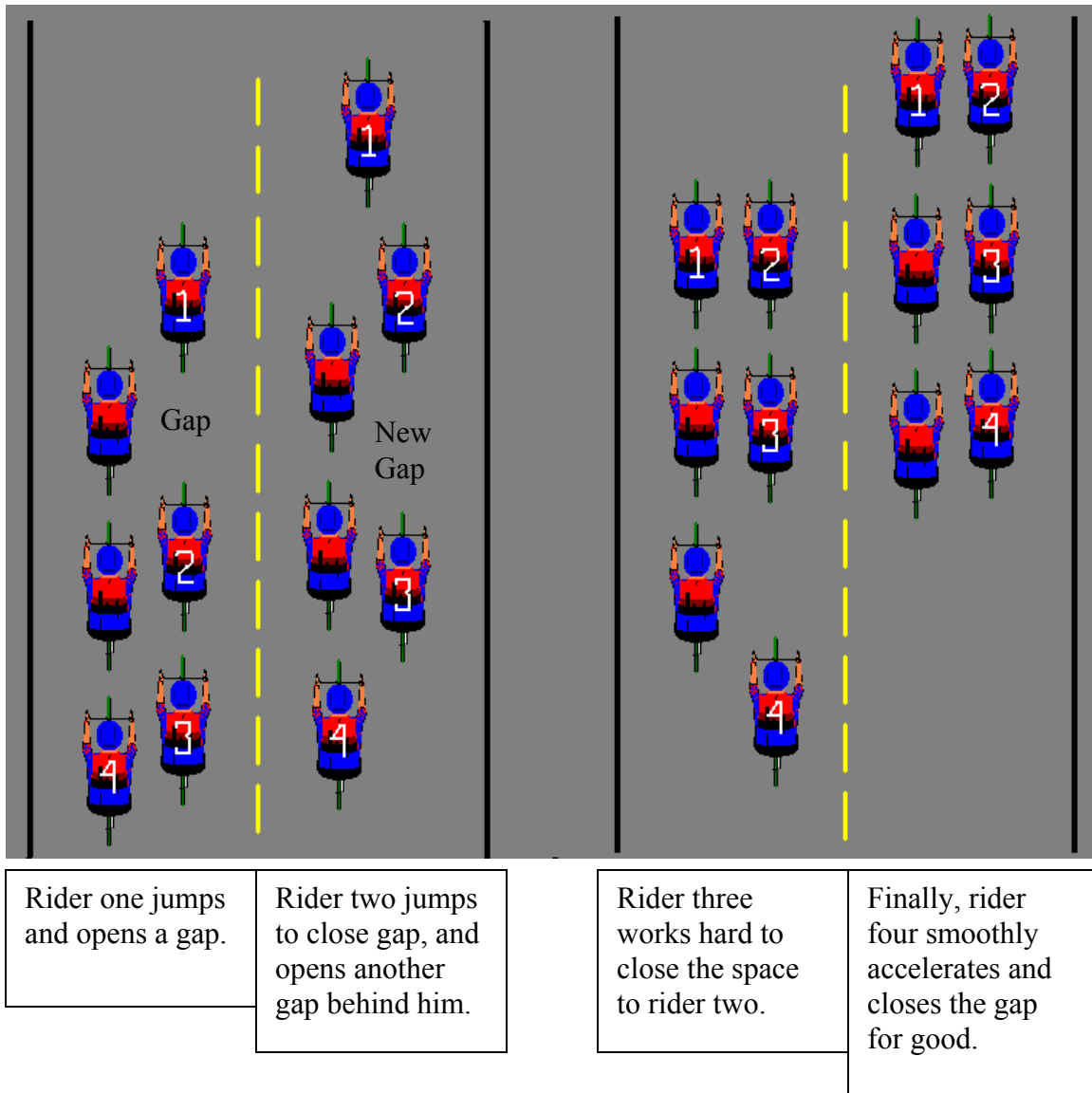


Figure 3.2 Gap Propagation in Pacelines

Changing Terrain

Hills should be ridden smoothly with momentum carrying the group as far as possible aided by timely shifting and the proper continuation of the paceline. Hills will require everyone in the group to increase his effort or the entire group will lose momentum over the hill. Groups that ride together often begin to understand the strategies necessary to ascend each type of hill as fast as possible. Communication is important so the team stays in close contact throughout the rise.

A rotation going over a small hill should continue to rotate as usual. Sometimes the length of time at the front is less than normal because the hill requires more energy to hold the same speed. If one person tries to pull the entire length of the small hill, the

entire team will lose momentum and the lead rider will probably be exhausted by the top. Small groups use a slingshot-effect to keep the team's momentum up the hill. The front rider pulls until the hill just starts to slow the group then pulls off allowing the rest of the group to carry some momentum up the hill.

The front rider is in danger of being dropped if he pulled too long because the rest of the team will speed ahead. A shorter than normal pull by the first rider will allow the team to maintain its momentum, and guarantees that the lead rider has enough energy to drift to the back of the line and reattach to the paceline.

Longer, gradual hills only require an adjustment at the start of the incline because once the transition is made the team can resume their regular rotation at a slower speed.

Long, steep hills, where the pace of the team drops to below ten or twelve miles and hour, are better tackled without the team rotation. At these slow speeds, the aerodynamic efficiency of the team diminishes making the front rider less important for [drafting](#) and more important for setting a consistent team pace. If the front rider were to move over as in a typical rotation and drift back to the end of the line, would need to accelerate to rejoin the group. Because steep hills already place each rider at their maximum effort, even the small acceleration to rejoin the paceline is a threat to the team. To remove this threat, it is typical for one or two riders to lead the team up a really steep grade keeping a pace that allows the entire team to remain connected. If the grade decreases for even a few hundred feet, the rotation could restart allowing a change in the front rider without the risk of weakening the group.

Rolling hills require a combination of the above skills and constantly require the paceline to adjust their speed, length of pull, and intensity to match the terrain. After learning the simple motions of a paceline, most paceline practices should cover rolling terrain to increase each rider's knowledge of different situations and challenge the team's abilities. A successful paceline depends on each rider having understanding the same strategy and the same definition for each hill size. If even one rider in the group fails to adjust their pull length to the terrain, the entire group will lose speed. If a rider pulls too long up a hill that rider risks being dropped from the group. Table 3.2 explains proper paceline efforts over hills. A team that practices on changing terrain will be able to work together and keep its momentum throughout any race course.

Table 3.2 Paceline Strategies for Terrain Changes

SIZE OF HILL	PACELINE ROTATION
Short hills	First rider pulls through the transition into the hill then moves over when momentum begins to slow. This rider rejoins the back of the paceline as the group nears the crest of the hill. Lead rider must save energy to maintain contact with paceline as it powers over the incline.
Medium length hills	First rider pulls through the transition while the second and third riders attempt to keep the group's momentum throughout the rest of the hill. All former lead riders must gauge the length of their pulls at the front, saving energy to regain contact with the paceline after pulling off.
Long, medium grade hills	The lead rider pulls through the transition into the hill like before then the paceline settles into a normal rotation for the rest of the hill.
Long, steep hills	Because of decreased aerodynamics, the lead rider maintains the front position up the entire hill. The lead rider is not providing much of a draft, but is really helping the entire team maintain a steady pace and focus during the maximum effort. If the incline drops, the rotation should restart allowing the team to change leaders.

As a paceline coasts down a steep hill, the front rider is no longer pedaling harder than the others, but instead is limiting the group's speed. The second rider is not affected by the wind that is slowing the first and may use the dead air between them to accelerate around the front rider while the rest of the group follows. The acceleration takes energy and should be a smooth effort so any other riders in the paceline stay in contact. Working together on a straight downhill grade a group can quickly achieve speeds in excess of fifty miles per hour.

Additional Paceline Techniques

The goal of every rider should be to ride in a smooth, straight line, inches from the wheel in front of them for an infinite number of miles. While maintaining this position, the rider should adjust to hills, and corners. Professional riders understand the advantages of these skills and use them whenever they ride in a group. Even with competitors from different teams, the knowledge of these skills allows any breakaway to succeed.

Two skills allow precise speed control in a group. The best method of controlling your speed in a paceline is to increase or decrease the pressure on your pedals while maintaining a high cadence. By soft-pedaling for a couple of revolutions the racer will slow slightly and avoid the person in front of him. Increasing the pressure on the pedals allows a rider to close a small gap. A more urgent but smooth slowing effect can be

created by applying the brakes while still pedaling. This method prevents a sudden change in speed and maintains the rider's cadence and position in the group.

The paceline should increase or decrease their speed only in small increments. If the front rider wants the group to speed up, she should slightly increase the pressure on her pedals. This increase will be sensed by the rest of the riders who will react by increasing or decreasing their pedal pressure accordingly. If the acceleration is smooth enough, no gaps will open up and the group will not waste energy trying stay connected while gaining additional speed.

The only way for stronger riders to help increase the speed of the paceline is to take longer pulls at the front. The stronger rider should move to the front, increase his speed slowly and stay at the front as long as he can maintain that speed. If a strong rider moves to the front and accelerates, he will open a gap and actually weaken the group by forcing other riders to close the separation.

Although the best riders can keep a smooth paceline for miles with only a few inches between them and the next rider, beginning cyclists should not attempt to keep the same small gap. Paceline safety relies on the front rider moving to avoid obstacles and for each rider to keep a straight, steady line. Trying to maintain a gap smaller than is comfortable will decrease the smoothness of any rider and is unsafe to the rest of the group. If you are a beginning rider, keep a distance from the rider in front of you that is comfortable allowing you to focus on riding smoothly and keeping a relative speed equal to the rest of the group. Eventually your experience level will allow a smaller gap.

Other mistakes affecting the efficiency of the paceline include:

- Pulling out too far from the lead position
- Not recognizing when to move to the faster line.
- Starting the move to the faster line too late.

The lead rider only needs to move over about a foot to allow the rest of the team to move up beside them. On small roads, corners, or other terrain variations it is tempting to pull over too far reducing the effectiveness of the group because no one is in your draft.

When a rider nears the back of the pack, she should start looking over her shoulder to see where the last rider in the other line is located. When that rider is even with her bicycle, she should begin accelerating and moving over to the other line. If she waits until the last rider has passed, there will be a gap there when she moves over and she will have to work harder to close that space, Figure 3.3. The worst situation is when a rider does not realize he should move over and has a very large gap to close or is dropped because he didn't rotate in time. This occurs most often when a rider has been dropped leaving an empty space where he used to be, or a rider has moved to the position of gatekeeper, explained later.

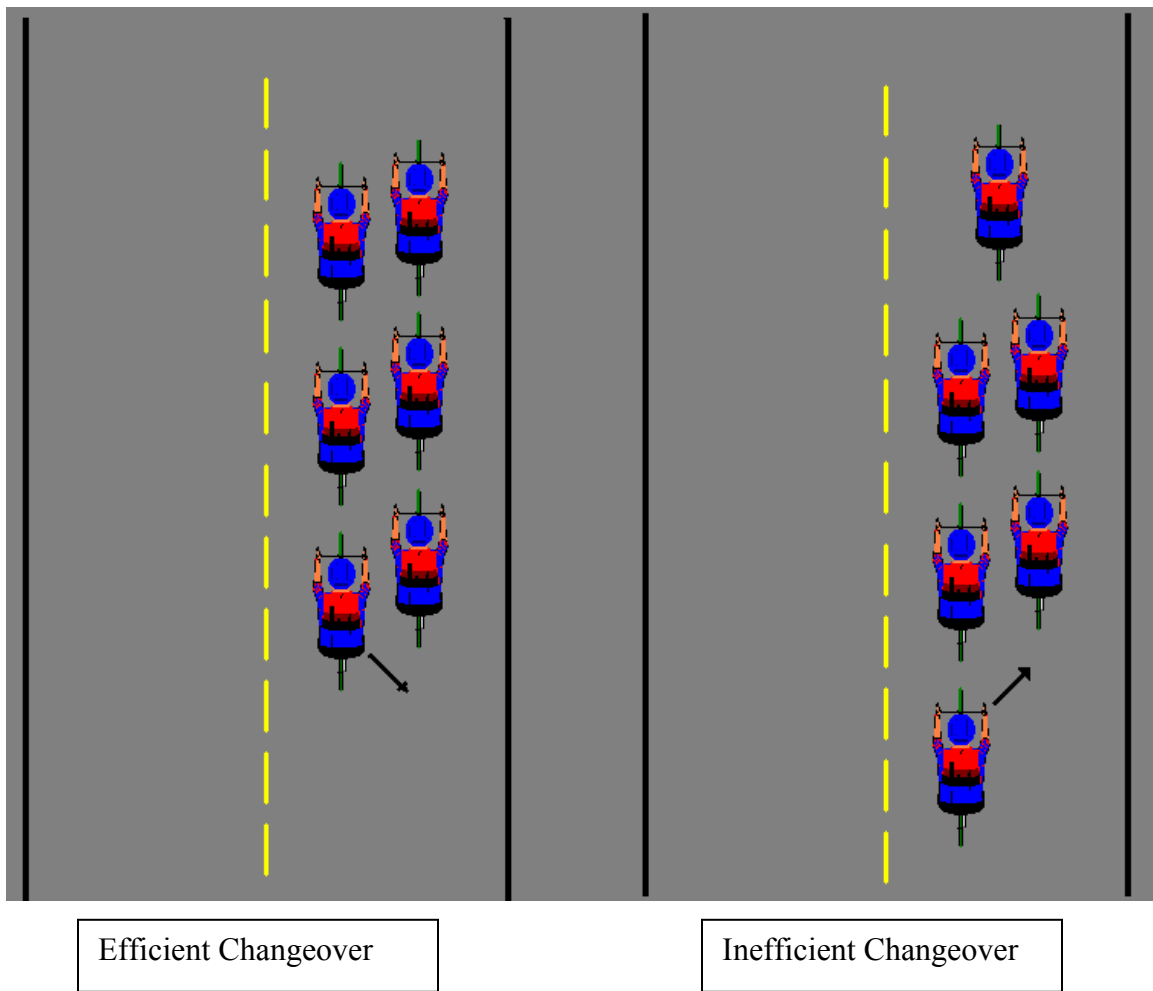


Figure 3.3 Proper Changeover at the Rear of the Paceline

To prevent dropped riders and to assist in the rotation, it is common for two communication channels to develop. Once the front rider has moved over and slowed slightly, she will often say, “clear,” when there is enough space for the next rider to move over and join the slower line. Similarly, the last rider in the fast line will yell, “I’m last,” as she moves forward indicating to the last rider in the slower line that it is time to switch lines. These comments are effective at any skill level for keeping the paceline moving and create a strong team atmosphere in the group.

The Gatekeeper

Once a rider gets tired, he will no longer be able to maintain his speed when confronted by the headwind at the front of the paceline. The tired rider will slow down when the wind hits him and this effect will slow down the entire paceline. The tired rider should recognize this situation and no longer attempt to stay with the main part of the paceline.

The rider can maintain contact with the paceline while resting by moving to the gatekeeper position. The gatekeeper is a rider who does not rotate with the paceline, but remains in the slipstream of the group by staying behind the riders changing lines at the back of the paceline, Figure 3.4. In order to help the riders maintain smooth changes from the slow line to the fast one, the gatekeeper should move over behind the slow line until the last rider recognizes that it is their time to accelerate and move over to the faster line. The gatekeeper can help the group by telling people when it is time to change lines, but otherwise should be nearly invisible to the rest of the riders.

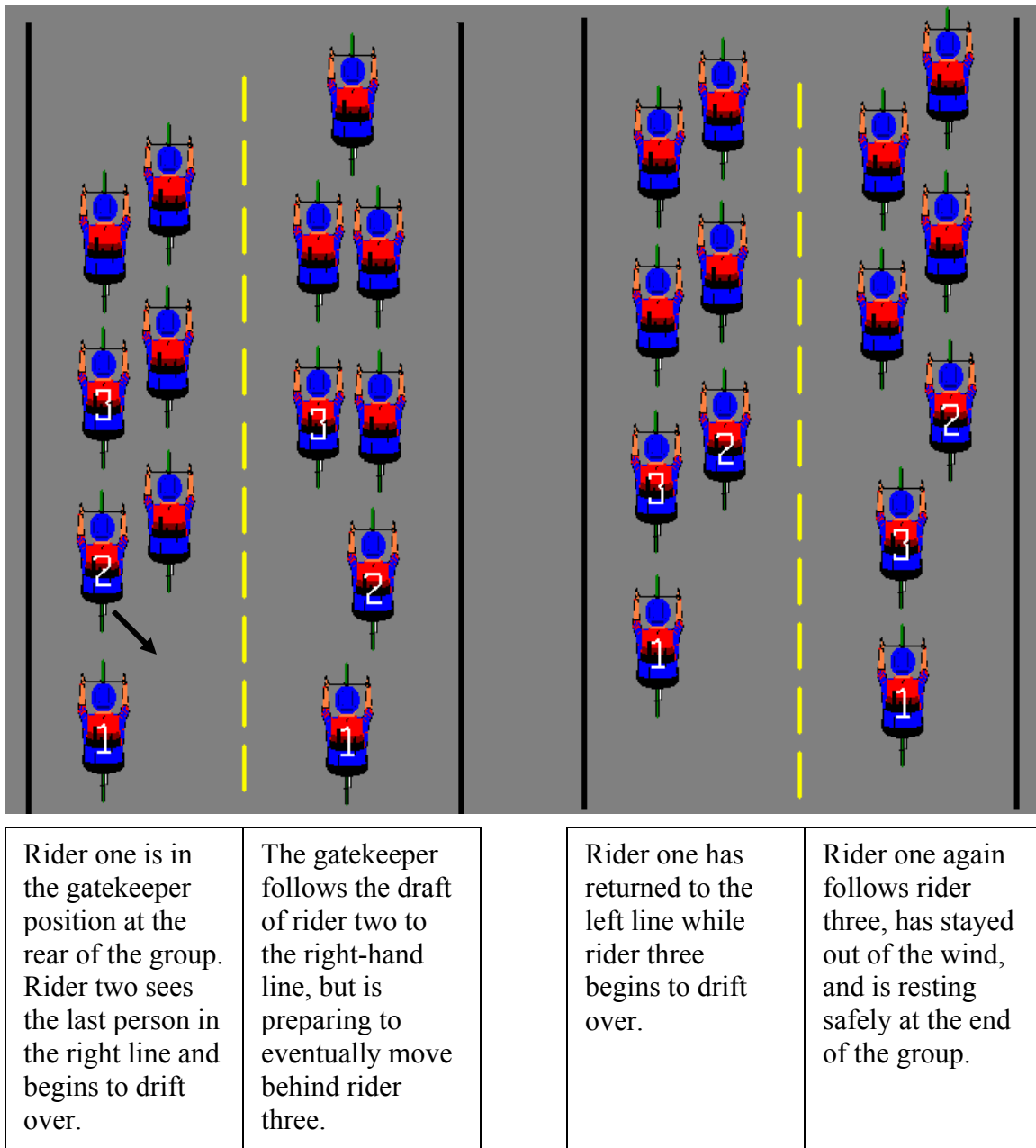


Figure 3.4 Gatekeeper Formation

Paceline Tactics: Applying the Paceline to Race Situations

In the peloton, the front four to twenty riders (depending on road width) control the speed of the rest of the pack. If the front riders want to go slow, the entire group will go slow. If a team wants to control the speed of the race or maintain a steady position at the front of the peloton, it must occupy the entire front section of the peloton.

A team interested in holding a rider in a specific position for a portion of the race, or a team that is trying to force a specific pace on the rest of the peloton must come to the front and create a paceline with a gatekeeper. This paceline at the front forces the rest of the peloton to match the paceline's speed. Having a gatekeeper or two will keep any intruders out of the paceline and the team will be in complete control. This is known as setting up a train at the front of the peloton. Trains rarely form in lower level pelotons because teams do not practice this simple skill. In comparison, all professional races have trains form at one time or another. Peloton speed control is a skill necessary to chase down a break or position a rider for an attack or sprint. If your team can develop this skill, and practice it a few times in the early season training races, a huge advantage will have been won.

Most teams choose a team leader for each race. This rider is supposed to save as much energy during the race to be used in a late attack or the final sprint. The team leader may or may not be the team captain, who is the person responsible for organizing the team. A team leader may be chosen because her riding skills are best suited for the particular race course or because she has the fastest sprint. The rest of the team works to protect the team leader from using energy. These riders are known as domestiques because they are not trying to win the race but are working to help the team leader. In a paceline train, the team leader is always the gatekeeper while the domestiques rotate in front and control the race.

A paceline train is a positional tactic. The level of effort required by the domestiques is so high that a team will use a train only during specific parts of a race. There are six ways that your team can use a paceline train to affect the outcome of the race:

- To hold the position of your protected rider (team leader) for a corner.
- To hold the position of your protected rider for a hill.
- To hold the position of your protected rider for a sprint (leadout).
- To crank up the pace of the peloton so other riders can't rest.
- To maintain a high speed in the peloton to chase down a break.
- To chase back onto the peloton from a flat tire or a dropped rider.

In all cases at the front of the peloton, the paceline is used with one or two people working as gatekeepers. The gatekeepers effectively work to keep the rest of the peloton out of the paceline, Figure 3.5. The high speed of the paceline and the increased distance

between the rotating team members up front and the angry peloton behind prevents other teams from interfering with the rotation.

It is only when positioning at the front of the peloton is important that a train should be formed. A train is not an attack because the entire peloton is drafting off of the efforts of your team. The faster the train moves, the less likely it will be bothered by other riders accelerating around it. As well, a fast train will stretch the peloton out into a double-file line with some riders being 100-200 meters behind the front of the pack.

The train, moving at full speed, only allows a two-person-wide draft behind it. If the train is moving fast enough, the rest of the peloton will be two people wide. Anyone that wants to move into a position towards the front of the pack will have to ride into the wind and equal the energy used by the domestiques. By locking the peloton into one position, the team running the train will maintain an excellent positional advantage entering a corner, starting a hill, or coming into the finish.

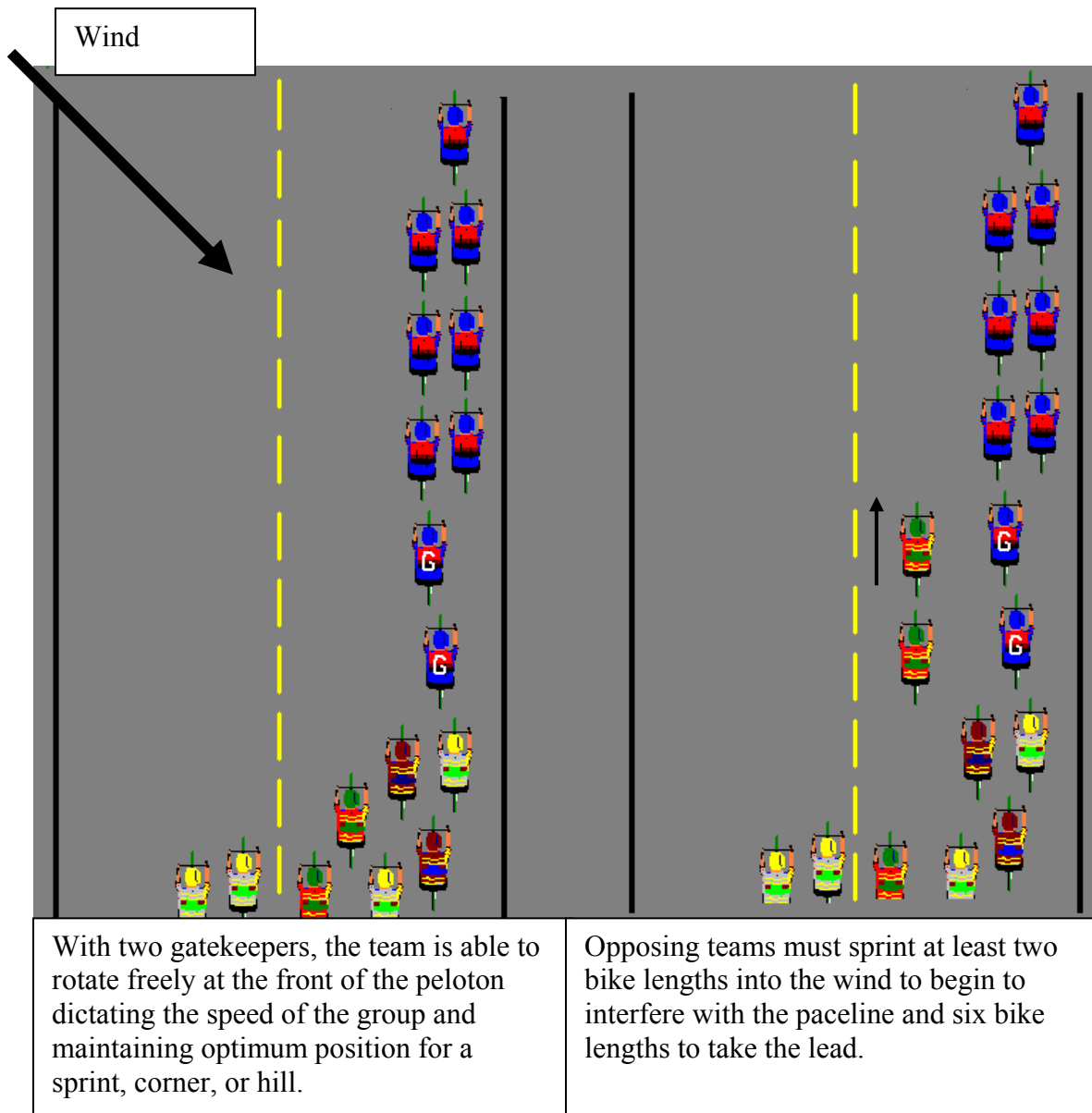


Figure 3.5 A Train at the Front of the Peloton

Protecting the Leader--The Paceline Train at Professional Stage Races

Entering the last stage Chris Horner was leading the **general classification** of the Fitchburg Longsjo Classic Stage Race in 2002. The last stage was a downtown criterium featuring a three-corner course about a mile long. From the start of the 63 mile race Chris' Prime Alliance (white jerseys with blue and white helmets leading in this picture) teammates immediately formed a train at the front of the field allowing Chris to avoid potential crashes, while saving energy in case a break left Chris in danger of losing his lead.



Photo courtesy of www.longsjo.com

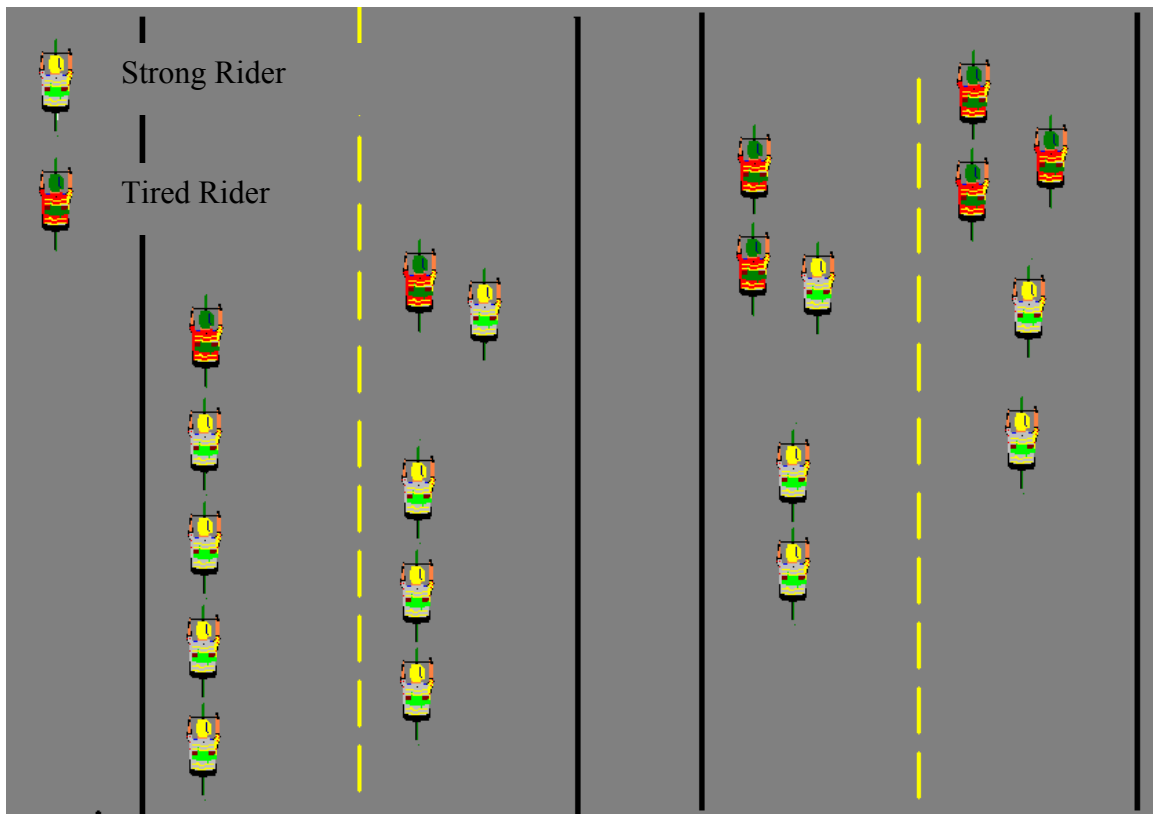
Six teammates lined up in front of Horner (orange jersey, seventh in line) at the front of the pack. They set a pace high enough that no one could attack, but slow enough they could maintain their position throughout the race. Every 10-20 seconds, the front rider pulled over and retreated back to a position just in front of Horner. With the team focused on Chris' success, they allowed other teams to come forward during prime laps, but immediately regained the lead after each sprint.

Without a team, Horner would have had to battle for a spot at the front, sprinted out of each corner, followed every attack, and avoided any crashes. With the strong, organized Prime Alliance team, Chris simply followed the wheels of his teammates to victory.

Pacelines Involving other Teams

Spontaneous pacelines form whenever all involved riders mutually benefit from a consistent, high pace. Attacks off the front need to paceline together to widen their gap to the peloton. All dropped riders will benefit if everyone works together to chase back to the main field. As long as everyone involved understands how a paceline works and how to keep things smooth the group will accomplish its goal.

Unfortunately, it is very rare for a paceline to form in lower level races. So few riders know how to start a paceline (by moving off of the front and going to the back of the line) that they stay at the front until the group speed drops. When the group speed drops, the second rider will accelerate ahead of the front rider, Figure 3.6. This acceleration will open a gap between the new leader and the rest of the group causing everyone to work harder. Eventually everyone is tired and the group slows causing the attack to fail, dropped riders to remain out of the race, or chase efforts to lose ground even on [solo riders](#) off of the front.



The first rider slows after being in front for 30 seconds and does not move over properly to start a group paceline.

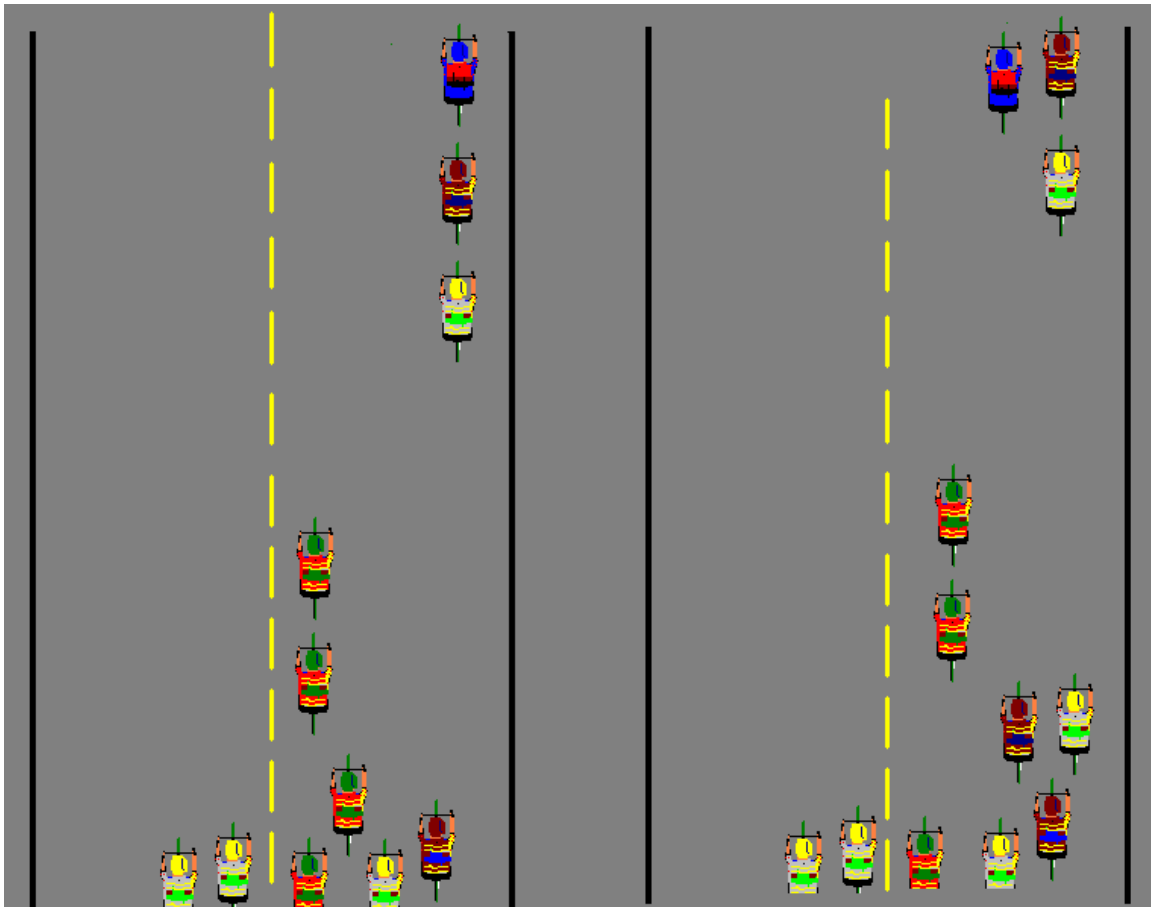
With the whole group slowing down behind the lead rider, the second rider jumps to the front to raise the pace. This forms a gap with the rest of the group causing them to work harder.

The third rider jumps to close the gap, opening another space in front of the forth and fifth riders.

Now everyone at the front of the group is tired, the pack's speed diminishes and the attack fails, or the break gets away.

Figure 3.6 Typical Disorganized Chase Effort

Riders that attack a peloton must immediately utilize a paceline formation to create a gap on the field, figures 3.7 and 3.8. The lack of paceline organization and skill is the number one reason attacks fail in the lower categories of amateur racing. Uneducated riders do not recognize that working together is an important part of bicycle racing. A team that practices pacelines and understands when to implement them, will have stronger attacks, will drop fewer riders, and can use a train to maintain excellent team positioning.



An attack by three riders from different teams will be chased by riders from other teams.

The riders in the break must immediately form a paceline and work together to keep their advantage over the larger peloton.

Figure 3.7 The Attack Forms a Paceline Immediately

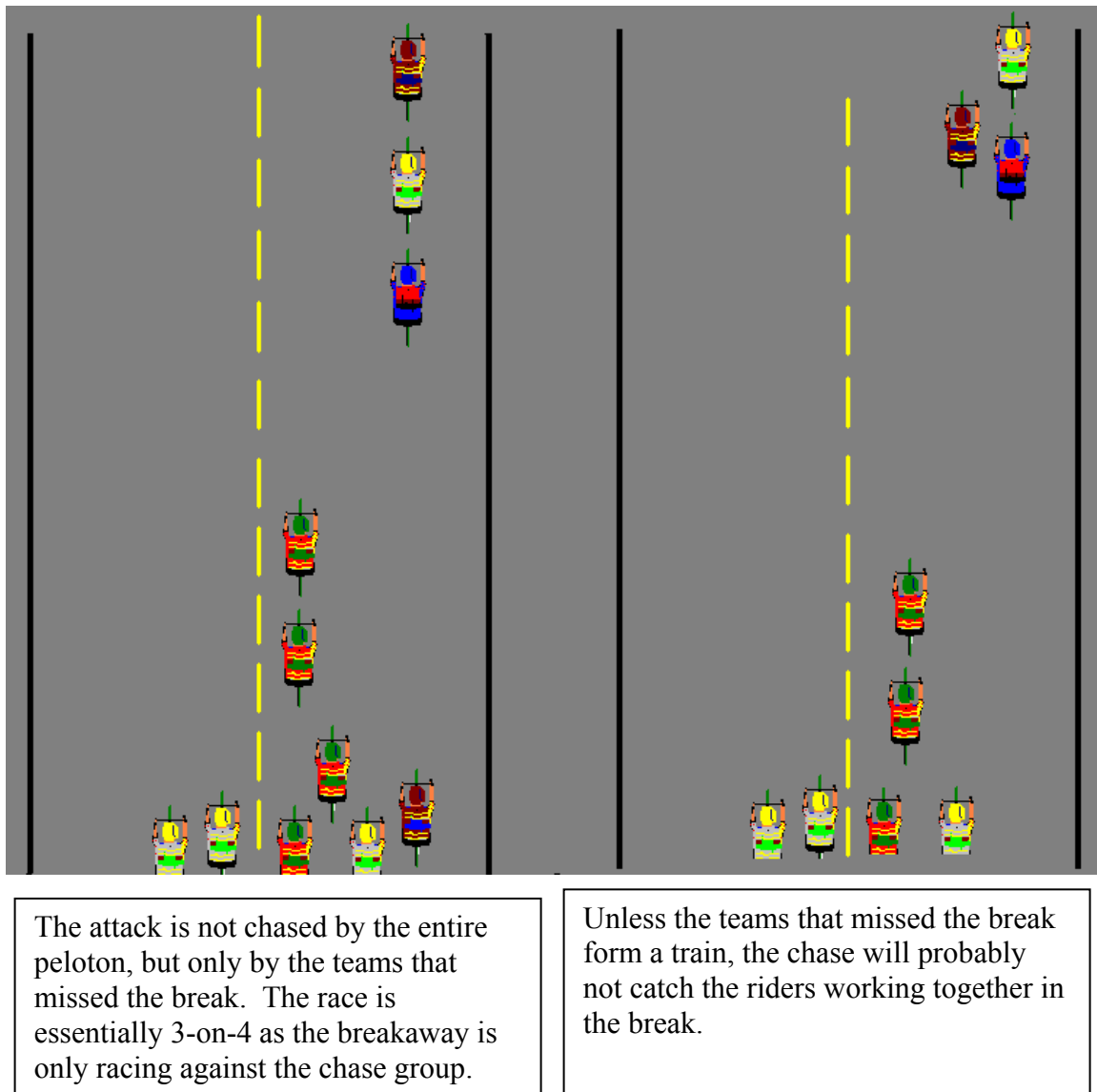


Figure 3.8 The Paceline will Defeat a Poor Chase Effort

Knowing that most amateur riders do not paceline correctly, an experienced rider will attack and gauge the paceline skills of the other attackers. The experienced rider going on the attack should reach a top speed then move over to start the paceline. If the other riders respond with a good pull, the attack might succeed. If the second rider slows down as well, or moves to the front and parks it there until the group speed drops by one or two miles per hour, then the attack is doomed because it is filled with incompetent racers. The experienced rider should immediately rest and return to the pack. Because of the low skill level of beginning rider it is often advantageous in an amateur race to attack as a team. Better to take your own paceline with you than trust your efforts to a bunch of beginners.

Another spontaneous paceline should occur (but rarely does) when a group of riders has been dropped from the peloton. As few as two or three riders working together should be able to catch onto the back of any peloton. An experienced rider understands that a solo effort never will catch the peloton while a group paceline would be effective. The experienced rider starts the paceline as soon as possible. If the other riders do not respond, the experienced riders will attempt to teach the inexperienced riders how to paceline in a last ditch effort to return to the peloton. It is funny to think about trying to learn a major skill such as riding in a paceline while racing, but it happens off the back of the peloton almost every weekend. If teams are not going to teach the skills, then experienced riders will.

Professional vs. Amateur—Race Support, the Caravan, and the Bus

Professional racers with flat tires immediately drop back and receive a new wheel from their team car. Once the wheel is fixed, the caravan of automobiles driving behind the main peloton allows the rider to draft slightly while working back into the main group. If a team has extra energy or the race is moving really fast, one or two domestiques will drop back and help the disabled rider regain the peloton.

During a long stage race such as the Tour de France, some domestiques will work for their team leader for only a portion of the race then drop off of the pace. A stage race typically requires that every rider finish each stage within a certain time period. Eventually, a group of tired domestiques from various teams forms behind the main peloton. The racers must work together to manage their efforts to make sure they finish the stage before the time limit expires so they will not be disqualified from the race. This group of riders from different teams is known as the bus.

Amateur racers often have neutral support to change flat tires, but without the race caravan and no teammates, a flat tire almost always signals the end of competition for the day. Riders with flat tires, mechanical problems, and poor fitness eventually form a bus of their own. Riding in this bus means that you are not going to win the race, but creates an opportunity to practice small group situations. Riders in this situation should always race against these riders as if you were in a breakaway off the front of the peloton instead of off of the back. Always try to beat whatever group you happen to finish with in the race and when you do finally enter a real break you will know how to win.

Riding Within the Peloton and Protecting a Team Leader

The perfect position for a rider trying to save energy is within the first two rows of the peloton. From this vantage point, the rider is out of the wind but is still close enough to respond to attacks or launch an attack of their own. Unfortunately, 124 other riders are trying to occupy this same spot and the front of the peloton is constantly changing due to attacks, hills, and corners. A well trained team can keep its protected rider near the front of the pack the entire race in preparation for an attack or an attempt to win the final sprint.

When a team is not using a paceline train to control the front of the peloton, the domestiques work to protect the team leader from the wind. Protecting the team leader requires at least one teammate to be in front of the team leader at all times. This effort does not require much organization and must be flexible to respond to changes in the peloton due to hills and corners.

It is possible for only one rider to protect another rider, making a team of two surprisingly effective. A team with two or three riders in the front line of the peloton will create a usable space for their team leader in more situations and further gains in efficiency will be created. If the race is fast or long, the front riders will tire and having more teammates is best. A group of six or more riders will be hard to keep connected in the shifting peloton, but may be possible on wider roads. My teams have had as much as twelve riders protecting one team leader on a wide section of road. Having more riders is always helpful so take as much space as the peloton will allow.

If more than one rider is available to protect the team leader multiple organization structures can be used to keep the domestique strong and ready to react to attacks on the peloton. Table 3.2 describes the basic methods of organizing the domestiques on a team.

Table 3.2 Organization of Domestiques

NUMBER OF DOMESTIQUES	TYPICAL ORGANIZATION
1	Domestique rides in front as long as possible.
2	The three riders stay in a line with the first rider continuously in front until tired.
2	The two domestiques alternate the work in front until both are tired
3-6	All domestiques trade pulls in front of the team leader and work to keep the entire team in position. Domestiques always remain near team leader.
6+	Two or three riders protect the team leader until tired. The other riders take over the domestique duties as necessary.

Protecting the chosen team leader of the day is the most basic team tactic available at the lower levels of racing. Having a group of riders in front of you at all times can save up to 30% of the protected rider's energy during a race. The additional strength of the team leader increases her ability to win a sprint or make a strong attack near the end of the race. The domestiques are always near the team leader, but their relative location changes as the peloton changes. A team leader led and surrounded by the domestiques is known as an **amoeba** because its form is always changing to react to the peloton's shape. The amoeba is a defensive tactic the train is a positioning tactic. The amoeba and the train are the two basic positions available to amateur teams.

Professional vs. Amateur—Protecting the Leader of the Tour de France

The Tour de France is approximately 80 hours long. A team leader races unprotected 6-15 hours of this time.

- 7 minutes in the prologue
- 1 hour in the team time trial
- 1 hour in each individual time trial (2-3 per tour)
- 1-2 hours in each mountain stage (5-7 per tour)

The rest of the tour is spent resting behind the domestiques. Once the race enters the mountains, the domestiques still carry most of the load by using a paceline train to control the pace as long as possible up the hill. In the 2002 version of the tour de France Roberto Heras led his teammate Lance Armstrong all the way to the finish of some of the mountain tops further reducing the amount of time Armstrong spent in the wind. Essentially the domestiques race for 65+ hours to help the team leader save energy for their 6-15 hours in the front. It is the effort during these few hours that the race is won.

Without a team, an amateur rider must spend a large portion of a local race in the wind to maintain a strong position within the peloton. If the race is 60 miles long, this translates into over two hours of work to maintain position. For comparison, a rider with a strong team may spend only ten seconds exposed to the wind—those ten seconds are the sprint that wins the race!

Protecting your Territory

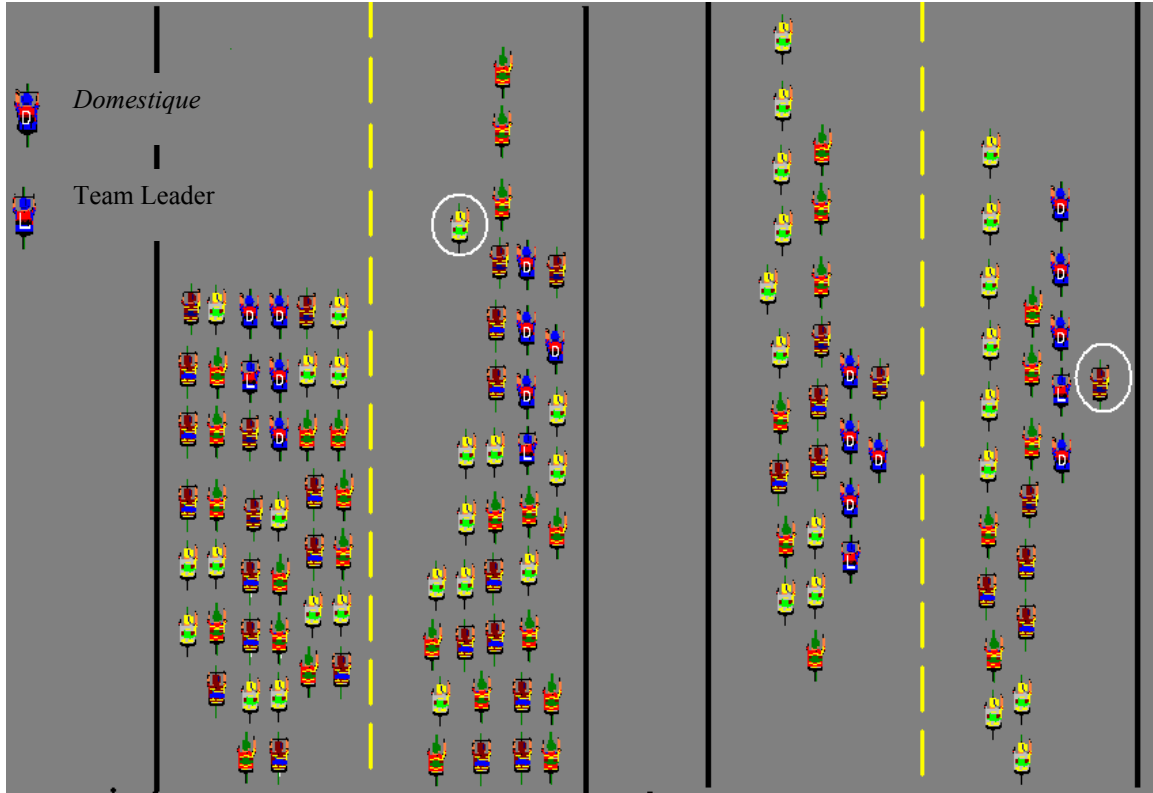
The group of similar jerseys riding at the front automatically will cause some riders to stay away from your protected position. The other rider thinks that if he intrudes, he will have to deal with a group of hostile riders and that alone will keep the opponent a safe distance away. If someone does begin to move over and take the protected spot, a quick, "that is my area," response may be enough to let the rider know he is welcome. Of course, there is no legal reserve put on that section of the peloton, but a little intimidation is enough at times.

Other riders will not care if this is your spot and will move into your space anyways. If you are the protected rider and this happens, tell your teammate Ryan, “Ryan, move left,” and have them move to protect you and not the intruder. If you fall back a spot or two to the third or fourth row of the peloton, also yell to your teammate, “Ryan, back one,” or, “Ryan, back two,” and Ryan will know to slip back a little to regain contact with you. The team leader is responsible for keeping the amoeba connected using these simple commands.

If the team finds itself away from the front of the pack, it is the responsibility of the domestiques to pull the group back to the front. In some cases when the pace is really high, this will require a major effort from the domestique, but is absolutely necessary to keep the team leader in position to win the race. A team with more domestiques in a race has a stronger ability to protect the team leader and keep perfect positioning throughout the race.

Reading the Peloton

The peloton can assume three different forms depending on the speed of the group. Each form has a spot that is most desirable for a team of riders protecting their leader, Figure 3.9.



The flat-front peloton is moving slowly and many riders are on the front row. The amoeba is positioned front and center with the team leader protected in the second row.	An attack extends the peloton moving the amoeba to one side of the road. The circled rider is forced to retreat back into the peloton while the team leader is held in place by the domestiques.	A counter attack on the opposite side of the road begins to rotate some of the positions within the peloton.	The amoeba escapes the rotation if the domestiques match the effort of the counter attack, holding the team leader out of the wind and at the front of the peloton.
			The circled rider is forced to match the attack alone, or retreat the rear of the group.

Figure 3.9 The Amoeba Changes as the Peloton Changes

A **flat front** peloton is formed when the group is going slow and there are no attacks. Because of the slow speed, many riders are willing to ride at the front. An attack quickly elongates the group as one rider moves forward and a couple of riders attempt to move into the attacker's slipstream. If the attack does not form a gap, then the peloton is in a long, skinny shape and the best place for the team to protect their rider is a few rows back and on the leeward (non-wind) side of the peloton. It is not necessary to follow the attack very close because once the attack is over, the peloton will return to a compact position, but hesitating too much will move the group towards the back of the peloton requiring a greater effort to regain position after the attack.

The most demanding situation is when a counter attack occurs on the far side of the road from the team's location. When this happens, everyone around the team is suddenly moving backwards (relatively) to the other side of the peloton on the attack. Here, the team must accelerate and move forward to create a solid position near the front. Each attack from opposing teams takes a little bit of energy out of the domestiques.

All of these situations also apply to riders without a team. If the solo rider is trying to keep a position near the head of the peloton, he too must react to counter-attacks and will need to ride into the wind to maintain his advantageous position. Just holding his position during an initial attack relies on finding a spot in the peloton behind another rider. If a spot in the very front of the group does not open up, the solo rider must drift backwards to find a good draft within the peloton. An organized group of domestiques will maintain a protected area for their team leader. Even if the domestiques must take the entire force of the headwind for a portion of the race, the protected rider will continue to save energy.

With a few team practices behind you and a team attitude created by riding all winter, the effect of the amoeba on saving the energy of the team leader is huge. Rehearse the amoeba in a few early season practice races and instead of having a scattered team worthless to your cause, your teammates will always be right there to help you win.

Professional vs. Amateur---The Amoeba?

Professional domestiques have many responsibilities. A portion of the team is always protecting the team leader while other riders go back to the team car for water, jackets, or food. If the team leader flats, the domestiques stop and give up their bike or help paceline the team leader back into the peloton.

Amateur races do not have team cars or race caravans. Most races will not even have a support car to help with flat tires. In this case the domestiques are mainly working to protect the team leader during the race. To simplify everything the two or three riders working together in formation to protect the team leader at all times is known as the "amoeba" because of its ever-shifting shape. This is simply a descriptive term that is easier to say than, 'protecting the leader.'

Changing the Amoeba into the Train

For most of the race, the team will be able to simply hold the protected rider near the front of the pack and gain significant energy advantages over the competition. Certain parts of the course and race situations require the use of the train to gain even more advantage by positioning the protected rider at the very front of the pack. The transition from amoeba to train should be practiced until it is perfected.

Corners in criteriums, [circuit](#) races, and near the end of road races, are tactically significant because of the accordion effect of the peloton. It is very hard for a peloton that is organized four or five riders wide to roll through a corner without slowing down and bunching up. As the front riders slow down slightly for the corner, the peloton compacts. The wide peloton can not make the corner all at once so the back riders must slow down even more to allow each rider through the corner. Because the first riders went through at nearly full speed, the last riders must accelerate to regain the group. The bunching up and extending out of the peloton is known as the accordion effect and being anywhere outside of the first few rows requires more energy.

Because we want the protected rider to have the easiest path to the finish, using the energy of the domestiques to pull the team leader into the very front of the peloton is a good idea. If the corner is near the finish line of a race or just before a hill, being one of the first ten riders through the corner is absolutely imperative to win the race or to save energy on the hill. In these situations the team must react.

A few hundred meters before the peloton comes to a tactically significant corner, the team leader should tell his or her teammates to move up into position by yelling, "Corner!" a few hundred meters before the turn. The group should respond by gradually accelerating (gradually, as to not open any gaps) and pulling either in front, or closer to the front of the peloton for the turn. If necessary, the group should form a paceline to hold the protected rider as close as possible to the front of the pack. If the group moves to the front a long way before the corner, the team leader will have to play gatekeeper and allow his or her teammates to return to the paceline after taking a pull at the front. With the paceline rotating, and the protected rider in the gatekeeper position, the team will be able to roll through the corner at full speed forcing the rest of the peloton to account for the accordion effect of the pack.

Coming into a large hill, the team in front will not have to worry about the reverse accordion effect as the front riders slow down and the back of the pack bunches up. Being at the front of the pack entering a long hill is also advantageous because the team already has a hundred meter lead on the last rider in a large peloton. Starting at the front for a large hill is almost mandatory to finishing the hill in the front group. Even if your teammates spend a lot of energy to keep you at the front, they too will be starting the hill in the lead and they also will have an increased chance of riding the hill with the front group.

As the team nears the large hill, the paceline at the front responds properly to the change in the road. If the hill is not steep, the paceline will continue to rotate while the protected rider acts as a gatekeeper. Lower angled hills allow higher speeds and therefore the aerodynamic advantage of the paceline is still effective. Keeping the pace high over rolling, or low-angled hills effectively attacks the peloton. With your protected rider perfectly placed and your train barreling over the hill, any riders left out in the wind or at the back of the pack are subject to being dropped by your efforts. If you do not take charge over the hill, it is more likely that the peloton will stay intact and fewer riders will be punished for poor positioning.

Much steeper hills require a different approach as speeds below twelve miles per hour reduce the aerodynamic efficiency of the group. A strong climber will benefit most from having his team set a consistent, fairly high pace to reduce attacks from other strong climbers. At low speeds, it is just as effective to have one person ride in front of the group for as long as possible (hopefully a few minutes) then completely drop off the pace. A rotating group on a steep hill will lose just as much energy as is gained by having the front rider slow down and go to the back. Once the former lead rider reaches the gatekeeper, she normally would accelerate back into the paceline. On a steep hill, this acceleration will cause the rider to fatigue and to drop off the pace.

A really strong amateur team will put eight or ten riders in the front allowing each rider to pull the entire peloton for a few minutes before dropping off the front group. By the end of the hill, some of the teammates will be left while the protected rider still has not seen the wind and has had a very easy ride to the top. If the teammates drop off before the top of the hill, the protected rider is still in a position of strength because she has not worked the entire race. At this point the team leader can attack or simply force the other riders in the front group to take up the pace making duties. If the other riders will not come to the front, they risk slowing down enough to allow the team leader's teammates back into the front group where the teammates can again go to the front and push the pace back to a high level.

Isn't this effort at the front during a large hill virtual suicide for the domestiques? Not necessarily. Two things determine where the rider is located at the top of a long hill: initial position and fitness. A team of riders not working together will start the hill at different positions in the peloton. The smart (or lucky) riders on the team will have found a position at the front, while newer riders will be found in the middle or the back of the pack. As the peloton climbs, each rider fatigues according to his own personal fitness level and eventually finds a group of riders similar in ability. These groups stick together the rest of the race.

A domestique working in the train to keep the protected rider(s) at the front of the field will guarantee that he starts the hill at the front of the peloton. Most likely, this spot is much better than if he had worked alone. If the rider then goes harder than normal for a few minutes during the climb, he is surely going to drop off the pace before the top. After going all out, the rider will begin to plummet back through the thinning peloton. Eventually, the rider will recover and most likely the group of riders at this point is a

better group than if the rider has been working alone the entire race. In addition to a better position in the peloton at the top of the hill, the rider has experience riding at the pace necessary to climb at the front of the pack. Knowing how hard the good riders climb and trying to achieve that effort for an amount of time increases the fitness and experience level of the domestique for the next race.

Most riders go through years of racing before they find themselves at the front of the pack on a large climb. When they do get there, they do not have a strategy to implement and do not know how to gauge their efforts. If a team looks at each race as a part of the larger picture, less experienced riders will be in the thick of the action, will gain valuable experience in their first races, and will push themselves to their own personal limits because they know the team needs their effort. With these advantages, your team will win more races by using favorable tactics, and your teammates will improve their riding skills much faster than riders from other teams.

The Chase

At sometime during the year a really strong rider will attack and form a break that your team was not able to join. If this happens, everyone on the team must automatically realize that a chase must occur.

The proper thing to do is to use your team paceline to go to the front of the peloton and push the group's pace to a consistently high level. Eventually this will bring back the break. A gatekeeper must be used to ensure the integrity of the paceline, but in this case, the gatekeeper should allow experienced riders from other teams to join the chase. The gatekeeper should not allow riders from teams in the break to join the paceline because they will work against the effort. Unknown riders should be treated as a risk because they may not work well with the group.

Most likely your team is the only qualified group to lead the chase. It will take a lot of energy, but if you leave the job to the other riders in the pack, the break will stay away because of a lack of organization. Although it is a risk to pull the entire peloton back to a break, it is better than simply waiting to sprint for something other than first place.

The Leadout

Coming into the finish a team is expected to hold the protected rider in the proper position for the sprint by performing a leadout maneuver. This is very similar to moving forward to gain the advantage in a corner or coming into a hill, but the speeds are often much greater. The team must also have the correct timing to come to the front and push the pace until the finish while keeping the speed high enough to keep the rest of the peloton from attacking and gaining the front positions.

The basic leadout is a combination of positioning (like a corner) and giving a hard effort then pulling off (like a steep hill). With a few miles to go the team moves to the front and establishes its paceline with a gatekeeper (or two) formed by the protected rider(s). This paceline must be fast enough that other teams can not move forward into the lead, but it also must be contained enough so your team does not tire out before the finish, Figure 3.10.

It is the gatekeeper's job to communicate speed requirements to the rest of the team. If the finish is a long ways off, the gatekeeper might yell, "maintain," telling the paceline that their speed is sufficient enough to control the peloton. If the peloton is crowding the gatekeepers, he should yell. "up," letting the paceline know to increase the speed. If the pace is too high and there is a danger of tiring before the finish, the gatekeeper could also yell, "slow it down," and the paceline should decrease the speed a little. These commands can be changed by your team as necessary. They were chosen for my teams because the number of syllables relays the information. With the peloton rolling and the wind creating a lot of background noise, it is not always possible to yell effectively to your team, so knowing that a one-syllable yell means to go faster is sometimes important.

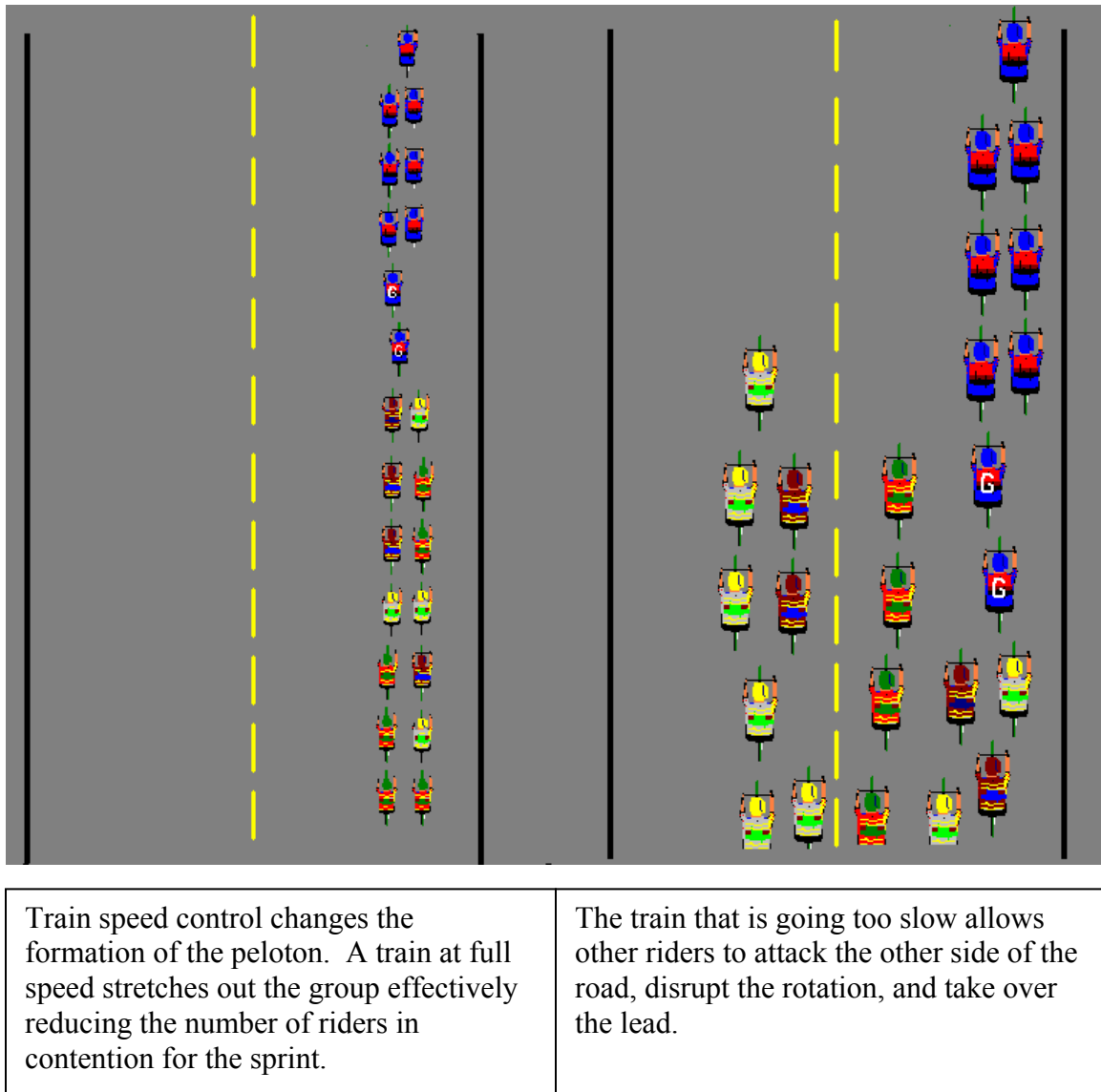


Figure 3.10 Speed Control in the Leadout Train

A good race will have signs near the road for 100, 200, and 500 meters to go in the race. If these signs are not present, the team must scout out the finish area before the race so everyone understands how to time the leadout. Knowing that a typical amateur reaches top speed and can sprint effectively for only a hundred meters or so, the final push to the line will depend on how many people are in the leadout train. A team leader with six teammates running a paceline in front of them will be able to start the final leadout with approximately 800-1000 meters to go. At this point the team should increase the pace somewhat gradually (again, to keep everyone together) or all at once at the command off, “now!” from the team leader. With the train accelerating away each teammate pulls for as long as possible at maximum effort, then moves away into the wind allowing the next teammate to sprint ahead at a higher speed, Figure 3.11. If the timing is correct, the protected rider and possibly another teammate or two will be ready to sprint

with about 100-200 meters to go. At this distance, the protected rider is at least a bike length in front and will be able to out-sprint anyone else in the field due to their greater energy reserves and excellent positioning at the front of the peloton.

Professional vs. Amateur--Leadouts

One major difference between professionals and amateurs is the ability to produce maximum power for an extended amount of time. When a professional rider puts their head down and decides to ramp up the pace, they are able to hold this level for minutes, not seconds. Professional team leadouts do not necessarily have to continuously rotate in a paceline train because each rider can pull the entire peloton for a minute or more.

Professional races in Europe are held on closed roads so the entire pavement is open and movement within the peloton is easy. The wide roads allow a sprinter's team to come forward to the front of the race at any time.

The center line rule in most amateur races narrows the road such that only a few riders may lead a race at a time. The narrow roads also limit the motion throughout the peloton and can lock a team into a poor position coming into the sprint finish. An experienced team will come to the front of the peloton when the course is wide enough to move the entire team at once then use a paceline train to maintain their position.

Amateur teams must use a highly organized paceline train to maintain their position because a single amateur rider can only pull the peloton at top speed for a few seconds. The power difference between the professional domestique and the amateur rider forces amateur teams to practice their leadout timing and coordination until perfected. As your team progresses, practicing a single-line paceline gives your team an additional advantage.

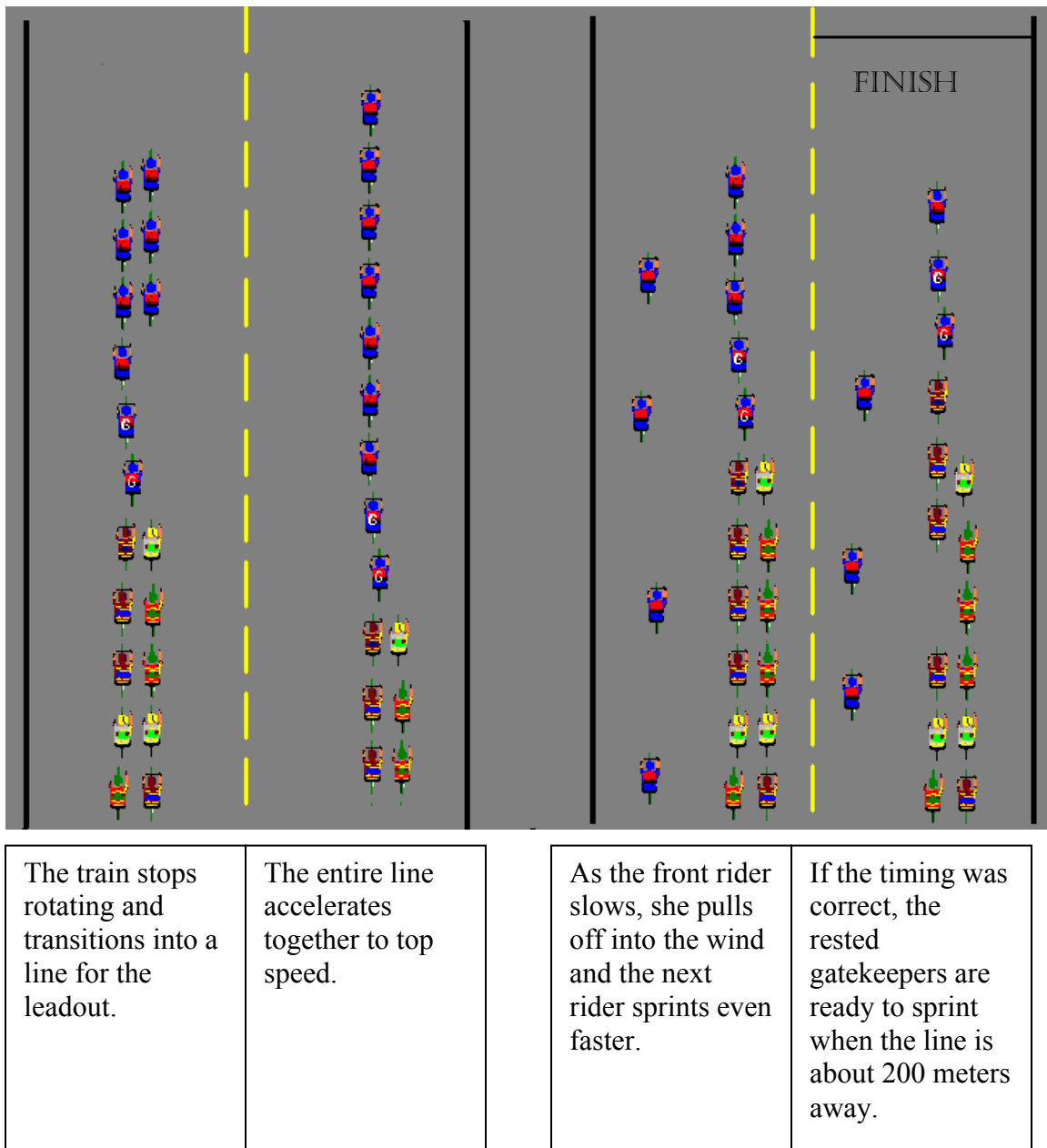


Figure 3.11 The Typical Leadout

It is vital that expended riders move off of the front of the peloton into the wind (to increase the draft for the next rider) and immediately raise their hand as if signaling a mechanical failure. After pulling off, the expended rider will rocket back through the raging peloton and is a hazard to the group if he does not signal in some way that he is slowing and not sprinting. Hopefully the road near the finish is wide enough that the rider can safely move a good distance away from the peloton.

In an amateur field, the sprint is typically won by the person in second place with 100-200 meters to go. It is the job of the leadout team to put the protected team leader into this position. The above strategy is the perfect method for achieving this goal. As we all know, things do not always go as planned.

The first mistake that teams make when preparing for a leadout is to get caught within the middle of the peloton just when they should be moving to the front. With narrow roads, hills, corners etc. it may be necessary to have the team move to the front many miles before the finish. A wide road is required for the entire team to move up into the front and there might only be a few sections of the course where this can occur. If the team waits too long the next chance to move up will be just a few miles before the finish when the speed of the peloton increases and starts to string out the group. With a long, skinny peloton, the team can move up, but not without sacrificing a couple of riders taking big pulls in the wind to get everyone into position.

The second most typical mistake is for the team to go too hard or too easy once they reach the front. Going too hard may actually be good because they might form a small break. This might be an especially effective plan if there is a small hill near the end of the race. Usually, though, the team goes too hard, burns out too early, and leaves the protected rider alone with a kilometer to go. At this point, the energy reserves and sprinting skills are all the protected rider has to try to win the race. Going too slow will allow the peloton to engulf the paceline causing enough confusion that the team disintegrates before the finish. Now the team leader is probably five or six rows back in the peloton and may be boxed out of the sprint.

Experienced teams can practice and use a [professional rotation](#) for technical corners. The professional rotation is one where the front rider pulls for 5-20 seconds regardless of how many riders are in the leadout. The group forms a single line because there is only one rider going backwards at a time after his pull. The long, skinny line is easier to maneuver through back-to-back criterium corners but the longer pulls require more energy from each rider.

Although it may sound difficult, putting your entire team into position for the leadout is not impossible. Remember, this is an amateur peloton full of untrained, unaware riders focusing on their own pain and their own chances of winning. Even though the other teams will see all twelve of your teammates lining up at the front, they will not know how to stop the rush, especially since they will be fatigued from riding alone for forty miles.

The acceleration and the synergy of a perfect leadout is quite an experience. The last members of the group coming into the line are usually sprinting 3-5 miles per hour faster than they could have achieved on their own. Looking down and seeing 35, or 40 miles per hour knowing that you are going to take the win after a perfectly executed leadout erases any memories of long, cold practice sessions earlier in the season.

Other Duties of the Domestiques

While protecting the team leader at the front of the peloton, the domestiques are also responsible for keeping the race under control. Throughout a race, opposing riders will attack the peloton requiring domestiques to leave their leader and mark the attack. If the domestiques do not mark an attack and a breakaway forms, the entire team must work together to chase down the breakaway. A team should decide to mark attacks or chase attacks before each race.

The layout of certain race courses diminishes the chances of breakaways forming from attacks. Short, flat courses without wind do not give an attacker much advantage over a large peloton. A team that saves energy by ignoring attacks by single riders is predicting that breakaways will not form. Hills, wind, and smaller pelotons allow riders to attack and split the group increasing the chance of breakaways forming. In this case, a team that marks each individual attack thinks that a breakaway eventually will win the race. By studying the race course and using experience from earlier years, a team will decide to mark each individual attack or simply wait for a break to form before the whole team chases it down.

Teams may also decide to mark only certain attacks. Stronger riders can force a breakaway even on flat ground and must be watched closely by the domestiques. It is typical for a team to choose to chase down breakaways together, but individually mark a small number of attacks by certain riders or other strong teams. Combining these strategies requires a watchful eye and strong knowledge of the competition.

In a one day race, the chosen team leader could change if necessary. Flat tires, tired legs, or motivated teammates force a team to be flexible in trying to win the race. If the chosen team leader has a flat tire, the team may just change leaders and continue the race. Often at the lower levels there is simply not enough time to go back and help the former team leader back into the race.

A stage race creates an entirely different story. Because the general classification of the race is based on the results from multiple races, the team leader highest in the standings may be the team's only chance at performing well. In this case, a flat tire by the team leader can not go unnoticed. The team leader is responsible for informing as much of the team as possible of the situation and for changing the tire with no delay. The domestiques for the day should immediately stop with the team leader in order to form a team chase back to the peloton. A team of six or more riders inevitably will catch a peloton during a road race while a stranded team leader would have lost the race.

Professional vs. Amateur—Who would ever want to be a Domestique?

A domestique on a professional team is motivated mostly by money and the opportunity to gain fitness by working harder in races. If their team leader wins, they get a pay bonus. A win also brings more exposure to the team and thus better sponsorship deals leading to better pay. Professional riders know that the extra training effect from riding in the front of the peloton all day gives them better fitness for their future careers.

An amateur rider that works all race to protect the team leader and helps in the leadout only to pull off and get 45th place must be crazy. With race fees climbing to above twenty dollars, why would anyone pay money to help someone else win? The answer is what allows an amateur bike team to win consistently.

The factors driving an amateur domestique include:

- A domestique one week could be the team leader the next week.
- Riding as a domestique is a better workout than being team leader.
- Giving a leadout is intense even if you are not the sprinter.
- A teammate winning a race because you helped is satisfying.
- Not all racers are talented at all race disciplines.
- Focusing on responsibilities other than winning the race is less stressful than trying to win every week.
- An inexperienced rider can immediately help the team as a domestique and the extra training hastens their rise to team leader.

A consistent team structure allows each rider to eventually work all three positions of attack team, domestique, and team leader. The responsibility of each member is different and leads to a different race experience. When the team wins because of a team effort, the entire group is able to share in the win.

A rider that does their best to understand their responsibilities in each race and performs these responsibilities to perfection is welcome on any team. The best domestiques turn into the best team leaders at the amateur level because their experience leads them to the most efficient position in any peloton. With most race seasons extending over twenty races, there is plenty of time for each member of a team to rotate through the team positions and learn to master all of the skills required in cycling.

Small Group Strategies

Small groups form at the front of the pack and off the back of the pack. Smaller races of 10-50 people can also behave like small groups and the tactical advantages of teamwork are magnified. At the lower levels, small groups usually form because of a large hill that breaks up the larger peloton into groups of varying fitness levels. If you are strong enough to survive the hill in the front group, your knowledge of small group situations can greatly increase your chances of winning. If you have a teammate or two in the group, your chances are even greater.

The goal of any small group situation is to come to the finish line by yourself after attacking and time trialing away from the rest of the pack. A solo finish like this is one of the most rewarding accomplishments in cycling. A peloton of 125 riders rarely will let a single rider away by themselves, while the smaller group of 5 or 10 eventually will fatigue, panic, and stop working together just as the strongest (or smartest) rider slips away for the win.

A successful breakaway goes through the following steps in every race:

- Attack.
- Immediate paceline with everyone working together to create distance from peloton.
- Pacing of the group to maintain distance from peloton.
- As the finish line nears, individuals begin to attack the small group.
- After a number of attacks, one rider slips away and is now time trialing against the rest of the small group.
- The solo rider is caught (and more attacks ensue), a group sprint occurs, or the time trialist finishes the race for a solo victory.

Again, the first skill necessary for small groups is the paceline. As mentioned before, any group attacking off of the front must immediately use a paceline to gain time on the peloton (see Figures 3.7 and 3.8). If the group does not organize and does not work together, the pack of 30-90 riders always will catch the group of 10. For a group off the back of the peloton, a paceline is also necessary to latch onto the leading pack. Initially, work as hard as you can to enlarge the distance between the peloton and your breakaway.

Once the break is significantly ahead of the peloton, each rider will begin to match their efforts to those of the other riders in order to gain an advantage at the end of the race. If one rider slows, the other riders may slow as well trying to conserve energy. Hopefully the officials in the race will update the group with the time gap to the peloton. Once the paceline has opened the margin to the pack, try to stay near the front of the small group. Do not spend too much time in the wind, but the front has many advantages. When other riders attack, you can respond immediately if you are in the front few positions. If you are in the back, it is likely that one rider in the group will not be able to respond to the attack, a gap will form two riders in front of you, and this will

leave you separated from the chase. It is a terrible feeling to lose a race because you were blocked out of a small group situation, so stay near the front.

When the combination of distance to the finish line, gap between the break and the pack, and the speed of the group are such that the breakaway is probably not going to be caught, then it is time to begin racing against the other riders in the group. This distance from the finish line of your attack is based on the terrain, your ability to time trial, and the strength of the group. It is typically within the last five to ten miles that a break will begin to attack itself.

If the group is still working together, continue to help the cause, but balance your efforts. It is important to continue working because the other riders quickly will notice a rider hanging on the back and will begin to attack you specifically. If there is a rider in the group that is not working, place yourself directly in front of that rider and do everything that you can to disrupt the paceline. Everything mentioned before: jumping when at the front of the group, pulling over too far at the front, not maintaining your momentum on a hill will make the person behind you work harder. As that person tires, or the rest of the group shows signs of fatigue, it is time to attack.

Attacks should come when you reach the front or near the front of the small group. Any time that you open a gap with the rest of the group, everyone in that group is forced to work harder than you to close that gap. The best time to attack is on a slight uphill or through a corner, especially if you can maintain your momentum after the group has lost its momentum. At this point, the group will have to accelerate much more than you and it will have to work harder to follow you up the hill.

A good attack causes the riders behind you to form a paceline and chase. Most often, the group is not organized enough to form a smooth paceline and each rider eventually must accelerate alone to match your efforts. Even though you are working into the wind, each other rider in the group is also riding alone in the chase and therefore is using as much energy as you. Your attack forced the strongest riders to work harder giving you a better chance of winning.

If someone else attacks the group, do your best to match their acceleration while trying to draft as much as possible off of the other members of the group. If the attack is strong, work with the remaining riders in a smooth paceline until the attacking rider is caught. Panic and chasing alone are detrimental to the chase effort. The only reason that a single rider should stay away from a group is if the group does not work together.

This is one of times in bike racing where the diversity of tactical knowledge within the peloton really allows the racing to be exciting. If everyone reacted perfectly, an attack on a small group would not cause panic as each rider in the group would understand that five people working in a paceline will catch anyone riding alone. The exciting part, especially at the lower levels, is that people are alarmed at each attack and immediately accelerate to chase the attacker. This quick acceleration destroys the paceline for a few moments until the group can start the train. Once the chase begins,

one of the riders will get to the front, stay there for longer than their strength allows, and slow down the entire group. The next rider may not have ever ridden in a paceline and will jump thinking that “pulling through” means pulling ahead. The third rider will try to match the jump and will be exhausted when he or she reaches the front. Finally, the last rider will pull through normally while trying to teach the rest of the group how to race. This calamity assists the rider who has practiced small group situations, is able to control their emotions when attacked, and can immediately start a smooth flowing paceline and work with the group until the attack is caught. At that point, the strong rider knows to attack on the slight uphill and pull away from the resulting mess of poor riding going on behind them.

Winning your group—the Massachusetts State Road Race

A race with a large hill almost guarantees that there will many small groups of riders finishing the race. I always told my team to, “win whatever group you are in.” Even if you were not in the first group over the top of the hill, treat 40th place as you would 1st and try to use small group tactics to ride away to a solo finish. In my last race in New England this was the case and I was racing for a place much lower than 40th. The group included about seven riders and was formed with over thirty miles to go in the race. Before the finish was another monster hill followed by some rolling hills to the line.

During the thirty miles before the last hill, our group would ride up to other packs or single riders dropped from the peloton in front. As we neared each small group, I would attack and try to bridge up to the rider in front of us. Whenever I attacked, the stronger riders would try to chase me down, but did not know how to start a paceline, and did not work together very well. Knowing this, each of my attacks took more energy out of the chasing riders than the short, controlled, time trial took out of me. I also had the advantage knowing that if I reached the group ahead of me, I could paceline with that group and use my new friends to stay away from my former adversaries.

As soon as I was caught, the rest of the group was ready to slow down and rest. The chase had caused each of them to accelerate and now their legs were burning. Soon after each time I was caught, another small group would appear a half mile in front of us. When we came to a small uphill or corner I would attack again going out for a short time trial while the other riders grumbled and took up yet another poorly organized chase.

By the time we reached the hill with five miles to go, most of the group was exhausted. I felt good because I simply had to time trial in front while the other riders worried and worked hard to catch me. Everyone in the group knew I was strong and many of them thought I would attack at this obvious tactical point in the race. Instead, I could use this to my advantage and waited for the tired riders to attack. One rider had the climbing skills to accelerate and I quickly latched onto his wheel using his draft and pace to pull me up the incline. After following this rider up the entire hill I accelerated away using the strength I had saved from drafting up the hill. Using small group tactics I used my energy in a way to tire the other riders while keeping my legs fresh. After my final attack with four miles left I was on my own for an exciting 60th place.

Here is a summary of how to win a small group breakaway situation without the help of teammates:

- Work as hard as possible with the group until the gap between the break and the peloton is insurmountable.
- Continue to pace yourself and the group until the finish line is within five to ten miles.
- Use poor paceline riding to force riders who are not working with the group to fatigue, or be dropped from the group.
- Stay near the front of the group so you can respond to any attacks.
- If you are attacked while in the group, latch on if possible, or stay calm and use the group's numbers to your advantage in the chase.
- Try to draft as much as possible while chasing and encourage the chase to use a paceline.
- If the attacker is caught, be prepared to launch or defend a counter attack.
- When you feel like you can time trial away from the group and win the race, attack on an uphill section and break the race apart.
- Gauge your efforts against the group and look back to see how organized the chase has become. Continue your advantage until the finish.
- When your attack succeeds, zip up your jersey, wave to the crowd, and enjoy every second of coming into the finish line alone.

Teammates in the Small Group

Having teammates in the small group can make for some really fun racing. With two or more teammates working together, one person can rest while the other works to keep the break alive. As the finish line nears, both of you can attack the group forcing the rest of the riders into chasing one or both of you at all times.

The number of teammates present creates different strategies shown in Table 3.3.

Table 3.4 Small Group Strategies

NUMBER OF TEAMMATES PRESENT	GENERAL STRATEGY
None	Work enough to guarantee the group's success and mark as many attacks as you can to avoid being left behind. Make one or two attacks when you feel the group tiring. If caught, positioning and timing during the sprint are essential to winning.
One	One rider works to guarantee the success of the group, then attacks as much as possible. The other rider rests. Either the attacking rider will get away or the resting rider will win the sprint.
Two or more	Riders can take turns attacking the group solo, in pairs, or all at once. One or more riders may also rest while the other two attack.

These tactics work best in a break of 5-10 riders. Similar plans can be created for larger groups and these tactics have proven effective in races up to 50 or 60 riders. In larger pelotons of 60+ riders, these tactics are less effective because the entire group will not become fatigued before the end of the race and breakaways are rare.

If your team has two riders in a break, they both should initially work as hard as possible to make the break successful. When the lead on the peloton is sufficient to begin pacing the break, one of the teammates should stop working and save energy for later in the race. Normally a person who is not working in a break is shunned for not helping, but with a teammate pushing the pace there should be no issues.

With five to ten miles to go and the peloton far behind, both of these riders can now work together to attack the rest of the pack. The basic strategies are:

- Sprint Method: one rider waits for the sprint while the other attacks. Either the attacker will get away or the sprinter will win the sprint.
- Group Attacking: both teammates attack at once immediately pacelining together to stay away from the field.
- Relay Attacking: one rider attacks first, forcing the group to pull the other rider with them in the chase. When the first rider is caught, the second rider immediately counter-attacks. With one rider attacking and the other rider not working in the chase, there is always someone from your team that is stronger than the other riders.

With multiple teammates there are also tactics within each strategy that increase the chances of a successful attack. One of these tactics is the de-acceleration gap. If both riders attack at once, the rest of the pack is forced to bridge the small gap to the attackers and chase. This is the Group Attacking method explained above. If the rest of the pack immediately latches on to the attack, the second rider can slow down (not by hitting the

brakes, but by coasting) causing a gap to open up between the front rider and the rest of the group. This de-acceleration creates a gap. In this situation, an attack is placed by slowing down, not by speeding up. Each opposing rider must ride around the slowing teammate and accelerate to mark the attack. The slowing teammate simply accelerates slightly to catch the last chaser and is pulled throughout the chase. Deceleration gaps are also very effective around corners and during the transitions into small hills causing the pack to lose momentum while the front rider can speed away from the group. Two teammates working together using a variety of attacking combinations creates one of the largest advantages known in bicycle racing.

Although it is frowned upon in professional races, if your teammate in the breakaway has attacked and is riding alone and the group has formed an actual paceline in a chase, your best defense is to take part in the paceline, but ride poorly. Do not ride aggressively, but simply take part in the paceline and jump when reaching the front, pull over too far putting the entire headwind into the rider following you, create gaps as much as possible by not responding to hills. Someone who has practiced riding in a paceline also will understand the weaknesses available in disrupting a paceline. Obviously, do not ride dangerously, but small gaps here and there can shift the focus from the chase, to you, effectively ruining the group's efforts.

If there are more than two riders from the same team, a combination of the above tactics will be highly effective depending on each rider's specific talents. If one rider is an excellent sprinter, allow them to rest the entire break while everyone else attacks. If everyone has strong time trial skills, have the entire group attack at once and immediately put all that team time trial work to use to sweep the top places at the finish. Having a podium full of teammates is the only win more exciting than coming into a finish alone.

Practicing

In the two months before the racing season a team needs to prepare by creating different scenarios that replicate the above situations and force each rider to make the correct decisions to win a race. Having a team that is extremely comfortable in a paceline, can protect a leader throughout a race, and understands small group tactics regularly will dominate the podium. It would take years of racing to see and react to all the situations that can arise in a race so organized teams can accelerate the experience level of their members with a couple of focused practices and practice races.

Team skills that need to be mastered for team tactics include:

- Pacelines on flat ground.
- Pacelines on rolling hills.
- Pacelines on extended hills.
- Downhill pacelines.
- Cornering in a paceline.
- Pacelines with gatekeeper.
- Riding in a peloton.
- Protecting a team leader / Amoeba.
- Chasing down a break.
- Leadouts/sprints.
- Small group situations.
- Communication.
- Attacking.
- Blocking.

The first six topics can be covered during any length team ride and are especially suited to long, slow, base mileage rides typical of early season training. It is a common misconception that pacelines need to be fast. A paceline is effective at any speed and at any intensity. Slower pacelines quickly build smooth, professional riding skills in new riders and help experienced riders become more efficient working as a group.

The next type of practice session is what I call a, “track workout.” Although it is not usually held on a track, it is a group of situations designed to be practiced at the team practice area. The team practice area is the office park, or other circular road without much traffic where the team can ride without worrying about stoplights or other interruptions. The different track events can be run at varying intensities and attempt to re-create the experience of riding in a peloton without entering a race. Mix and match the following workouts to suit the training needs of your team.

Professional vs. Amateur---Team Camps

A professional team only really needs to practice their team time trial, and some professional teams might not even do that. Bike racing tactics are so simple that once everyone knows how to participate in a paceline train and how to protect the team leader, they do not need much practice to integrate their basic riding skills with new riders.

An amateur team is the complete opposite. Few riders on the team have heard of a paceline train and don't know anything about when or why one should form. The same is true for a leadout. An amateur practice session moves everyone on the team to the same page, educates its riders on when and why each tactic should be used, and clears up how the team is going to perform leadouts and trains. The first year is a constant battle while later seasons may only require a quick refresher during a practice race to bring the new group together like a professional team.

Track Practice

Before the practice starts, ride at least 5-10 miles as a warm-up. Usually the team will ride to the practice area, or just do a few laps to get ready. In between each event a couple of slower laps should be ridden to balance the intensity of the events. A good ratio is two or three slow laps for each event lap.

Each workout is designed for at least ten riders. It may be possible to redesign each workout for more or fewer participants. If your team does not have ten or so riders, recruit other riders (masters, cat 1,2,3 etc.) from your racing club or invite members of rival teams as necessary.

Changing or modifying each basic workout allows a team to fortify specific weaknesses and work on skills required by certain races in their race calendar. Practicing these skills on the actual race course to be used later in the year is another method to build the team's experience and knowledge of bicycle racing.

Similar workouts can be applied to local group rides. Be careful not to disrupt the group or create dangerous situations on the open road. Group rides do not allow much time for team evaluation so do not rely on them to provide the best team training available.

Here are nine fun events that test teamwork and build the experience level of each cyclist.

TEAM TIME TRIAL	
Split into groups of 2 or more. Pick a time trial course to ride once a month in the early part of the season to promote paceline excellence and teamwork. Keep track of times throughout the year, and compare to other years.	
Focus:	Pacelines, time trial strategy, teamwork, length of pull, paceline efficiency.

AMOEBAS TO TRAIN—THIS SKILL ALONE WILL HELP WIN RACES	
Divide the group into groups of 4 and 6. Starting with a train in front of the small peloton, the group of 6 is going to change from train to amoeba and back every lap. The group of 4 should work as individuals to attack the amoeba and to stop the train.	
Focus:	Builds the essential skill of starting and stopping a paceline with gatekeeper train on command. Eventually is necessary for leadouts.

TRAIN PERFECTION	
Divide the group into 5 and 5. One group of 5 tries to run a train for as long as possible while balancing the speed necessary to keep the train in front. The other riders act as if they are the peloton. If the train tires or goes too slow, it will be attacked by the other riders. A good train will go fast enough to keep the other riders behind the gatekeeper but not too fast to wear out the team. Once the train is passed by two of the opposing riders the exercise is over.	
Focus:	Encourages riders to monitor the speed of the train in comparison to the energy of the peloton.

CHASING DOWN A BREAK / TEAM PURSUIT	
Break into teams of 4 and 6. The team of 4 starts the race with a small advantage (10-20 seconds). The group of six chases until the group of four is caught. Divide riders into teams of varying abilities to promote everyone working together. Riders with less fitness will take shorter pulls and have to decide if he needs to rest at the gatekeeper position. Setting a time limit or lap limit controls the length of the intensity. Changing team sizes encourages development of different paceline skills from two person pacelines to rotations.	
Focus:	Working in a smooth paceline is the only effective method of chasing a break. Promotes paceline skills and builds confidence in chasing abilities even among various fitness levels.

ONE LAP CRITERIUM	
Divide into teams of 4 and 6. The team of four is allowed to organize and do a leadout while the other riders are required to sprint alone. Any rider can attack at any time forcing the pace. The team of 4 should experiment with early lap pacelines for positioning and practice timing on their leadout train. Use the amoeba if necessary. Solo riders are encouraged to attack early and often then try to follow wheels in the sprint.	
Focus:	Amoeba to train, leadout timing, leadout strategies, leadout speed management, pack sprinting, attacking.

MATCH SPRINT	
Break into groups of 2-4 riders with similar abilities. Winner is first to cross the line after one lap.	
Focus:	Extreme small group situations, no teammates, sprinting.

NOT SO TEAM PURSUIT	
Just like the team pursuit break into groups of 4 and 6 or 5 and 5. Set a lap limit and give an initial head start to the smaller team. Smaller team must work together to build a lead, then outwit each other for the final win. The second group must chase as a unit, then may attack each other once everyone is together. Winner is the first rider across the line.	
Focus:	Rivals working together, rivals attacking each other, when to attack a small group, small group situations, bridging from one group to the other, sprinting.

BASIC SMALL GROUP	
Form a team of 2 and a team of 3 or 4. Set a lap limit for the race. The team of 2 are teammates, the rest are individuals from “separate” teams. The team of two should adopt a small group strategy to win the race. Small team wins if either of them cross the finish line first. The individual riders win if the small group fails to win the race.	
Focus:	Small group strategy, relay attacking, group attacking, small group leadouts.

MODIFIED OLYMPIC SPRINT	
Teams are created of 2, 3, 4, or 5 riders. Each team does one lap and the fastest time wins. Race timing can be changed to suit the needs of the team: timing the first to cross the line promotes a good leadout, timing the last to cross the line promotes a consistent effort from all team members.	
Focus:	Teamwork, paceline skills, length of pulls for varied fitness levels.

A great way to practice these events is to have the team ride out to the practice area and do one or two events before continuing a typical group ride. This limits the amount of high intensity effort as well as allows for a complete warm-up before going hard. A team that is not used to working out in an organized manner will be more receptive to this workout schedule.

The opposite method is to schedule the entire ride for team practice. The team can warm-up at the practice area and do specific events for the entire workout. Devoted workouts such as these keep everyone in a focused training mood and allow the team to truly build the new skills necessary for later races.

In both cases more excitement can be added to the routine by creating teams that last all season or preseason. Create a scoring system that allows each team to score points for winning a particular event. The four to six weeks of organized practices do not have to be boring or non-competitive. Creativity might also make track workouts a highlight of the season.

Finally, having small group workouts and inter-team competitions allows every member of your team to experience winning. Only one person can win a bike race and it is common for beginning riders to wait two or three years before coming in first. The small group nature of a team practice ensures that everyone will come in on top once in awhile and a team time trial allows four or five riders to win each race. Although team practices are designed to build teamwork and coordination, having a team that knows how to win is equally important.

A new rider will take years to experience all of these situations just by racing. Racing never will develop the team skills of pacelines, leadouts, and the amoeba required to utilize team tactics. Stopping for a few weekends a year and practicing specific situations in a controlled environment allows your team to gain advantages in every race you enter later in the year. It is unlikely that other teams will bother to match your efforts in practicing, and therefore will not be able match your efforts on the course.

Training Races

Another chance to work on team tactics is the training race. Often held in the early season, training races have few prizes for the winners, low entry fees, and liberal rules about dropped riders. If a team enters a training race it is free to practice new techniques and try new strategies without the fear of being dropped and losing the race. A team that does not create a specific plan to develop a skill in a training race is losing the opportunity to gain racing experience.

A training race creates an excellent opportunity to work on skills requiring a peloton to practice effectively. The amoeba, train, attacking, and blocking are hard to replicate without true competition. In a training race, your team will need to focus on

developing a specific skill and not worry about what the rest of the peloton is doing or thinking.

If the training race is a criterium, find out if there are rules about dropped riders returning to the peloton. If riders are allowed to return to the peloton, the team can attack without the fear of tiring and getting dropped. A no-drop rule is a great opportunity to allow your team to let loose and ride aggressively.

A practice road race gives the team an excellent opportunity to perfect the simple amoeba technique. The communication necessary to effectively protect a team leader will be tested while the protected rider can begin to feel the benefits of riding the entire race out of the wind and near the front of the pack. A road race leadout is more about positioning than speed so the team can go to the front early and test how well the team can keep a train moving while still having energy for a strong leadout.

A five-race training series will allow your team to practice the following situations:

PRIME SWEEP

A training criterium with multiple prime laps offers the opportunity to practice leadouts. The team should use the amoeba to conserve energy during regular laps. When the bell rings for a prime lap, the team should form a train and hold a high pace at the front of the peloton until the leadout begins. If the race has a no-drop rule, encourage the team to use as much energy as necessary for each leadout, then drop out of the race for a lap. Winning every prime as well as taking the race win is a sweep.
--

Focus:	Amoeba to train, leadouts, team sprinting.
--------	--

ATTACK AT WILL

A training criterium with less than 40 riders gives a team of 6 or more rider an excellent opportunity to practice attacking, forming a paceline, and working with other riders to create a breakaway. Riders relay attack the small peloton trying to form a paceline with any other riders that chase the attack. As soon as the first attack is caught, a second teammate attacks. This continues until the peloton tires of chasing and a break is formed. Without the fear of being dropped, each rider can attack as hard as possible. If a break forms, the team can practice blocking. If a break forms without a teammate present, the team can practice chasing.
--

Focus:	Attacking, paceline skills, blocking, chasing, small groups.
--------	--

TRAIN	
The team can go to the front from the beginning of the race and attempt to maintain a train for the entire race. Success will depend on the team coordination, team fitness, and effective gatekeeping and speed control.	
Focus:	Teamwork, paceline skills, train, gatekeeper, fitness.

AMOEBAS TO TRAIN—THIS SKILL ALONE WILL HELP WIN RACES	
The group changes from the amoeba position to a train each lap, every two laps, or at each corner. The transition from amoeba to train takes a high level of teamwork and coordination and is the foundation of advanced team tactics such as positioning and leadouts. If your team is good at one skill, make it this one!	
Focus:	Builds the essential skill of starting and stopping a paceline with gatekeeper train on command. Eventually is necessary for leadouts.

BLOCK AND CHASE	
A training race is inevitably going to have multiple riders looking to attack the field and try to create a breakaway. Whenever an attack by an opposing team occurs, your entire team jumps to the front and works together to try and disrupt any chase attempts, helping that attack succeed without working in the break. Even if one rider attacks the field she should suddenly have the support of your entire team. Once the breakaway has formed and developed a 30 second lead, the team should stop blocking and put together a chase effort. When the breakaway is caught, the team again waits for another break to form.	
Focus:	Team blocking, team chasing

Before the training race, remind your team that winning a training race means executing the chosen game-plan as well as possible. Since a training race doesn't offer prizes to the first across the line, the team is better served by trying new ideas, testing new skills, and working on the perfect execution of essential techniques such as leadouts, the amoeba, pacelines, and trains. Working on these techniques within a competitive peloton is the fastest method for developing these skills within your team, building trust between each rider, and winning more races later in the year.

Whether your team only attempts the simple amoeba, or the more complex leadout train, team trust and individual tactical savvy is built by practicing basic skills. Every opportunity in the early season to practice, evaluate, and perfect team skills will create more advantage in real races later in the year. Once each rider understands the benefits of each technique he or she will become excited about using their new advantage on the unsuspecting amateur peloton.

Chapter 4 Communication

Professional racers rely on small radios with earpiece microphones to talk with their teammates and with a race director following them in the team car. It is not impossible for an amateur team to invest in a set of small radios to help each other communicate during a race, but it is absolutely not required for team success.

Racing without radios relies on the following three things to work smoothly:

- Everyone should understand how and why each team tactic creates an advantage.
- Everyone should understand the basic strategy outlined by the team leader a few days before the race.
- The entire team needs to be able to execute the basic maneuvers of pacelines, gatekeepers, chases, leadouts, blocking and attacks.

If people do not understand that a leadout must start at a wide section of road where the entire team can get to the front, the team is likely to be boxed in coming into the finish line. If the team does not know how to execute a paceline with a gatekeeper for a few miles, it is likely that the protected rider will not start a major hill in the very front of the pack. If only half of the team knows the plan for the day, the paceline at the front will be meager and ineffective. Most likely the entire team will scatter throughout the peloton and not be available to create any tactical advantage. Practice and a general understanding of basic tactics leads will reverse all of these issues.

One rider can ruin the advantage gained by an entire race of tactical excellence. The level of coordination for team tactics to be effective is fairly high, but is easily learned in a couple of team practice sessions. Most amateur teams just do not spend enough time perfecting these skills because of the extra organizational effort required to create a team practice. However, if your team can get together with a tactical focus, the rest of the peloton doesn't stand a chance.

The communication necessary to hold the basic position has already been explained. The protected rider simply needs to keep a running verbal commentary with the domestiques in front of him or her to maintain an advantageous position within the peloton. Yelling, "right," or "left," is the most common form of communication as the peloton is always shifting. These commands are vital to the domestiques. It takes enough skill to maintain a position near the front of the pack without having to constantly look backwards to see if the protected rider is in position.

Amateur Tip:

If you are the protected rider for the day, **over-communicate** with your domestiques so they can maintain full concentration on keeping you in position. Say something every minute or so just so they know you are there. If you focus on communication, your domestiques can focus on riding well.

Not all corners or hills require your team to go to the front and paceline. A corner into a wide section of road will be just as easily maneuvered by staying in the front twenty or so riders in the peloton. It is absolutely imperative to be the first team through a set of two or more corners in a technical criterium, or through a corner leading to a hill in a road race.

The team leader should make the tactical decision for each terrain change. The protected rider is in control of the race and has the best vision of what is ahead while the domestiques have their head down punching through the wind. If the team leader wants to move forward a simple command of, “corner,” signals the team to move to the front. This command also is not distracting to the peloton because it is common for the front to talk about corners coming up, especially in a road race. With this in mind, it is absolutely necessary to use the lead domestique’s name when shouting a command. It is much better to say, “Ryan, corner,” when Ryan is the person in front of you. Your voice combined with his name will ward off other riders interfering with the team communication channels.

Another situation that requires a distinct command is the leadout. Starting the leadout may be planned in advance because of the size of the road or the terrain. Again, a wide road allows the entire team to move to the front while a narrow road or corners will leave the team split throughout the pack. With the leadout maintaining his position at the front, the team leader is in charge of controlling the team speed with the, “up, maintain, and slow it down,” commands. As the finish nears, the team needs to change from the rotation to the final sprint. Yelling, “switch,” may be enough to stop the paceline and line your team up for the sprint. A short time later, a command such as, “now!” allows the team to accelerate together and open a small gap on the rest of the peloton.

Having teammates in a small group situation is fun. The chances of one teammate winning the race has just increased dramatically and hopefully the two of you can dominate the rest of the riders in the pack. But how can you coordinate your strategies without tipping off the rest of the small group? By having one word names for each strategy such as, “Sprint,” “Group,” and, “Relay” is an easy method to coordinate the attacking scheme. Since both of you have been riding together for a few months in preparation for the race, it is usually obvious who is going to be the sprinter or the

protected rider for the day. If the teammate is new, the experienced rider will need to address when the attacks will occur, who is going to attack first, and who is in charge of a pack sprint.

Becoming separated from your teammates occurs during every race. The rider in front will not be able to find the rider in back without talking or yelling their position. If a race is winding down, locating the leadout team for the sprint is vital to the team success and a few extra lines of communication indicating the position of a straggling teammate can mean the difference between leading out your protected rider and leading out the rest of the field. A simple method utilizing a clock face was developed by a couple of riders from the Boston Road Club while I was racing for them. If your teammate is three riders in front of you and you want to let them know your position, simply yell, "Mike, Six Three!" This tells Mike that you are directly behind him (imagine, being in the center of a clock face parallel to the road, the twelve would be directly in front of you and the six would be behind you), and also indicates how far (three riders) behind him you are positioned. The clock position always refers to the clock associated with non-yelling rider. If you are two riders on the right of your teammate, you would yell, "Mike, three, two," so Mike would think of his clock face, look towards the three and expect to find you a couple of riders over.

The following chart can be used to create a set of communication terms for your own team. Adjust the signals to whatever you like and change the code as necessary as other teams learn your methods. More likely, other teams will be so confused by all the talking in the peloton that they never will suspect that they are about to be attacked and destroyed by riders working together.

Table 4.1 Team Communication List

[TEAM NAME]	[YEAR OR DATE]
Command	Meaning
Basic Position	
[name], left	<i>Domestique</i> moves left to account for change in protected rider's position.
[name], right	<i>Domestique</i> moves right to account for change in protected rider's position.
[name], back, number	<i>Domestique</i> drifts back a number of rows to account for change in protected rider's position.
[name], corner	<i>Domestiques</i> accelerate and move the protected rider into position for a technical corner. May not be necessary if all corners are technical.
Paceline Commands	
Up	Paceline should increase speed to stay in front of the peloton.
Maintain	Paceline speed is enough to keep peloton at bay. Should be repeated regularly even if the conditions do not change.
Slow it down	Paceline can slow down slightly because it is dropping the gatekeeper or the peloton is not attacking.
Switch	Switch from a rotation to a suicide paceline for a hill or leadout where each rider pulls at the front for as long as possible and then quits the race.
Now!	This command comes after the switch. With a full line of riders in front, the sprinter starts the leadout with this command. On this command, the entire team should stand up and accelerate towards the finish line hopefully opening a gap between the gatekeeper and the field.
Small Group Coordination	
Protect, [name]	The named rider is the designated sprinter and the rest of the teammates should focus on getting away from the break or leading out the sprinter.
Relay	Each team member should attack as soon as another team member is caught. This guarantees that someone from the team is on the offensive while the others rest.
Group	The entire group is going to attack in an attempt to paceline away from the rest of the pack.
Clock-face method	
[Name], clock number, distance to teammate	i.e. Matt, six, four means that the yelling teammate is four riders directly behind Matt.

There are only a few times when racers need to communicate during the race. Having a set list of commands creates a team that can work through any circumstance and can change the overall team tactics as the race progresses. If the team has a solid understanding of when to use team tactics, and is guided by a general knowledge of where on each race course team tactics will be effective, the team is ready to pull together, and pull ahead in each race.

Professional vs. Amateur—Before Race Radios

Race radios are only a recent invention. Before radios became widely available to professional teams, each rider had to know before the race their responsibilities and also how to react to changes in the race strategy. A racer with enough experience to be a professional already knew that if his team missed the break they had to go to the front and chase. Everyone had to be aware of who made the break so no one ever chased down their own teammates. A pre-race meeting with the team director outlined the team strategy for the day. If things changed, one rider might go back to the team car to hear new instructions, but if the race was fast each rider must adjust his responsibilities accordingly. The best teams could react in unison without specific commands from their coaches.

Amateur racers who are educated in team tactics automatically understand the needs of the team in every race situation. By studying the course before the race, every teammate knows exactly when to move the team leader into position for a hill, a corner, or the final sprint. Once the race starts, the group must act as a loosely connected team thinking as much about each other as for themselves. Decisions should be made with the overall team goals in mind, yet individual actions such as attacks, bridge attempts, and blocking must originate from individual actions.

If each race is replayed and discussed over email later in the week, the team begins to understand how each rider responds to race situations. This knowledge leads to more wins and more fun.

Chapter 5 Winning the race

A small team without much training can use the amoeba to protect one rider, create a paceline train to move that rider to the front then let the protected rider use his stored energy to win the final sprint. Experienced teams will try to save energy by using a train to lead the peloton through strategic corners and to hold a protected rider at the front of the peloton before large hills. Additional advantages are available to larger and even more organized teams. Attacking and blocking can be used to make the rest of the peloton chase your team while marking the attacks of other teams assures you that every break will contain a rider from your team.

If two, three, or four riders are strong enough to protect the team leader the entire race, the rest of the team is free to attack and try to create a breakaway from the peloton. The riders not actively protecting the team leader, known as the attack team, can race aggressively raising the pace of the peloton to an extremely high level. The attack team riders must lead attacks, mark attacks, and stay at the front of the pack to punish the peloton whenever it slows down to rest.

Eventually the aggressive riding of the attack team will create a breakaway gaining two advantages for the team. The rider in the break is assured a top finish and is now racing against a smaller group creating better odds of winning the race. In addition, the team is not responsible for chasing the break, saving energy for further attacks if the break is caught.

If the attacking group is successful at creating a break, the protected rider and the domestiques can begin to work together and interfere with the chase by blocking. Blocking just means using poor paceline skills to make sure the peloton does not form a train at the front of the group creating a consistent chase effort.

Rider Responsibilities

A team can be broken into different responsibilities which may or may not change throughout a race. Again, smaller teams should stick with simple defensive tactics for every race, while larger teams can be more aggressive. As the team grows in size and fitness, more riders may be used in offensive tactics without endangering the protection of the team leader. Table 5.1 lists rider responsibilities for a typical team of ten riders while Table 5.2 lists the advantages of each position.

Table 5.1 Rider Responsibilities

NAME	NUMBER OF RIDERS	RESPONSIBILITY
Domestiques	4	Always ride in front of the team leader. Work with Attack Team to chase down breakaways. Uses blocking positioning during breakaways. Is responsible for leadout at end of race.
Team Leader	1-2	Rides behind domestiques throughout the race. Acts as gatekeeper when necessary. Is responsible for climbing a long hill in the front group, winning the final sprint, or attacking at the end of the race. Leads all communication throughout the group.
Attack Team	4-5	Marks all early attacks from other teams. Uses relay or group attacking to create breakaways. Works with domestiques to chase down other teams' breakaways and helps with other positioning tactics. If still in the race, works with the domestiques in the leadout.

Table 5.2 Team Rider Benefits

NAME	BENEFITS
Domestiques	Instead of sitting in the pack and waiting for the sprint domestiques consistently ride hard throughout the race to protect the team leader. This additional training leads to greater fitness later in the season. Being involved in race tactics builds knowledge of peloton dynamics.
Team Leader	Obviously, due to their protected position in the peloton, the team leader greatly increases his or her personal chances of winning the race. The fitness gains of this position are not as great as the attack team or the domestiques.
Attack Team	Gains significant training in acceleration and attacking by focusing on creating a breakaway throughout the race. If a breakaway is formed, the member(s) involved have an excellent chance of winning.

Domestiques and attack team members work much harder throughout a race than the protected team leaders. By riding in the wind or attacking, these riders quickly gain the fitness necessary to win races. This training is an important part of a successful training program. The experience of being involved in the entire race and working as a team to accomplish one goal increases each rider's knowledge of peloton dynamics and tactical advantages increasing the gap between the organized team and racers competing alone.

If each team member focuses on their own set of responsibilities the work required throughout a race is spread throughout the entire team. A rider without a team must perform all of these duties throughout the race and after attacking the group,

marking other attacks, and gaining a strong position at the front of the peloton will have a less effective sprint than the team leader. A team that does not understand the different requirements for winning a race, or does not divide these needs throughout the team can not gain the full advantage of team organization.

Defensive Tactics

The basic defensive tactic is to use the amoeba formation to protect a chosen rider for the entire race. The energy that the team leader saves can be used later in the race in the final sprint or a late attack up a hill. Defensive tactics should be present at all times in every race. Teams of two or three riders should use this tactic exclusively because it gives them the greatest advantage in every race.

Marking attacks is an additional defensive technique that saves the rest of the team from chasing an eventual breakaway. Marking an attack simply means to follow any attacks made by other riders making sure they do not create a break without involving your team. The marking riders simply follow any attacks and do not necessarily work in the attack but latch onto the wheel of the attacker and stay in the draft as long as possible. Everyone except the team leader are responsible for marking attacks. Sometimes a domestique will have to leave the front of the team leader to mark an attack.

The team should commit as many riders as necessary as domestiques so the team leader is protected at all times. Any other teammates not protecting the leader should be marking attacks, or filling the duties of the attack team. The ultimate goal of every team is to protect the team leader first, then begin attacking to create a breakaway.

Positioning Tactics

Corners, hills, and sprints create situations in each race where it is advantageous to be in the very front of the peloton. In order to maintain this position, the amoeba formation should reorganize into a paceline train with the team leader as the gatekeeper, accelerate enough to keep the peloton from attacking the train, and hold that effort throughout the tactical situation.

By switching off the front domestique in an organized manner and using the team leader as a gatekeeper, the group can maintain a commanding position at the front of the peloton while still guarding the energy level of the team leader. The team may stay at the front as long as necessary to maintain the tactical advantage of the front position, but should manage the energy reserves of the team throughout the race. Once the tactical advantage is no longer present, the team should drop back into the less aggressive amoeba formation and let other teams work at the front of the pack.

The most important tactical position comes during the final sprint. The winner of any race is usually the person in second place with 100-200 meters to the finish. A leadout train is designed to place the team leader in this position at every race. By going to the front of the peloton and maintaining a high pace, the leadout train forces the entire peloton to spread out behind the train. If anyone wants to pass the train, he must accelerate around three to ten riders and stay there the rest of the race. Comparatively, the team leader simply needs to follow the train until he is ready to sprint. If perfectly timed, multiple teammates will be leading the race with 100-200 meters to go giving your team the best chance of winning.

Offensive Tactics

With a relaxed, flexible amoeba formation protecting the team's lead rider, any other teammates are available as offensive weapons to wear out the peloton. A team with only the amoeba tactic is forced to simply hold their ground no matter how fast or how slow the peloton decides to travel. Using teammates above and beyond the protective amoeba forces the peloton to react to your team's choice of speed, while keeping one rider ready for a final sprint.

It is imperative that before any attacks are staged, the amoeba is created and at least one rider is protected for a final sprint. If the team is scattered throughout the peloton, any team attacks are just as likely to tire out the rest of the team instead of fatiguing the opposition. With a protected rider, the attacks may wear out the domestiques, but the team leader will come to the finish line with full power.

One basic law of a cycling peloton is that the group's intensity is dictated by the first 20 or so riders. With the limited width of any given road, the riders in the front can go as fast or as slow as they commonly choose while other riders behind must adjust to the chosen speed. If at any point in the race the front 20 or so riders are tired, they can slow the entire group down to a snail's pace. If one or two riders feel like working harder, their attack forces the rest of the top 20 to accelerate and chase.

During most lower-level races the peloton will travel at a fairly consistent pace that is fast enough that everyone is working hard, but not fast enough to really wear anyone out before the end of the race. The consistent, medium pace allows weaker riders to save energy during the race for the final sprint. Using offensive tactics prevents the peloton from resting and regaining strength forcing any weaker riders to drop from the front pack. While a race without teams will finish with 75-100 riders, a good team can keep the race intensity high enough so only 20-40 riders enter the final sprint. A strong attack team makes the team leader's job in the final sprint much easier.

A race with multiple organized teams never rests. If the peloton rests, an attack will occur, a break will be formed, and teams missing the break will be forced to chase. The break that forms is known as a controlling break. Each team represented in the break

is in an advantageous position and can rest as long as the break is away. Teams that miss the break have lost control of the race and must work in the wind to chase down the break.

The attacking team has two choices: either to attack as a group, or relay attack where one person attacks each time a teammate is caught. Choosing which method to use is simple. A single rider attacks first. If riders from other teams are feeling aggressive and attempt to latch onto the attack, relay attacking should be used. If the lone attacker is allowed to leave the peloton alone, then that rider should return to the group (no use attacking solo with the entire race to go) and the whole attacking team should go as soon as the teammate is caught. Simply, if other teams are willing to attempt a breakaway, allow them to help. If no one wants to help, your whole team should attack as soon as possible.

Attacking with two, three, or even four riders is likely to evoke a chase from the peloton. This is great, the pace of the group will be very high and everyone is wasting energy chasing the attacking riders while the protected rider maintains a position directly behind a couple of teammates saving energy for the end. If you are extremely lucky, the large group attack by your team will create a gap allowing the attack group to immediately begin a paceline (as practiced in the preseason) and team time trial away from the peloton hopefully never to be seen again.

If a gap is formed and the group of five or more riders are riding in an organized paceline, the peloton must also become organized to match the efficiency of the small group. Considering that the amateur peloton consists of riders with different skill levels, fitness levels, and paceline abilities making an organized chase nearly impossible. Without an organized chase, a five rider rotation can outrace a 100 rider peloton on almost any course.

To open the all-important gap, the best times to attack the peloton are:

- When the group is tired.
- When the peloton does not expect an attack.
- After a prime in a criterium.
- On an uphill.
- During a corner.

Each situation can allow the gap to open up and let the attacking group form a paceline to get away. Two unique events occur in the lower levels to indicate the peloton is tired. After a flurry of attacks, if front of the peloton returns to a flat formation the peloton has decided to rest. If the front of the peloton goes flat, your team's immediate attack will punish everyone for being so lazy.

The second situation is really funny to watch. Again, after a flurry of attacks, the peloton will fatigue and one final solo attack goes off the front. This rider gets about 200 feet in front of the peloton before finding out that he is alone. The rider continues to hang off the front of the peloton while everyone else just sits there and watches him

work. Rarely does the peloton actually chase this rider because they are tired and they know that a single rider barely off the front is just wasting their energy. Eventually that rider returns to the peloton because he fatigues or gives up the attack. A good rule of thumb for attack team members is, “a single rider in front of a large peloton always needs company.” Your team should quickly attack and [bridge](#) to the solo rider, form a paceline, and ride away. This aggressive reaction punishes the front of the peloton for resting.

If the attacking team wants to have the best chance of creating a successful break, it should wait until the peloton is tired. The attack team doesn't need to begin working at the very start of the race, attacks from other teams and from solo riders eventually will wear out the peloton. During the race, when the peloton shows either of the above situations, the attacking team should immediately push ahead and force the pace. Once the peloton has slowed to rest, an attack will immediately create a gap and that gap will need to be closed by a physically and mentally tired group of riders.

At certain times in every race the peloton collectively will believe that an attack is not going to occur. During the first few miles of a long road race, attacks may not be marked because the peloton believes any attack will tire before the end of the race. An excellent example of this situation is the Tour de France. Most of the breaks seen on television occur in the first kilometer of a 150 km race. The break is allowed to go because the rest of the peloton knows they can catch the break before the end of the long day.

Criteriums have special laps (prime laps) where the first rider across the line wins a prize. Usually, this prize is some money or free merchandise. Not everyone in the race will sprint for the prime so the peloton naturally splits into two smaller groups. The group that sprints gains about 100-200 meters on the second group, but after crossing the finish line slows from their sprinting efforts. The second group simply keeps riding at a steady pace and eventually the two reconnect a little while later. Using the fact that the groups are smaller should signal that small group tactics are available for your team. If you have some teammates join the sprinting group, but do not actually contest the sprint, they will have some leftover energy and some residual speed to use for an attack. Because the small group was going for the prize, there are very few riders available to chase the attack and a gap will open up rather quickly. This is one of the most effective counter attacks available in any race.

Finally, any attack should come on an uphill section of the course if possible. Although it has worked before, a downhill attack is not commonly successful in creating a break because of aerodynamics. An acceleration on an uphill requires a lot of energy, but the chase acceleration requires the same amount of effort due to the decreased aerodynamic effect at slower speeds. It is also more likely for a rider to create a gap if the chasing riders take a long time to react to the attack. By the time the chase riders decide that they are being attacked, think about how much it is going to hurt to follow the attack up the hill, then stand up and accelerate, the gap has already been created.

Attacks that do not end up in a breakaway also help the team. Whenever an attack occurs, a domestique from the opposing teams will need to mark the attack taking one person away from the opposition's amoeba. To maintain their position near the front of the pack, the rest of the domestiques from the opposing team must also match the acceleration of the attack. Although these adjustments are small, each acceleration fatigues the rest of the peloton.

The benefits of attacking the opposing teams is obvious: a fatigued team of domestiques can no longer mark attacks, maintain the position of their team leader, or effectively form a paceline train to chase a breakaway. By the end of the race, a team that has been attacked effectively will not have the energy to create a leadout train and will be forced to sprint from the middle of the pack instead of from the front. Your team, however, has either created a breakaway or saved enough energy to form a leadout and win the final sprint.

The concept of tiring out the other team's domestiques is the central theme of aggressive team cycling. If your team is strong enough, large enough, or coordinated enough to relentlessly attack the peloton you will force other domestiques to chase you, tire, and drop out of the race. Without a full team of domestiques, the other team leaders must sprint or climb for themselves against your entire team. If there are no other coordinated teams in the race your attacks quietly exhaust the competition making the final sprint a guaranteed win.

Professional vs. Amateur—Attacks and Breaks

Two differences allow breaks to occur in almost every stage of the Tour de France but almost never in the local amateur peloton: length of race and teamwork. Professional races are often six or eight hours in length. Even the best professionals can race at full speed for only two or three hours so races start with teams vying for a spot in the breakaway.

A professional race typically has eighteen teams with nine riders on each team. Let's look at the division of the peloton.

- 18 team leaders, one per team
- 90 domestiques, five per team
- 54 attack team members, three per team

The 108 domestiques and team leaders are only interested in the final two hours of the race. The 54 attack team members hold the responsibility of creating a break so the domestiques do not have to help chase. During the first hour of the race the attack team members fight to create a break with attacks occurring and teams immediately responding with chase efforts if they miss the break. Finally, enough of the attack team members are tired and a clean break starts to gain time on the peloton. The teams that missed the break form a train and take the lead of the main peloton.

Eventually, the combination of a tired break and an efficient chase effort brings the entire field back together for the last section of the course. At this point, the team leaders are the only ones left with energy and they begin attacking hoping to form another small break or use their leadout trains to gain positioning to win the final sprint.

Amateur pelotons do not face six hour races and do not have a division of labor as organized as professional teams. A break can not be guaranteed to form in shorter races because all of the riders can go hard for the entire time. In longer races, breaks may form as the peloton tires or starts saving energy for the sprint. Teams that effectively divide their labor into an attack team forces individual riders to use energy throughout the race giving additional advantages to the protect team leader, while possibly creating a breakaway guaranteeing a strong finish.

Blocking the Chase

After a break has formed and the peloton is beginning to organize an effective chase, the protected team leader and the domestiques have new assignments. A domestique should mark each attack to make sure no one bridges up to the break. This domestique can foil a bridging attempt by not working with the small group or, conversely, work with the bridging effort to add another teammate to the break. By positioning themselves near the front of the peloton, the domestiques may also slow the momentum of the main peloton helping the break open a larger gap. The individual efforts of each domestique combine to reduce the effectiveness of opposing chase efforts and increase the chances of a successful breakaway.

Moving to the front and not attacking is especially effective going into a corner or a hill. The lead person through a corner in a criterium has complete control of the peloton's speed so a blocker coasting through the corner can add up to five seconds to the break's lead. In a criterium with four or five corners a lap, this can easily balloon the gap by ten or twenty seconds during each loop. The same is true going into a hill. If the pack is being aggressive, their momentum easily will carry them over a small hill keeping the peloton's speed high. If a couple of blockers are at the front going into the hill and do not aggressively attack the hill, the entire peloton will be in their small gears trying to crest the top. In each case, just having a rider in the front will disrupt any chase efforts and weaken the chance of the breakaway being caught.

An excellent way to block a chase effort in a road race is to keep one rider in second, fourth, and sixth place at all times. With this formation, the only effective attack that will not also carry one of your teammates is an attack from deep within the peloton. Also, when the rider up front tires and moves over, the teammate in second place can immediately slow down the peloton's pace allowing the break to gain time. If an attack does come from a group within the peloton, at least one teammate must try to join the attack. By joining the attack it guarantees that if the attack does bridge up to the break, at least it carried a teammate up there too.

The more you think about it, blocking is simply using the same skills that cause a paceline to lose speed. Jumping and creating gaps, not maintaining the group's momentum, slowing down through corners, all of these are the same principles used throughout team tactics. If you are trying to go faster, ride smoothly and maintain your momentum. If you are trying to slow other riders down do the opposite. The effect of blocking is not always visible. The few seconds gained by slowing down the peloton in one corner are made up by the chase group in the next. Slowly each blocking effort wears out or disorganizes the chase group enough to allow the breakaway to maintain their lead and win the race.

With a larger team of ten or more people combining the effects of protecting one rider while attacking with all the rest is an excellent way to wear out a peloton and create a break or increase your chances in a final sprint. Adding smaller responsibilities such as marking other attacks and blocking add to the advantages a team may gain through team

tactics. In every case the advantage is small. An effective team that utilizes every available opportunity increases their chances of creating a break and guarantees that their team leader reaches the end of the race with fresh legs and a great position at the front of the peloton.

Race Progression

The goal of each team is to increase the chances of winning by either having the most energy at the end of the race or by creating a breakaway and winning a small group situation. Every race goes through a predictable progression that leads to one of the two successful endings. Every racer must understand how a race should progress and use this knowledge to know when to apply different tactics to help the team. If individual riders recognize situations such as an attack, a breakaway, or a chase situation and act accordingly, team communication becomes less complicated.

Every race starts with the peloton together. To make things easier, the members of your amoeba (the domestiques and the team leader) should do everything possible to start the race near the front. This will save energy throughout the beginning of the race and is absolutely essential to success in a technical criterium. The attacking team should also attempt to start at the front of the peloton in order to be ready to mark early attacks and possibly lead an early break.

Throughout the race the amoeba simply stays near the front of the peloton as much as possible and works to save the energy of the team leader. The amoeba may form a small train if necessary for strategic reasons and eventually will form a train for the leadout, but otherwise hovers near the front of the race.

The attacking team marks all early attacks in an attempt to be a part of any breakaway that forms. If the team fails to mark an attack and a breakaway forms without any riders from your team, **the attack team, the domestiques, and the team leader must gather, go to the front, form a train, and chase until the breakaway is caught.** If your team misses a break it is worthless to continue saving energy because there is no chance to win the race. The most common communication breakdown occurs when a break forms and your team does not recognize that they are not represented. A chase will not be effective with the participation of only a couple of teammates. A chase requires three or more cooperating teammates recognizing the situation and moving to the front of the peloton at once.

If your team has one or two strong time trial riders an alternative known as the bridge may be used to control a break. If your team does not have a person in the breakaway group, a member of your team may attack the peloton in an attempt to bridge to the break. If successful, your team gains the advantage of having a person in the break and does not need to expend the energy of the entire team in an organized chase. If the bridge fails, the team should start chasing as a group.

If one of your team members starts a break or marks a breakaway, the team is in the best position possible. The domestiques can continue to maintain the amoeba and protect the team leader while the rest of the team works to block the chase of the break. The rider in the breakaway is guaranteed a strong finish and has an excellent chance of winning the race because a small group sprint is much easier to win than a large pack finish.

If your team has someone in the break, bridging to the break can gain additional advantages. Again, a strong time trialist may attack the peloton and bridge up to the break after a long effort. The goal of this attempt is to add another person to the break, not pull the entire peloton across the gap. If the attack does not form an immediate gap on the peloton, the person should slow down stop their effort. It would ruin the break to attempt a bridge with the entire peloton in your draft.

At no time should a team actually chase down a breakaway that contains a teammate. A breakaway guarantees that one of your teammates finishes near the front of the race. If you chase down that breakaway your team is back to a large sprint finish which doesn't guarantee a top position.

Every race goes through the following progression:

- The peloton starts as a group.
- Someone attacks the peloton.
- Teammates attempt to mark the attack.
- If a break is formed, do you have any teammates in the break?
 - Yes—good, maintain the amoeba and block any chase groups.
 - No—do you have any riders strong enough to bridge?
 - Yes—bridge to the break.
 - No—the entire team goes to the front to chase.
- Break stays away for the rest of the race.
Or
 - Peloton catches break and sprints for the finish.

Again, the critical part of each race for team coordination is the break. If the team misses the break it is absolutely critical that the entire team works together for the chase. Breaks succeed because of a disorganized chase more often than any other reason. If your team is represented in the break, blocking can increase the chances of the break succeeding. At the same time, the amoeba should be maintained in case the break is caught.

The goal of the attacking team is to create a break and win the resulting small group. The domestiques and the team leader are simply saving energy for a possible sprint finish or for marking a late attack. By dividing the team's responsibilities and understanding when to chase, when to attack, and when to block, the team can guarantee a strong finish in every race.

Understanding and Beating Individual Racers

Racers not involved in an organized team have few options for gaining tactical advantages. One rider has a limited amount of energy available for attacking, marking other attacks, positioning, chasing, and sprinting and therefore can not perform each task perfectly. Exploiting this weakness allows teams to form breaks and guarantee top finishes.

Most individual riders enter a race expecting a sprint finish. The individual riders spend the entire race focused on positioning themselves near the front of the peloton and trying to stay out of the wind as much as possible. The single rider is forced to use his own energy to maintain his position during attacks, hills, and corners. Near the end of the race an individual rider attempts to use the rest of the peloton as a leadout before the sprint. By moving into tenth position with a mile or two to go and allowing the front riders to tire, the individual rider occasionally reaches the magic spot of second place with 150 meters to go leading to a strong sprint and possible win.

Some riders will become bored during a race and attempt to balance defensive tasks by attacking a couple of times. Unless the attack forms a gap and also includes other riders who understand paceline techniques, the attack only serves to expend precious energy that could have been saved for the sprint. Some attacks go unmarked leaving the attacker alone about 200 meters in front of a peloton that has lost interest in chasing. The lone rider usually doesn't have the fitness to time trial the rest of the race at 25+ mph so the break only fatigues an aggressive rider. One person will not be able to attack a peloton enough to tire it out and create a break further reducing the effectiveness of riding an aggressive race.

The individual racer eventually learns that the only strategy available is to simply train more than everyone else and develop the ability to time trial away from the peloton or overcome poor positioning by learning to out-sprint five or six riders at a time. After a couple of race seasons a rider will gain enough fitness to time trial or sprint alone and eventually he wins a few races and upgrades to the next level.

Races full of individual riders are extremely boring. Since everyone is saving energy for the sprint and no one wants to go to the front and push the pace, the race becomes one long parade around the course. The people that choose to attack fail miserably because no one understands that working together is an essential part to road racing. Other attacks fail because no one is willing to risk "race suicide" by attacking so hard that they might be dropped later. A sprint is guaranteed in every race because that seems to be what everyone wants.

Compare this strategy with the advantages available to an organized team. Half of the team is thinking about the sprint while the other half is free to attack and form a break. The team leader is focused solely on the sprint and is guaranteed to have the most energy left at the end of the race. The domestiques work to protect the team leader and eventually position the team leader perfectly by using a leadout. The attack team is

allowed free reign on the front of the peloton making sure the competition tires before the finish. Working together guarantees that if the attack team goes as a group, the small paceline will be able to create a gap, form a break, and maintain a serious threat to the rest of the peloton. If a break forms, half of the peloton senses that there may not be a sprint finish and abandon their energy-saving plans and sprint alone after the break (further reducing their energy). The other half do not worry about the break because they are focused on positioning and saving energy at all costs (reducing the chances of a chase succeeding). In either case, the peloton chases with an inefficient single-rider mentality that is easily beaten by an organized paceline in the break.

Savvy individual riders will work with your team to help guarantee a strong finish. Once your team is known for attacking and creating breaks, stronger riders will mark your attacks. These riders have raced enough to understand pacelines and know that working together helps a break gain ground and benefits everyone involved. Eventually your team will not need to attack as a group, but will be able to create a break utilizing strong riders from other teams similar to professional races. A group of six teammates attacking at once creates a different reaction in the peloton than a group of riders from different teams. Six riders from the same team is a serious threat and even amateur pelotons will work hard enough to end the break, while a mix of teams might go unnoticed allowing a gap to form.

If a couple of organized teams form in your area the attacking squads from each team can work together to form a break in every race. Once races start ending with breakaway wins, more riders will join teams and practice their team skills because simply waiting for the sprint will no longer be an option. When teams race against teams attacks continue all race, breaks consistently form, chases will be organized, and if there is sprint, leadout trains guarantee that the last few miles are fast and furious.

Specific Strategies for Typical Race Courses

Although each race course is different, they all can be described by one of the following examples:

- The flat road race.
- A road race with rolling hills.
- A road race with a medium hill every lap combined with flats.
- The road race with one big hill.
- The technical criterium.
- The wide/fast criterium.

The Flat Road Race

Protecting your best sprinter is an essential tactic for a flat road race. Without a major hill the race is most likely going to come down to a pack sprint and you want your lead rider to be there with plenty of energy. Using your whole team to protect one or two

riders the whole race is not abnormal. If the course includes corners or narrow roads make sure the team moves to the front to keep the protected rider in the first few rows.

If a flat road race is open to wind the tactics change with every turn. If there is a headwind, the group will need to work hard at keeping the protected rider up front, but the team does not necessarily need to be forcing the pace. The same is true with a tailwind. Riders will tend to attack with a tailwind because it feels like they are going fast. Let these riders attack while your team's domestiques work to keep the team in the top ten or twenty.

If there is a crosswind everything changes. Your team should go to the front and paceline with all its worth. The protected rider(s) should play gatekeeper to keep other teams away from the front. The train should position itself at the extreme windward side of the road; if the wind is coming from the left, move to the left. This is essentially an offensive tactic that forces the maximum number of riders from other teams to ride into the wind and waste energy trying to keep the pace. If the crosswind section of the course is long enough, the peloton will split into smaller groups known as [echelons](#). If your team is leading the front group, you have avoided the split.

The final item required for a flat road race is an excellent leadout. With a flat course, there are few reasons for people to be dropped from the peloton and the entire 125 person field is often coming into the sprint at once. The leadout team should be able to go to the front early and keep the pace for a couple of miles to guarantee a perfect position for the protected rider(s).

Rolling Hills

Rolling hills wear out the peloton and provide the inclines necessary to create breakaways. No longer should the team focus exclusively on protecting a couple of sprinters, but now a section of the group should be part of the attacking team. The peloton will fatigue over each rolling hill and longer hills may break the group apart. The attacking team can be aggressive on each hill forcing the peloton to chase them over the top leading to a break.

If a race has a major hill every lap, followed by a long downhill and some flat sections, every team tactic skill will need to be utilized. As always, the protected rider must have a couple domestiques to keep him or her out of the wind and ready for the final sprint. The domestiques must be strong riders to pull the protected rider up the hill near the front of the peloton each lap, while battling for position on the downhill and flatter sections of the course. The attacking team must force the pace up the hill every lap. By doing so, the entire team will be at the front of the pack for the long downhill and will never be stuck at the back of the peloton. Also, each hard trip up the hill will drop a few opposing riders off of the back, whittling down the size of the peloton. Sprinting against 60 riders is always better than sprinting against 125, especially if 10 of the 60 are your teammates. Finally, the entire team should be available for the leadout at the end.

Alternative Strategy—The Sterling Classic Road Race

One of the oldest races in the United States is the Sterling Classic Road Race in Sterling, MA. The course starts with a half-mile fairly steep hill followed by a two mile gradual incline before a long winding descent. After the drop, the course turns onto a wide highway for a three mile return to the town center. The race ends with a hard right-hand corner leading to the finish line at the top of the steep hill. The forty five mile distance in the category four race is not enough to guarantee a small group at the finish.

Forcing your team to attack and pull the entire peloton up the hill each lap uses a lot of energy, but is the only way to create a substantial break. Unfortunately, the wide section of road up the valley floor allows the peloton to see a break up the road and effectively chase each separation. Instead of forcing a break, an alternative tactic is to have your team start the hill in the front, but use as little energy as necessary to climb the slope. The whole team should be at the rear of the peloton at the top and throughout the descent. The long, wide road at the bottom allows the team to quickly regain a position at the front of the peloton before the next trip up the hill. By using a small train to move the team to the very front of the pack, everyone can fly through the hard corner at the bottom of the hill and save even more energy during the climb.

This tactic, a mixture of defense and positioning, allows everyone on the team to save as much energy as possible throughout the race. The large hill by itself serves to tire out and thin the rest of the peloton. The wide road to the finish allows every teammate to move into position and create a huge paceline train on the last lap leading to an invincible leadout and a guaranteed win up the steep climb.

The Big Hill

A road race with one very large hill is simple: use all of your team to deliver the strongest climbers to the base of the hill first. The hill will create a break automatically and also will drop a fair portion of your team from the lead group. Use your entire team to lead from the beginning and hold your climbers at the front of the pack. Just before the climb, the team should go to the very front and drive up the pace forcing the peloton into a long line and giving your climbers as much lead over the back of the pack as possible. When the climb starts your climbers will not have to deal with narrow roads and the accordion effect of the peloton. The faster riders of the team should set the pace up the hill according to basic paceline strategies, while the team leader maintains an advantageous position behind as many other teammates as possible. The goal of the domestiques is to set a pace high enough to prevent attacks that force your team leader to

accelerate. A smooth, fast tempo keeps the team leader in position and protected. If a domestique can no longer hold the pace he or she should move over and let other teammates lead even if the tired domestique drops out of the front pack. Hopefully, by starting the hill with your whole team in front there will be multiple riders in the front group at the top.

At the summit of the hill, the small group that is left in the front should contain at least one rider from your team. The teammates left should follow normal small group strategies to ensure their group is not caught on the downhill or on a later flat section of the course. Dropped teammates should not assist in any chase efforts unless they are sure there are no teammates in the front group. Because the protected climbers were able to climb the hill from the front and did not have to hold their position on the run-in to the hill, they should have as much energy as possible to attack the small group and gain a solo victory.

The Mental Aspect of Climbing

Climbs longer than a mile or so forces each rider to establish a pace that is comfortable. The level of this pace is dependant on the person's fitness and pain threshold and may be required to change in response to other racers. A rider starting in the front group with a high level of fitness has the best chance of reaching the top first as long as they are mentally able to focus and maintain his or her highest pace.

When a competitor attacks the small group, each rider must quickly decide weather they are able to accelerate to mark the attack. Once a rider is in their comfort zone and personally believes they are going as fast as possible the thought of accelerating is not appealing. Great riders always know they can go a little faster while climbing and respond to attacks while riders unfocused on winning the race are not able to mentally break out of their comfort zones. When a rider's mind tells them that they can not go any faster it is said that they have cracked and they lose their position at the front of the field.

The speed of a group racing up a climb will constantly change if no one is willing to force a consistently high pace. In an attempt to gain the lead, individual riders will chose to attack the group as it slows forcing everyone to constantly adjust their speed and mental comfort zone. Every attack provides the opportunity for strong riders to mentally lose their focus and fall off the pace.

Having a domestique from your team control the pace of a long climb forces every rider in the group to the limit of his or her comfort zone. This constantly high pace assures the team that their leader is not attacked. If the team leader is not attacked, they can find an acceptable mental comfort zone and ascend the climb without fear of losing their position. Competitors that cannot maintain this pace are dropped.

Technical Criterium

All team tactics in criterium races are dictated by the corners. Because the first riders through a set of turns can maintain top speed, every other rider in the peloton must work harder to stay with the group. The protected rider should take as many of these turns near the front of the pack as possible. This can be performed using either the amoeba formation or the train. These two strategies should be combined to both save as much energy as possible and yet keep the protected rider near the front at all times. If the team is regularly putting all ten or so riders through a set of corners in front, it is a great tactic to have the front five riders attack through the corner while the second five riders coast and create an instant gap. This is a very hard maneuver to coordinate, but also very effective if the peloton is being lazy.

Before a criterium it should be decided as a team if primes are important or if the race win is what the team wants. If the race win is important, just keeping your rider near the front and reorganizing after each prime is the best strategy. If winning primes is worth more to the team, a leadout should be formed for each prime lap from the amoeba near the front of the peloton.

Primes versus Race Wins

As a Kansas racer attending the Nebraska State Criterium Championships a race win meant very little to me. There wasn't any money at stake and the state title went to the first Nebraska racer across the line. The four corner criterium course was wide and fast so positioning was simple and the corners meant that only a couple of racers would contend for each prime. The twenty lap race would have four primes before the final finish.

Knowing that the end of the race meant nothing, I decided only to race for each prime. Twenty laps later I had won a new helmet and two cases of Powerbars while the race winner was holding a cheap bronze medal.

Wide and Fast Criterium

The best race to learn team tactics is a wide and fast criterium. Here there is plenty of room for your teammates to get into position at the front. There are also plenty of opportunities to practice leadouts, attacks, and blocking. Attacking can come at any point in the course and the prime laps create enough chaos to allow gaps to form and to break up chase groups. These races are really fun for organized teams.

The Art of the Training Race—Wells Avenue

Every Sunday during the Spring, Summer, and Fall for over twenty years there has been a bicycle race at the Wells Avenue Office Park in Newton, MA. The three races held each week have attracted the attendance of Tyler Hamilton, Frank McCormack, and a host of other professional racers. The Sunday morning time slot allows most racers in New England to race twice a weekend while the low-key nature of the event allows teams to build their skills each week.

The course has only one real corner. The rest of the one mile loop is traffic controlled and wide enough to allow anyone to move throughout the peloton. These conditions allow any team to attempt a leadout in pursuit of the famous \$5 primes, a can of Shaklee drink mix, a pound of Peets coffee, or some Slick Willy bike lube.

Six weeks into the race season the category 5 team from the Boston Road Club decided they could use team tactics to win every prime in the Wells race that weekend. On Saturday the team went to Wells to practice their leadout coordination and timing. In two hours the team felt like everyone knew what was necessary to win each sprint.

Riders without a team often hesitate when a prime lap is announced. A single rider is typically not strong enough to take the lead, hold it for an entire lap, then win the sprint. An organized team has an opposite reaction. The BRC guys sprinted to the front as soon as a prime was announced. With the rest of the peloton hesitating, the attack by the team forced everyone in the pack into last place and required they work hard and chase the train formed by the BRC. The rotating train allowed the BRC to keep the pace high enough that every other rider could only hang on the rest of the lap. In the final corner, the eight BRC riders had formed a line and each member started to sprint. The energy saved by the group combined with the attack a mile before the finish let the BRC win every prime and eventually the race---all with just two hours of practice and perfect execution.

Chapter 6 Rider Upgrades and Team Success

After four weeks of team preparation including two track practices and a couple of training races your team dominates the first race of the season using attacks to successfully force the other riders to chase your team. The break didn't last long (need to develop more attacking speed) but the amoeba protected your strongest rider the entire race. The leadout into the final sprint functioned perfectly placing your rested sprinter close enough to the front to take the win.

The excitement after a team win is phenomenal. Everyone is happy and even the new riders were able to contribute an early attack and help the amoeba at times. The race next weekend has been mentioned already and improvement ideas to the team strategy are apparent to all. The team captain's organization brought the group together and every rider learned valuable lessons on the art of road racing.

Handling Each Win

The race win brought in \$40 in cash and some merchandise to the team leader. Standing on the podium to accept the prize brings a smile to the rider's face as all the years of training combine together for this moment. With the team's help, the team leader experienced something that very few riders ever accomplish—a race win.

Although everyone on the team celebrates, the same question is on everyone's mind: "what is the team leader going to do with the \$40?" Ouch. The team that just worked so hard for each other may also drive itself apart.

If the team captain created a purse strategy at one of the initial team meetings the moment would clear itself up quickly. If nothing has been discussed, the team captain should bring up the sticky subject as soon as possible to prevent the team from disintegrating.

When a team works together for a race win the person (or people) winning prizes should voluntarily distribute the money and prizes to the rest of the team. Sometimes, though, the team doesn't provide a leadout, fails to attack the peloton, and a single rider from the team happens to win with their own ability. Should the race winner distribute the winnings now?

All of these questions create a tough balance for a team. Some typical solutions are:

- Merchandise is kept by the race winner.
- Merchandise is kept by the team and sold at the end of the year.
- Merchandise is distributed as best as possible.
- Cash is distributed to all team members evenly.
- Cash is used to buy lunch after the race.
- Cash is used to buy dinner later in the week.
- Cash is deposited in the team bank account and distributed at the end of the year.

The easiest method of organizing the purse strategy is to have the winner keep all merchandise and use any money awarded to pay for lunch immediately following the race. Usually merchandise isn't worth a whole lot (a couple of tubes, or a pump) or the race winner already has the product at home. The winner might take what she wants and gives the leftovers to whoever could use it on the team. If the team wins a large amount of merchandise throughout the year, the garage sale is a good option to convert the goods into cash which is easier to distribute among the members.

Cash is usually awarded as a check so it is hard to divide the winnings evenly among the riders on the team immediately after the race. Saving the money until the end of the year sounds fair, but this rewards the rider that doesn't go to each race. Buying the team lunch is a nice gesture that makes the team even more motivated to win every weekend.

A team leader that chooses to keep all the winnings even if the team helped win the race quickly loses the support of the team. If one race leader gives everything back to the team and another one keeps it all to themselves, which team leader is going to get more support in the next race? Most professionals give all their winnings to their team in order to keep the team working as hard as possible in the next race. The leader gets the glory, the domestiques get the cash.

Choosing the Next Team Leader

If the race schedule from the beginning of the year is intact, the race leader should already be chosen for the next race. If one rider is winning and riding well, changing team leadership is an option that may disrupt the team effort. Forcing a rider that just tasted a win to be a domestique the next week works only for extremely team-oriented racers.

Three things affect how this situation should be resolved. Team experience is the first item the team captain should consider. If the entire team has a similar level of experience, each rider should be allowed to be the team leader for the races that suit their talents the best. If the entire team has the same ability to win races, the team leadership should change more often.

The second consideration is team performance. If the team leader scheduled for the next week is not riding well or did not work with the team this week, that rider may want to exchange the team leader role from next week for one later in the season. Because each human body reacts differently to training, a rider may not be ready to peak physically as originally scheduled in February. Switching up the team leadership role for a couple of races later in the season is easy and good riders will understand the change.

The next week's race may be a completely different race than the first. A criterium demands different abilities than a road race and if one rider wins a sixteen mile technical criterium he or she may not be the best choice for team leader in next week's sixty mile road race. The team captain should point out each rider's individual talents and make sure that rider is set to win the other races in the year that suit those skills.

It is always possible to have two or three protected riders. Having more riders increases the chance of winning a group sprint if there are enough domestiques to protect everyone.

If the team has a wider range of experience levels the most experienced riders probably should be protected each week. Less experienced riders will gain more fitness by protecting the team leaders and attacking than if they were working alone. Eventually, the inexperienced riders will be strong enough to win races or the team leaders will gain enough points to upgrade.

The more experienced riders should also give up their position as team leader in lower level or practice races as often as possible. Two riders can not stay in race shape all season and if they occasionally give up their position and work for the team they will be rewarded by stronger team efforts in the next race. Again, the more the team leader gives back to the team by working in smaller races and distributing their winnings, the better the team will work together and the more wins everyone will achieve.

Rider Upgrades

A strong rider that has raced for a couple of years usually desires an upgrade to the next level of racing. Racing with riders that do not know how to paceline, do not sprint in a straight line, and take corners slowly is not much fun after a couple of years. When the strong rider reaches the point where another good finish is enough to upgrade he or she often will demand to be placed in the team leader position.

If the team has two or three other strong racers ready to upgrade the team leader must decide which one should be given the chance to upgrade first. Each rider has worked hard for the team and shared the team leadership position until now and what should the team captain do to work this out?

One method is for the team captain to put both riders on the attack team and encourage them to attack together and try to create a breakaway during the race. A break will guarantee that both riders finish in a top position and achieve the points for the upgrade. Because both riders are talented, driven, and in excellent condition, they will possess the power and ability to sprint ahead of the peloton and create a strong break that continues to the finish. Weather the break wins or fails, both riders will have put enough effort into the race that they will be satisfied.

A second choice is again to pick the race where a particular rider is most talented and let that govern the decision. If the next race is a criterium, chose the criterium master as the team leader. Encourage the other rider to be the main leadout rider to make sure the team leader finishes at the top of the field. Hopefully there will be two races that weekend and both riders can take turns being the team leader and both can upgrade at the same time.

Success

A team captain sitting at the lunch table after a big race win actually deserves the most credit for the success. Without the early season meetings and long, foul weather bike rides the team wouldn't have built up the paceline skills to take control of the peloton. Without the annoying practice sessions forcing everyone to pick up their speed and intensity and try new tactics, the team would not have created the break or performed the perfect leadout. Finally, without the extra organization of knowing the race course, the basic tactical outline and the driving directions each week the team would arrive at the line disorganized and out-of-sync for the race.

When the team comes together and everyone has fun at the race the effort is suddenly forgotten. Putting together a new team every year is as much of a challenge as winning the big race. Instead of training alone this year, give teamwork a chance and try to learn more about bicycle racing. The skills formed by working together only add to the experience of joining the peloton. Learning how to gain every strategic advantage is a skill that never will leave you and only builds from the initial effort.

Appendix 1 Team Questionnaire

This example questionnaire can be printed directly from the book and used at your first team meeting. Having a written detail of each member's needs, a team leader can create team features to meet these needs. Understanding the motivation of each rider allows a team leader to help dissolve rider conflicts and maintain team unity. Allowing each rider to voice their opinions about the team makes everyone comfortable with where the team is going and how the team is organized.

Team Questionnaire

Name: Address: Phone: Email: Color of Bicycle:	<table style="width: 100%; border: none;"> <tr> <td style="text-align: right;">Racing</td> <td></td><td></td><td></td><td></td><td></td><td></td> </tr> <tr> <td style="text-align: right;">Experience</td> <td>1</td><td>2</td><td>3</td><td>4</td><td>5+</td> <td>years</td> </tr> </table> <table style="width: 100%; border: none;"> <tr> <td style="text-align: right;">Road Race</td> <td></td><td></td><td></td><td></td><td></td> </tr> <tr> <td style="text-align: right;">Category</td> <td>5</td><td>4</td><td>3</td><td>2</td><td>1</td> </tr> </table> <table style="width: 100%; border: none;"> <tr> <td style="text-align: right;">Cyclocross</td> <td>A</td><td>B</td><td>C</td> <td></td> </tr> </table> <table style="width: 100%; border: none;"> <tr> <td style="text-align: right;">MTB</td> <td>Expert</td> <td>Sport</td> <td>Beginner</td> </tr> </table>	Racing							Experience	1	2	3	4	5+	years	Road Race						Category	5	4	3	2	1	Cyclocross	A	B	C		MTB	Expert	Sport	Beginner
Racing																																				
Experience	1	2	3	4	5+	years																														
Road Race																																				
Category	5	4	3	2	1																															
Cyclocross	A	B	C																																	
MTB	Expert	Sport	Beginner																																	
List previous team experience:																																				
What did you like about these teams?																																				
What would you change about these teams?																																				
Do you have a coach? A brief description of your training philosophy or schedule:																																				
How many hours a week do you expect to train:																																				
Are you willing to ride in the mornings? What time to you absolutely have to be at work?																																				
Can you ride on the weekends? Circle the times that are good for you: Saturday: Morning Afternoon Sunday: Morning Afternoon When would you prefer the training rides to start: Where would you like training rides to meet:																																				

What are your strengths: (circle all that apply)

Time Trials	Hill Climbs	Criteriums	Road Races	Team Tactics
Rolling Hills	Long Hills	Corners	Leadouts	Sprints

What would you like to improve about your riding?

What are your two most favorite races of the year?

1:

2:

Why do you like these races?

Appendix 2 Winter Riding

Promoting a long, slow ride in the depths of winter may be the hardest job a team captain faces. Everyone who has joined the team has an obvious desire to ride their bike accompanied by an excuse why it's too cold. At the same time, each person has a weakness that will get them out on the road. Bringing the whole team together relies on identifying excuses and advertising the benefit of each ride.

Common rider weaknesses:

- Social. Once you have a couple of riders going out each weekend, the rest of the team eventually will feel guilty for not attending.
- Clothes. Use the clothes list below to get everyone set for winter riding.
- Fun. Winter rides are great because you can talk the entire time. Without sprints and other wackiness to get in the way, these can be some of the best rides of the year.
- Race benefits. Races are won with base miles and aerobic training. After a strong winter campaign, fewer sprinters will be dropped hills and hill climbers can hang on in during leadouts.
- Candy. I never eat healthy granola sports-oriented mush in the winter. Existing on candy bars, cookies, and even soda is only possible during a calorie burning winter ride. A water bottle full of hot chocolate or coffee isn't bad either.
- Epic rides. Six hours, eight hours, go long or go home. Build endurance by going out for long rides and make the most of every good day. When a rider learns how much training the group is doing each weekend he or she will want to join the procession.

It took a year in Boston to realize that it was even possible to ride a bike in horrendous weather. Once I found out that forty to sixty people gathered each weekend for an eighty mile tour no matter what the temperature, I found the thickest tights I could buy and headed out on my bike. Tricks like upside down water bottles, thick non-freezing sports drinks, and a bottle of hot tea in the back pocket make winter riding great fun. Because the weather doesn't get above freezing from November to late April in New England, riding and racing in the wintry mix is an important skill.

There are no fashion shows in the winter peloton. Beating the elements requires a lot of clothes. Here is a typical clothes list for temperatures in the teens.

Feet require range of motion, sweat control, and insulation. For really cold days, add a handwarmer packet to the system.

- Thin, wool or polypropylene ski socks near the skin. Half of my team wore electric socks that had a built-in battery-powered heater.
- Shoes that are not tight.
- If it is wet outside, add a grocery bag outside the shoe for water protection.
- Oversized wool ski socks outside of the shoes.

- Neoprene booties.

Legs also need range of motion, but fewer layers.

- The thickest, warmest tights you can find. Do not mail order these tights. Go to the store and feel the fabric making sure it has a fleece layer and possibly some windproof panels for extra warmth. These will cost over \$100 but are an important investment for winter riding.
- Wind pants help if there is slush or precipitation on the ground.
- Regular cycling shorts go over or under the tights.

Upper body

- Non-cotton turtleneck.
- Various layers of fleece.
- Oversized cycling jersey mainly for the pockets.
- Quality cycling jacket that is water-proof and breathable.
- Small water bottle filled with warm water or tea for back pocket.

Hands

- Ski or snowboard gloves. Cycling specific gloves are usually more flexible, but rarely have enough warmth to maintain warmth. Lobster gloves with two finger pockets are the best, but some mittens work well and some gloves are just as good. Make sure the gloves fit a little loose or finger circulation will be an issue.
- Extra fleece gloves in back pocket allow you to change to warm gloves as necessary.
- Hand warmer packets are excellent for really cold temperatures.

Head and Neck

- Fleece neck warmer that pulls over mouth keeps air warm as it enters your body.
- Under-helmet hat that covers ears is a necessary investment.
- A lot of riders combine the neck warmer and hat by wearing a balaclava.
- Sunglasses keep the cold off of your eyes and protect more of your face.

It took me over ten minutes to completely dress for a cold weather ride but I was very comfortable on the bike. If things got a little cold or boring, the team would just stop in a gas station to warm up and grab some food. Eating and drinking are important to staying warm and sometimes we would stop every hour. I cannot remember a time when a ride was completely canceled because of extreme temperatures. If it snowed too much, everyone would just go skiing instead of boring inside workouts.

Appendix 3 Glossary

Clicking on an entry shows the word in context.

Amoeba The most basic defensive tactic where the domestiques maintain a position near the front of the peloton with the team leader continuously riding in the draft behind them.

Breakaway If a small group is formed more than a few seconds in front of the main peloton it is known as a breakaway or break.

Bridge Moving from the peloton to an existing breakaway is known as a bridge. The bridge may be a single rider or a small group of riders attempting to join the break.

Categories Racers in the United States are divided into five skill-based racing categories. The lowest level is category five and includes riders with less than ten races of experience. Levels three and four are intermediate levels with longer, faster races. Levels one and two often race together and are sometimes grouped with professionals.

Chase Chasing occurs when a breakaway forms or a rider is dropped from the peloton. In each case the chase occurs until the peloton regroups. A breakaway is chased by the main peloton until it is caught. A dropped rider chases the peloton in order to regain the benefits of drafting within a large group.

Circuit Race Any race held on a course 2-6 miles in length with multiple laps. These races are more spectator friendly than a road race, but are generally longer and less technical than a criterium.

Criterium A road race held on a circuit of generally less than 2 miles, often with multiple sharp corners.

Defensive Tactics Any teamwork designed to conserve the energy of the team leader. By conserving energy, the team leader has an advantage during a large hill climb or the final sprint at the end of the race.

Domestique A rider designated to use their energy in any way to conserve the energy of the team leader. This usually includes riding in front of the team leader, working in a train to chase down breaks, marking attacks from other teams, or participating in a leadout to position the team leader for the final sprint.

Drafting A person riding behind another rider is drafting, or gaining the aerodynamic advantage of the slipstream formed by the first rider. In a crosswind, the rider may not be directly behind the front person, but is still saving energy.

Dropped riders If the peloton holds a high pace for a long time or over a hill, some riders will fatigue and will not be able to maintain contact with the peloton. These riders are said to be dropped from the peloton until they recover and chase back to the main group.

Echelon A diagonal formation of riders caused by a crosswind. Since the draft of each rider in a crosswind is no longer directly behind that rider, a paceline will no longer be straight and will form a diagonal across the road. Since a road has a limited width, a large group of riders in an echelon may split into multiple groups to stay out of the wind.

Flat-front peloton A peloton going at a slow speed has a flat front caused by the large number of riders willing to ride in the wind to maintain their position.

Gatekeeper A rider at the back of a paceline who does not participate in the paceline is the gatekeeper. In a train, this rider allows the paceline to continue without interference from the rest of the peloton.

General Classification The winner of a stage race is the rider with the lowest accumulated time. The results of each stage are added together and the rider with the lowest time is said to be leading the general classification.

Leadout A positioning tactic where a train is used to win a pack sprint. The train is formed before the sprint begins and eventually each rider accelerates to their top speed and pulls off putting the team leader in position to sprint for the race win.

Marking an attack Someone following an attacking rider is marking that attack. If an opposing rider attacks it is usually advantageous for one of your teammates to accelerate and follow the attack so the team does not need to chase the breakaway later in the race.

Offensive Tactics An offensive tactic is any team effort designed to wear out the peloton by forcing them to chase an attack.

Paceline A paceline is a systematic method of conserving energy where each rider rides at the front of the group for a small amount of time before he goes to the back of the line to rest.

Peloton The largest group of riders in a bicycle race is known as the peloton.

Positioning Tactics Any teamwork designed to put a specific rider in position to win a race. These include leadouts, preparations for a big hill, and technical corners.

Prime Pronounced “preem.” In a race with multiple laps, the winner of certain announced laps will get an additional prize. The lap is often indicated with the ringing of a bell the lap before the award is given.

Professional Rotation A form of paceline where the front rider pulls for 5-20 seconds, then goes to the back of the line. This formation requires less coordination and movement from the group, but also requires more energy because of the longer pulls.

Relay Attacking An attacking scheme where one rider attacks the peloton first, followed by other attacks. Once the first attacker is caught, a teammate immediately counter-attacks. The team keeps attacking like this until a breakaway is formed. Relay attacking forces the peloton to continuously chase a member of your team and doesn’t allow the group to rest.

Riding in the wind Riding at the front of a group takes more energy due to the aerodynamic effects of the atmosphere. Someone in the front of the peloton is riding in the wind.

Road Race Any race on a course over 2 miles in length with few technical corners and often traveling between multiple towns. Similar to a circuit race but usually has a longer loop and fewer laps.

Small Group When the peloton is smaller than 40 or 50 riders it is a small group. More aggressive tactics are effective on groups of this size. Smaller groups of 5 or 10 riders allow additional team advantages.

Solo Rider A person in a race that is not supported by a team and is working alone to maintain their position in the peloton.

Team Captain Person responsible for team organization, calling practices, organizing meetings and running the team is known as the team captain. Not necessarily the team leader.

Team Leader The team leader is the rider chosen for each race to conserve energy by riding behind the domestiques. This rider is expected to perform gatekeeper duty throughout the race and to attack near the end of the race or sprint during the leadout.

Team Time Trial A time trial where small groups of riders are allowed to work together while riding a set distance. The team with the lowest time wins.

Time Trial A race where riders leave the start line alone and are not permitted to work together. The person with the lowest time around the designated course wins.

Train A paceline with a gatekeeper that intentionally rides at the front of a large peloton to control the peloton's speed and to maintain the team's position near the front of the race.

Training Race Any race with low entry fees and few prizes where your team will be more focused on practicing skills than winning the race. Some of these races have a no-drop rule where a dropped rider can rejoin the peloton at any time. These races are excellent for beginning riders and team trying to work on teamwork skills.